Online Appendix

The Unrealized Potential of a Parent's Promise:

Cosigned Mortgages and Access to Credit in the United States

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This appendix is referenced from the main paper and is not meant to be read on its own. Section and exhibit numbers are prefixed OA throughout, so a reference to Table OA1 is always to this document and a reference to Table B1 is always to the printed appendix.

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OA1 When the Rate Reads the Menu

The paper's pricing rule reads the lowest score on the file, the child's on a solo loan and the lower of the two on a cosigned one. This section states the model when the rate reads the menu as well, the extension of Remark 1 with full weight on pool quality. With full weight on the pool the score drops out of the rate, so the score penalty of Section 3.1 has no counterpart here; at lower weights it enters through the posted component $\bar{\pi}(y_{\min})$ of Remark 1. The rule here reads the cosigned pool at the child's income, averaged over parental incomes, as Proposition 5(ii) assumes. The environment, payoffs and timing are those of Section 3.1 and Appendix B.1.1, with one change: the lender posts a schedule $r_k(y)$ that must be consistent with the pools it reads, so the lender becomes a player and the equilibrium a fixed point. Every $r(y)$ of the paper is read here as the menu's own rate $r_k(y)$. The equal rate pass-through of Section 5 is evidence against this rule; it is kept here as the natural alternative and because Corollary 1 refers to it.

OA1.1 Equilibrium Definition

An equilibrium is a quadruple $(y_G^{pool}, y_B^{sep}, \{r_k(y)\}, \{\mu_k(y)\})$ consisting of two income cutoffs, a rate schedule, and a belief system such that:

(i) **Parental optimality (participation):** Parents offer a guarantee if and only if the participation constraint (B.10) holds at the schedule, $\beta\,\Delta V(y,\theta;\, r_{co}(y), r_{so}(y)) - (1-\pi_\theta)\,\Delta u_p \geq \eta_p$, yielding offer propensities $q_\theta(y, y_p)$ and integrated posteriors $\bar{q}_\theta(y)$.

(ii) **Belief consistency:** Lender beliefs $\mu_k(y)$ are derived from Bayes' rule given the observed menu choice k, income y, and the equilibrium offer propensities $\bar{q}_\theta(y)$ wherever Bayes' rule applies. Off the equilibrium path, beliefs are admissible and deter profitable deviations.

(iii) **Competitive pricing (lender participation):** Lenders set rates by (B.17), the menu-reading rule of Remark 1, given beliefs $\mu_k(y)$, which breaks even on the occupant's own payments in each active pool; counting the parent's cover, the cosigned pool earns the rent of Proposition 1, so the lender's participation constraint $\Pi_k \geq 0$ binds on the solo menu and is slack on the cosigned one.

(iv) **Borrower optimality (incentive compatibility):** Borrowers choose menus to maximise expected utility (3) given the rate schedule $\{r_k(y)\}$, so a type-θ borrower with $o = 1$ takes the cosigned menu if and only if $\Delta V(y,\theta) \geq 0$ in (4).

OA1.2 Regularity for the Menu-Reading Rule

Beyond (A1)–(A4) of Appendix B.1.3, the menu-reading equilibrium uses four conditions on the taste shock and the rates: (M1) and (M2) make the belief–rate map a contraction, (M3) ties the income slope of the belief schedules to the prior's, and (M4) extends the boundary conditions to the pooling and separating rates.

(M1) **Boundedness:** $\sup f_\eta \leq \bar{f} < \infty$.

(M2) **Sufficient Dispersion:** The variance of F_η is large enough that $\bar{f}$ is small relative to $\mu_{co}(y)/(1 + r_{co}(y))$, ensuring $\partial ER/\partial r_{co} > 0$ globally.

(M3) **Belief Smoothness:** The dispersion of F_η is large enough that $\bar{q}'_\theta(y)$ is uniformly small, so that $\bar{\pi}'_o(y) \approx \Delta\pi\, \lambda'(y)$ for $o \in \{0, 1\}$ and the income sensitivity of both belief schedules is driven by the prior gradient $\lambda'(y)$ alone.

(M4) **Boundary Conditions under the Menu Rates.** (A2) holds also under both $r_{co}^{pool}(y)$ and r_{co}^{sep} of (OA1)–(OA2).

OA1.3 Rate-Dependent Cutoffs

Two rates are central to the analysis:

$$r_{co}^{pool}(y) \equiv \frac{1}{\bar{\pi}_1(y)} - 1, \tag{OA1}$$

$$r_{co}^{sep} \equiv \frac{1}{\pi_B} - 1. \tag{OA2}$$

The pooling rate $r_{co}^{pool}(y)$ prevails when both types are present in the cosign pool and the bank updates beliefs via Bayes' rule using (B.12). The separating rate r_{co}^{sep} prevails when the cosign pool contains only Bad types, so the bank's belief is $\mu_{co} = \pi_B$. Since $\bar{\pi}_1(y) > \pi_B$ for all y, I have $r_{co}^{sep} > r_{co}^{pool}(y)$: the separating rate is strictly higher than the pooling rate. Define the two rate-dependent cutoffs:

$$y_G^{pool} \equiv \inf\{y > y_F : \Delta V(y, G; r_{co}^{pool}(y), r_{so}(y)) \leq 0\}, \tag{OA3}$$

$$y_B^{sep} \equiv \inf\{y > y_G^{pool} : \Delta V(y, B; r_{co}^{sep}, r_{so}(y)) \leq 0\}. \tag{OA4}$$

The cutoff y_G^{pool} is the income at which Good types are indifferent under the pooling rate; it marks the exit of Good types from the cosign pool. The cutoff y_B^{sep} is the income at which Bad types are indifferent under the worst-case separating rate.

The equilibrium quadruple the cutoffs of (OA3)–(OA4) belong to is defined above; the maintained conditions are (A1)–(A4) of Appendix B.1.3 and (M1)–(M4).

OA1.4 Single Crossing, the Fixed Point and the Three Regions

Lemma OA1 (Type Cutoff Ordering under the Menu Rates)**.** *Under* (A1)–(A3) *and* (M1)–(M4)*, and for any bounded pair of menu rates, the indifference thresholds are ordered $\tilde{y}_G < \tilde{y}_B$: Good types are the first to leave the cosigned menu.*

Good types are the ones for whom cosigning stops paying first, because the insurance component $\Delta\underline{u}$ they weight least is exactly the component the riskier type values most, at every income. Proofs of both lemmas are in Online Appendix OA2.

The trap is not automatic. It has positive measure if and only if Bad types still prefer the cosigned menu at y_G^{pool} once repriced at the separating rate, $\Delta V(y_G^{pool}, B;\, r_{co}^{sep}, r_{so}(y_G^{pool})) > 0$, which is equivalent to $y_B^{sep} > y_G^{pool}$. I call this the *survival condition*.[1]

The pooling rate (OA1) is defined against beliefs that are themselves formed at that rate, since r_{co} enters ΔV and hence the offer propensities (9). It is therefore a fixed point rather than a formula, and it exists.

Lemma OA2 (Belief–rate fixed point)**.** *Fix $y < y_G^{pool}$ and let $\mathcal{R} \equiv \big[1/\pi_G - 1,\ 1/\pi_B - 1\big]$. Define*

$$T_y(r) \;\equiv\; \frac{1}{\pi_B + \Delta\pi\, \Lambda_1(y; r)} - 1, \tag{OA5}$$

where $\Lambda_1(y; r)$ is the posterior (B.12) evaluated at the offer propensities $\bar{q}_\theta(y; r)$ that (9) generates when the cosigned rate is r. Under (A1), (A3), (M1) *and* (M2)*, T_y is a continuous self-map of the compact interval $\mathcal{R}$ and therefore has a fixed point, which is the pooling rate $r_{co}^{pool}(y)$ of* (OA1)*. If in addition $\bar{f} \equiv \sup f_\eta$ is small enough that $|T_y'| < 1$ on $\mathcal{R}$ — which is what the dispersion condition* (M2) *delivers — the fixed point is unique and best-response iteration on* (OA5) *converges to it from any starting rate in $\mathcal{R}$.*

The two halves do different work. Existence needs almost nothing, because beliefs are bounded: whatever parents do, $\mu_{co} \in [\pi_B, \pi_G]$, so the posted rate cannot leave $\mathcal{R}$. Uniqueness needs the offer margin to be insensitive enough to the rate, and it is the same condition that keeps the naive iteration from cycling. Without it the map can oscillate — a higher rate drives out the good types' parents, which raises the rate — which is the Akerlof cobweb in its dynamic form, and it is why the quantitative model of Section 6 damps the corresponding update rather than iterating it directly. Nothing here claims uniqueness for that model; the claim is about the stylized one, under *(M1)–(M3)*.

Proposition OA1 (Three-Region PBE under the Menu-Reading Rule)**.** *Suppose* (A1)–(A4), (M1)–(M4) *and Assumption 1 hold, and let the pooling rate be the fixed point of Lemma OA2. What follows characterises the sorting equilibrium given that schedule. The corresponding fixed point of the full life-cycle economy, which is a system in the whole premium matrix rather than a scalar at each income, is solved numerically in Section 6.*

Three Regions. *If the survival condition holds, there exists a unique PBE* within the class of equilibria satisfying (A1)–(A4), (M1)–(M4) and Assumption 1, given the rate schedule*, with feasibility*

[1] If it fails, Good types exit and Bad types follow immediately: Region 2 collapses to a point and the equilibrium reduces to two regions with the single cutoff y_G^{pool}. Statement and proof in Online Appendix OA2.

threshold y_F and income cutoffs $y_F < y_G^{pool} < y_B^{sep} < \infty$. The equilibrium rate schedule is:

$$r_{co}(y) = \begin{cases} r_{co}^{pool}(y) = \dfrac{1}{\bar{\pi}_1(y)} - 1 & y < y_G^{pool}, \\ r_{co}^{sep} = \dfrac{1}{\pi_B} - 1 & y \in [y_G^{pool},\, y_B^{sep}), \\ \textit{(menu infeasible)} & y \geq y_B^{sep}, \end{cases} \tag{OA6}$$

where r_{co}^{sep} is constant in y since the belief $\mu_{co} = \pi_B$ is pinned by full revelation. Beliefs and sorting are as follows:

- ***Region 1*** *($y < y_G^{pool}$): Both types choose $k = co$ if $o = 1$. Lender belief $\mu_{co}(y) = \bar{\pi}_1(y)$ via Bayes' rule (B.12). The parental screen is active: $q_G(y, y_p) > q_B(y, y_p)$ by Lemma 2.*
- ***Region 2*** *($y \in [y_G^{pool},\, y_B^{sep})$): Only Bad types choose $k = co$. Menu choice $k = co$ fully reveals $\theta = B$, so $\mu_{co}(y) = \pi_B$ by Bayes' rule. Bad-type parents cosign actively; Good-type parental offers are payoff-irrelevant since Good types choose $k = so$ regardless of o.*
- ***Region 3*** *($y \geq y_B^{sep}$): Both types choose $k = so$ regardless of o. No cosign pool exists.*

The full proof of Proposition OA1 is in Online Appendix OA2.

Region 2 is the trap, and its sharpest testable implication is that the cosigned–solo rate differential should be widest over the intermediate income range $[y_G^{pool}, y_B^{sep})$, where the cosign belief sits at its lower bound π_B while the solo belief still reflects the full population mix. The gap $\bar{\pi}(y) - \pi_B = \Delta\pi\, \lambda(y)$ is maximal there.

OA1.5 Pool Quality and Sensitivity Decoupling

I characterise pool quality in the three-region equilibrium of Proposition OA1. Let ω_{co}^1 and ω_{co}^2 denote the population shares of cosign borrowers in Regions 1 and 2 respectively, with $\omega_{co}^1 + \omega_{co}^2 = 1$, and let $f_{co}^1(y)$ and $f_{co}^2(y)$ denote the corresponding income densities, each normalised to integrate to one over their respective regions. On the solo side, let $f_{so}^{lo}(y)$ denote the income density of solo borrowers with $y < y_G^{pool}$, and split the high block at y_B^{sep}: $f_{so}^2(y)$ on $[y_G^{pool}, y_B^{sep})$, where the cosign menu is open but only Bad types take it, and $f_{so}^3(y)$ on $[y_B^{sep}, \infty)$, where the cosign menu is infeasible. The corresponding population shares are ω^{lo}, ω_{so}^2 and ω_{so}^3, summing to one, with $\omega^{hi} \equiv \omega_{so}^2 + \omega_{so}^3$.

The Region-2 solo pool is *not* priced at the unconditional benchmark. There every Good type chooses solo, while a Bad type does so only when no guarantee was offered, which happens with probability $1 - \bar{q}_B(y)$. Bayes' rule therefore gives

$$\Lambda_2(y) \;\equiv\; \Pr\Big(\theta = G \mid y,\, k = so,\, y \in [y_G^{pool}, y_B^{sep})\Big) \;=\; \frac{\lambda(y)}{\lambda(y) + \big(1 - \lambda(y)\big)\big(1 - \bar{q}_B(y)\big)} \;>\; \lambda(y), \tag{OA7}$$

and hence $\bar{\pi}_2(y) \equiv \pi_B + \Delta\pi\,\Lambda_2(y) > \bar{\pi}(y)$. The same revelation that pins the Region-2 cosign pool at π_B must cleanse the solo pool there: what leaves one pool has to be accounted for in the other. Above y_B^{sep} no cosign menu exists, so the solo pool carries the unconditional $\bar{\pi}(y)$.

Under the rate that reads the menu, (B.17), a one-unit increase in g raises the posted rate by $1/\mu_k(y)$. Writing $\beta^k_{rate}(y) \equiv 1/\mu_k(y)$ for this rate pass-through, a worse pool passes more through:

$$\mu_{co}(y) < \mu_{so}(y) \iff \beta^{co}_{rate}(y) > \beta^{so}_{rate}(y). \tag{OA8}$$

Under the pricing rule (2) each menu passes through $1/\bar{\pi}(y_{\min})$, equation (11) in the main text, the same on both where the parent's file is at least the child's; the ordering here and in Propositions OA2 and OA3 concerns the menu-reading rule.[2]

Proposition OA2 (Pool Quality Decomposition)**.** *Under the three-region equilibrium of Proposition OA1, the pool average qualities are:*

$$\bar{\mu}_{co} = \underbrace{\omega^1_{co}\int_0^{y_G^{pool}} \bar{\pi}_1(y)\, f^1_{co}(y)\, dy}_{\text{Region 1: pooling, parental screen active}} + \underbrace{\omega^2_{co}\int_{y_G^{pool}}^{y_B^{sep}} \pi_B\, f^2_{co}(y)\, dy}_{\text{Region 2: pure adverse selection}}, \tag{OA9}$$

$$\bar{\mu}_{so} = \underbrace{\omega^2_{so}\int_{y_G^{pool}}^{y_B^{sep}} \bar{\pi}_2(y)\, f^2_{so}(y)\, dy}_{\text{Region 2: Good types plus unoffered Bad types}} + \underbrace{\omega^3_{so}\int_{y_B^{sep}}^{\infty} \bar{\pi}(y)\, f^3_{so}(y)\, dy}_{\text{Region 3: no cosign menu}} + \underbrace{\omega^{lo}\int_0^{y_G^{pool}} \bar{\pi}_0(y)\, f^{lo}_{so}(y)\, dy}_{\text{Region 1: unoffered only}}. \tag{OA10}$$

Adding and subtracting the unconditional benchmark $\bar{\pi}(y)$ inside each interior integral decomposes the quality gap into six economically distinct terms,

$$\bar{\mu}_{so} - \bar{\mu}_{co} = \underbrace{B}_{>0} + \underbrace{D_1}_{<0} + \underbrace{D_2}_{<0} + \underbrace{S}_{>0} + \underbrace{R}_{\lessgtr 0} + \underbrace{A}_{>0}, \tag{OA11}$$

stated in full in Online Appendix OA2. Each carries an interpretation. B is the base effect: solo borrowers above y_G^{pool} have no cosigned counterpart, and they are the good risks who left. $D_1 < 0$ and $D_2 < 0$ are the two screen *terms, and they work against adverse selection: $o = 1$ lifts the cosign pool above the unconditional benchmark and $o = 0$ depresses the low-income solo pool, both by Lemma 2. $S > 0$ is the mirror image of the trap on the solo side: in Region 2 the Bad types who hold an offer leave for the cosign menu, so the solo pool they leave behind is* better *than the unconditional benchmark, by* (OA7)*. It is the accounting counterpart of A — the same exit, seen*

[2] Pass-through is stated pointwise, $\beta^k_{rate}(y) \equiv 1/\mu_k(y)$, rather than as a pool average $\mathbb{E}[1/\mu_k \mid k]$, and the choice is deliberate: the two pool averages are *not* ordered by the ordering of the pool means, since Jensen's inequality bounds each expectation below by the reciprocal of its own mean and says nothing about which is larger. The gap it leaves is within-pool dispersion, and the pools differ in dispersion as well as level, μ_{co} being constant at π_B over Region 2 while μ_{so} varies over the whole support, so the pool with the lower mean can carry the smaller $\mathbb{E}[1/\mu]$. The pointwise statement is also what the design needs: the discontinuity of Section 5 is local to the conforming limit and prices it with a scalar pool quality.

from the other pool — and it widens the measured gap rather than narrowing it. R is a reweighting term that the maintained primitives do not sign, because it compares mass-weighted integrals of the same benchmark over sub-pools whose masses differ; its contribution is an empirical question, absorbed into the pool-quality gap the NSMO measures directly. $A > 0$ is the trap: Region 2 cosigners enter the average at the worst admissible belief π_B rather than at the benchmark $\bar{\pi}(y)$ — indeed rather than at the still higher pooling belief $\bar{\pi}_1(y)$ they would have carried without the good types' exit — so the trap depresses cosigned quality and widens the gap. That channel is absent from any single-cutoff model and is what the three-region structure adds.

Adverse selection therefore dominates, $\bar{\mu}_{co} < \bar{\mu}_{so}$, exactly when $B + S + A$ outweigh the two screening terms, and $B+S+A > |D_1|+|D_2|+|R|$ suffices. Whether it does is an empirical question, settled by the NSMO pool-quality evidence rather than by the primitives: the decomposition says which forces are in play and with what sign, not which of them wins. Taking that sign from the data, (OA8) *then orders the rate pass-through at any income where the priced gap holds, and it is that* ordering *— higher rate pass-through set against lower rejection pass-through — which Section 5 evaluates, as a bound on the first and a sign on the second.*

The enforcement hazard and the compensating factor. Underwriting nets a compensating factor off the raw ratio before applying a common standard, rather than applying two different standards: this is the hazard shift (12) of Section 3.5, where H is increasing, differentiable and convex on the range of *effective* ratios the two pools occupy, and $\gamma(y_p) \geq 0$ is increasing in the guarantor's income. A solo application carries no compensating factor because there is no second borrower to compensate with; a cosigned one does, because the automated track evaluates the blended file within an overall risk assessment and imposes no occupant-only ceiling (Section 2). Leniency is therefore a matter of *degree* in one primitive, the weight τ the underwriter puts on the guarantor, rather than a difference of standard.

Proposition OA3 (Sensitivity Decoupling under the Menu-Reading Rule)**.** *Let the rate read the menu,* (B.17) *in Remark 1. The rejection pass-through for pool k is* (14) *of Section 3.5, with $\sigma_k = L/(Y_k\,\mu_k(y)) = DTI_k$, the average of the borrower-level DTI sensitivity weighted by the slope of the enforcement hazard at each borrower's position. (Under the sharp-cap special case this reduces to the familiar threshold-density formula $\mathbb{E}^k[\sigma_k \mid DTI_k = \overline{dti}] \cdot f^k_{DTI}(\overline{dti}; g)$.) The model predicts $\beta^{co}_{rej} < \beta^{so}_{rej}$ despite $\beta^{co}_{rate} > \beta^{so}_{rate}$ when the following two conditions hold:*

(C1) ***Quality Dominance:*** *$\bar{\mu}_{co} < \bar{\mu}_{so}$, established in Proposition OA2 under the three-region equilibrium. This implies $\mathbb{E}[\sigma_{co}] > \mathbb{E}[\sigma_{so}]$: cost shocks pass through more aggressively to the riskier cosign pool on the intensive margin.*

(C2′) ***Sufficient compensating-factor weight:*** *$\tau \geq \tau^*$, where τ^* is the unique weight at which $\mathbb{E}^{co}[H'(DTI_{co} - \tau\gamma)\,\sigma_{co}] = \mathbb{E}^{so}[H'(DTI_{so})\,\sigma_{so}]$. Under* (12) *the left-hand side is strictly decreasing in τ, so τ^* is unique where it exists and is finite whenever $H'(d) \to 0$ as the effective*

ratio falls below the cap. At $\tau \geq \tau^*$ the leniency ratio

$$\frac{\mathbb{E}^{co}[H'_{co}(DTI_{co})\,\sigma_{co}]}{\mathbb{E}^{so}[H'_{so}(DTI_{so})\,\sigma_{so}]} \;\leq\; 1 \tag{OA12}$$

follows, *and it is strictly decreasing in τ: the condition restricts a primitive rather than the object the proposition compares. The institutional basis for $\tau > 0$ is the two-track rule: the occupant-only* 43% *ceiling applies on the manual track only, while the automated track evaluates the blended ratio of all borrowers within an overall risk assessment (Fannie Mae, 2020). Empirically the rejection hazard's slope through and above the* 43–45% *range is several times flatter for cosigned than for solo applications (Online Appendix OA5.9), which is an estimate of the left-hand side of* (OA12) *and so of τ relative to τ^*.*

When (C1) and (C2′) both hold, the leniency channel dominates the quality channel on the rejection margin: although cosigners face higher rate pass-through, the effective ratio the underwriter reads for them sits far enough down the hazard that $\beta^{co}_{rej} < \beta^{so}_{rej}$. Note the resolution of an apparent tension: cosigned applications concentrate at and above the cap on their raw *ratio, because families stretch on joint income, while sitting in the flatter part of H on their* effective *one. That is what a compensating factor is.*

The derivation of (14) and the proof that (C1) makes (OA12) a non-vacuous restriction are in Online Appendix OA2.

OA2 Proofs of Theoretical Results

Proof of Proposition 4 (Two Margins, Only One Affected). Both parts rest on one feature of (B.8): the loan-to-value test's only menu-dependent object is $\overline{\text{ltv}}^{k}$, an institutional constant that is a function of neither rate nor income, so the guarantee cannot slacken the loan-to-value test and weakly tightens it.

(i) Fix a household in $\mathcal{E}(\overline{\text{ltv}}^{co})$, i.e. with $a < (1-\overline{\text{ltv}}^{co})P$. Its loan-to-value test fails identically in r_{co}, so by (B.8) $\mathcal{F}_{co} = \emptyset$ for every $r_{co} \geq 0$ and its contribution to A_{co} is zero at every rate. Differentiating the zero function gives $\partial A_{co}/\partial r_{co}|_{\mathcal{E}} = 0$. The economics is that the two tests are conjunctive: a price operates inside the feasible set and cannot enlarge it along a dimension the price does not enter.

(ii) Raising $\overline{\text{ltv}}^{co}$ to $\overline{\text{ltv}}^{co}+\varepsilon$ lowers the deposit threshold to $(1-\overline{\text{ltv}}^{co}-\varepsilon)P$, so the households newly passing the loan-to-value test are those with $a \in [(1-\overline{\text{ltv}}^{co}-\varepsilon)P, (1-\overline{\text{ltv}}^{co})P)$, a set of positive measure for every $\varepsilon > 0$ whenever F_a has positive density there. They are admitted if they also clear the income test at the enlarged loan $L^* = \overline{\text{ltv}}^{co}P$, which is the qualification in the statement; the mass of admissions is then strictly increasing in $\overline{\text{ltv}}^{co}$. □

Proof of Proposition 1 (The Unpriced Guarantee) and Corollary 1. Write $K > 0$ for the lender's gross cost of a dollar lent, $K = 1$ under the normalization of (2) and $K = 1+\rho+g$ in the mapping

of Online Appendix OA5.1; the algebra is kept general in K. Write $\mu \equiv \mu_{co}(y, y_p)$, $\bar{\pi} \equiv \bar{\pi}(y)$, $\bar{\pi}_m \equiv \bar{\pi}(y_{\min}) \leq \bar{\pi}$ and $D \equiv \mu + (1-\mu)\bar{p} = \mu(1-\bar{p}) + \bar{p}$, so that $1 + \tilde{r}_{co} = K/D$ by (6). Since $\bar{p} \geq 0$ and $\mu < 1$, $D \geq \mu$, strictly when $\bar{p} > 0$.

Proposition 1. Under (2), $1 + r_{co} = K/\bar{\pi}_m$ and $1 + r_{so} = K/\bar{\pi}$, so

$$W(y, y_p) = \frac{K}{\bar{\pi}_m} - \frac{K}{D} = K\,\frac{D - \bar{\pi}_m}{\bar{\pi}_m D} = \underbrace{K\Big(\frac{1}{\bar{\pi}_m} - \frac{1}{\bar{\pi}}\Big)}_{S(y,y_p)} + \underbrace{K\,\frac{D - \bar{\pi}}{\bar{\pi} D}}_{r_{so}(y) - \tilde{r}_{co}(y,y_p)}, \tag{OA13}$$

which is (B.15), the closed form of (7), with its two parts. Hence $W > 0$ iff $D > \bar{\pi}_m$, i.e. iff $\mu + (1-\mu)\bar{p} > \bar{\pi}_m$, i.e. iff $\bar{p} > (\bar{\pi}_m - \mu)/(1-\mu) = \bar{p}_g(y, y_p)$, which is (8). Since $\bar{\pi}_m < 1$ the numerator is strictly smaller than the denominator, so $\bar{p}_g < 1$; it is non-negative iff $\bar{\pi}_m \geq \mu$ and negative, so that every cover overpays, where $\bar{\pi}_m < \mu$: either $\mu > \bar{\pi}$, a pool safer than the average child at y, or $y_p < y$ and $\bar{\pi}(y_p) < \mu \leq \bar{\pi}$. Monotonicity in coverage: $\partial W/\partial \bar{p} = K(1-\mu)/D^2 > 0$. Monotonicity in risk: $\partial W/\partial \mu = K(1-\bar{p})/D^2 \geq 0$, so W is weakly decreasing in the pool's default risk $1-\mu$, strictly when $\bar{p} < 1$: the posted rate does not move with the pool, while the fair rate rises as the pool worsens. Monotonicity in y_p: for $y_p \geq y$, $\bar{\pi}_m = \bar{\pi}$ is constant and $\bar{p}$ is non-decreasing in y_p, so W is non-decreasing, strictly where the cover rises; for $y_p < y$ the first part of (OA13) falls in y_p, through $\bar{\pi}(y_p)$, while the second rises, through $\bar{p}(y_p)$, so the sign of $\partial W/\partial y_p$ is not determined. Since W is increasing in D, it is largest where D, the lender's expected receipt counting the parent's payments, is largest. Substituting the posted rate into (B.14) gives the lender's realised profit on a cosigned loan,

$$\Pi_{co} = D\,\frac{K}{\bar{\pi}_m}\,L^* - KL^* = KL^*\,\frac{D - \bar{\pi}_m}{\bar{\pi}_m} = W\,D\,L^*,$$

positive exactly when W is.

Corollary 1. Under (B.17), $1 + r_{co} = K/\mu$ and

$$W(y, y_p) = \frac{K}{\mu} - \frac{K}{D} = K\,\frac{(1-\mu)\bar{p}}{\mu\big[\mu + (1-\mu)\bar{p}\big]} > 0$$

at every $\bar{p} > 0$. From $W = K(1/\mu - 1/D)$ and $dD/d\mu = 1 - \bar{p} \geq 0$,

$$\frac{dW}{d\mu} = K\left[-\frac{1}{\mu^2} + \frac{1-\bar{p}}{D^2}\right] < 0,$$

because $D \geq \mu$ and $1 - \bar{p} \leq 1$ with at least one inequality strict for $\bar{p} > 0$, so W is strictly increasing in the pool's default risk $1 - \mu$. Finally, $\tilde{r}_{co} < r_{so}$ iff $K/D < K/\mu_{so}$, i.e. iff $D > \mu_{so}$, i.e. iff $\bar{p} > (\mu_{so} - \mu)/(1-\mu) = \bar{p}^*$, which is (B.16); under $\mu < \mu_{so} < 1$, $\bar{p}^* \in (0, 1)$. The posted premium $r_{co} - r_{so} = K(1/\mu - 1/\mu_{so})$ is then positive while the fair premium is negative for $\bar{p} > \bar{p}^*$: the rule charges a positive premium on a pool that, netting the guarantee the lender actually receives, warrants a negative one. □

Proof of Lemma 1 (A Riskier Child Gains More). By (4), $\Delta V(y, B) - \Delta V(y, G) = (\pi_B - \pi_G)\Gamma +$

$(\pi_G - \pi_B)\Delta\underline{u} = \Delta\pi\,(\Delta\underline{u} - \Gamma)$, the cost of signing cancelling, so it suffices to show $\Delta\underline{u} > \Gamma$. Write $c_{co} = y - (1 + r_{co})L_{co} < y$ for consumption on the cosigned menu and h_{co} for its house. Then

$$\Delta\underline{u} - \Gamma = \big[u(y, h_{co}) - u(c_{co}, h_{co})\big] + \big[u(c_{so}, h_{so}) - u(y, h_r)\big].$$

The first bracket is strictly positive because u is strictly increasing in consumption and $c_{co} < y$: a miss costs the child no consumption. The second is non-negative by the participation condition in (A1): a solo buyer prefers her bundle to renting the floor at full income. Hence $\Gamma < \Delta\underline{u}$ at every income and under any pair of rates, since the rates enter only through the bundles. □

Proof of Lemma 2 (Informative Participation). By (9), $q_\theta(y, y_p) = F_\eta\big(\beta\,\Delta V(y, \theta) - (1 - \pi_\theta)\,\Delta u_p\big)$. Since F_η is strictly increasing on its support ($f_\eta > 0$ on $\mathbb{R}$), $q_G > q_B$ holds if and only if the argument is larger for the Good type:

$$\beta\big[\Delta V(y, G) - \Delta V(y, B)\big] \;>\; (1 - \pi_G)\Delta u_p - (1 - \pi_B)\Delta u_p \;=\; -\,\Delta\pi\,\Delta u_p. \tag{OA14}$$

From the decomposition (4), $\Delta V(y, \theta) = \pi_\theta\,\Gamma + (1 - \pi_\theta)\Delta\underline{u}$, so

$$\Delta V(y, B) - \Delta V(y, G) = (\pi_B - \pi_G)\Gamma + (\pi_G - \pi_B)\Delta\underline{u} = \Delta\pi\,(\Delta\underline{u} - \Gamma). \tag{OA15}$$

Substituting (OA15) into (OA14) and dividing by $\Delta\pi > 0$, the condition $q_G > q_B$ becomes $\Delta u_p > \beta\,(\Delta\underline{u} - \Gamma)$, which is (B.11) positive and exactly Assumption 1. Hence $q_G(y, y_p) > q_B(y, y_p)$ for all (y, y_p) is equivalent to Assumption 1, and holds under it.

Integrating the pointwise inequality over $F(y_p \mid y)$ preserves it, so $\bar{q}_G(y) > \bar{q}_B(y)$. By (B.12), $\Lambda_1(y)$ is strictly increasing in the likelihood ratio $\bar{q}_G/\bar{q}_B > 1$, giving $\Lambda_1(y) > \lambda(y)$; the analogous argument with complements $1 - \bar{q}_\theta$ gives $\Lambda_0(y) < \lambda(y)$. Therefore $\bar{\pi}_1(y) > \bar{\pi}(y) > \bar{\pi}_0(y)$ for all y. □

Proof of Proposition 2 (Three Regions under the Pricing Rule). Under (2) the rates read incomes only, so the pools enter no choice: the equilibrium is the parents' offers and the borrowers' menu choices at those rates, with the pools read off them. As in Appendix B.1.4, each loan is the lower of the desired loan and the cap of its menu, so no loan depends on a rate. Fix y_p and write $\Delta V(y, \theta; y_p) \equiv \Delta V(y, \theta;\, r_{co}(y_{\min}), r_{so}(y))$, continuous in y by (A1), the continuity of λ and that of $y_{\min} = \min\{y, y_p\}$; write $\Delta V(y, \theta)$ for the gain at equal rates, $r_{co} = r_{so}(y)$, and $\tilde{y}_\theta$ for its crossing.

Cutoffs. By (A2), $\Delta V(\cdot, \theta; y_p)$ is positive at y_F and crosses zero at most once; by Lemma 3 it is negative for large y. Let $\tilde{y}_\theta(y_p) \in (y_F, \infty)$ be the crossing, so that $\Delta V(y, \theta; y_p) > 0$ for $y < \tilde{y}_\theta(y_p)$ and $\Delta V(y, \theta; y_p) < 0$ for $y > \tilde{y}_\theta(y_p)$. By Lemma 1, which holds for any pair of rates, $\Delta V(\tilde{y}_G(y_p), B; y_p) > \Delta V(\tilde{y}_G(y_p), G; y_p) = 0$, so $\tilde{y}_G(y_p) < \tilde{y}_B(y_p)$. Hence $y_F < \tilde{y}_G(y_p) < \tilde{y}_B(y_p) < \infty$, and Region 2 has positive length at every y_p. The same argument at equal rates gives $y_F < \tilde{y}_G < \tilde{y}_B < \infty$.

The parent's file. The cosigned rate enters ΔV only through repayment-state consumption, and

$\partial \Delta V / \partial r_{co} = -\pi_\theta \, u_c(c_{co}, h_{co}) \, L_{co} < 0$ by (4). For $y \le y_p$ the two rates coincide, so $\Delta V(y, \theta; y_p) = \Delta V(y, \theta)$. If $y_p \ge \tilde{y}_\theta$, the two gains agree at $\tilde{y}_\theta$, so $\Delta V(\tilde{y}_\theta, \theta; y_p) = 0$ and, by single crossing, $\tilde{y}_\theta(y_p) = \tilde{y}_\theta$. If $y_p < \tilde{y}_\theta$, then at $\tilde{y}_\theta > y_p$ the cosigned rate exceeds the solo rate, so $\Delta V(\tilde{y}_\theta, \theta; y_p) < \Delta V(\tilde{y}_\theta, \theta) = 0$, while at $\max\{y_p, y_F\}$ the gain is positive (at y_p because $y_p < \tilde{y}_\theta$ and the two gains agree there, at y_F by (A2)); single crossing places $\tilde{y}_\theta(y_p)$ in $(\max\{y_p, y_F\}, \tilde{y}_\theta)$. Finally, for $y_p' < y_p < \tilde{y}_\theta$, $r_{co}(\min\{y, y_p'\}) \ge r_{co}(\min\{y, y_p\})$ at every y, strictly for $y > y_p'$ since λ is strictly increasing, so $\Delta V(\tilde{y}_\theta(y_p), \theta; y_p') < 0$ and $\tilde{y}_\theta(y_p') < \tilde{y}_\theta(y_p)$: the worse the parent's file, the earlier the exit.

Sorting. Fix y_p. For $y < \tilde{y}_G(y_p)$, $\Delta V(y, B; y_p) > \Delta V(y, G; y_p) > 0$, so every borrower with an offer cosigns. For $\tilde{y}_G(y_p) \le y < \tilde{y}_B(y_p)$, $\Delta V(y, G; y_p) \le 0 < \Delta V(y, B; y_p)$: good types take the solo menu (at $y = \tilde{y}_G(y_p)$ they are indifferent, a null set) and bad types with an offer cosign. For $y \ge \tilde{y}_B(y_p)$, $\Delta V(y, G; y_p) < \Delta V(y, B; y_p) \le 0$ and nobody cosigns.

Offers and beliefs. Parents offer with the propensities (9) evaluated at the posted rates, which no offer and no belief moves, so the offers are unique. In Region 1 the cosigned pool at (y, y_p) is the offered pool, and Bayes' rule (B.24) gives $\mu_{co}(y, y_p) = \pi_B + \Delta\pi \, \Lambda_1(y, y_p) = \bar{\pi}_1(y, y_p)$, which exceeds $\bar{\pi}(y)$ by Lemma 2. In Region 2 every cosigner is a bad type, so $\mu_{co}(y, y_p) = \pi_B$. In Region 3 the cosigned menu is unused, and since no rate reads beliefs, no off-path belief changes any choice. Offers, choices and on-path beliefs are each pinned down, so the equilibrium is unique. □

Proof of Lemma 3 (The Gain Vanishes at the Top). As $y \to \infty$ the desired loan exceeds L^*, so the cosigned loan is the cap and $c_{co}(y) = y - (1 + r_{co}(y_{\min}))L^* \to \infty$ at every y_p, with $h_{co} = h^*$ fixed, and $c_{so}(y) \to \infty$ with $h_{so}(y)$ in the compact range between $h_{so}(y_F) \ge h_r$ and its cap $\alpha \, \overline{\text{ltv}}^{so} P$; the posted rates lie between pricing at π_G and at π_B, whichever income the cosigned one reads, so neither consumption level is held back by them. By (A1), $u(c, h) \to \sup u$ as $c \to \infty$ at every $h \ge h_r$, and since u is continuous and increasing in c the convergence is uniform over h in that compact range (Dini). Hence each of $u(c_{co}, h^*)$, $u(c_{so}, h_{so})$, $u(y, h^*)$ and $u(y, h_r)$ converges to $\sup u$, so $\Gamma(y, y_p) = u(c_{co}, h^*) - u(c_{so}, h_{so}) \to 0$ for every y_p and $\Delta\underline{u}(y) = u(y, h^*) - u(y, h_r) \to 0$. By (4), $\Delta V(y, \theta; y_p) \to -\phi < 0$ for both types. □

Proof of Lemma OA1 (Type Cutoff Ordering under the Menu Rates). Fix the pair of menu rates. By Lemma 1, which holds for any pair of rates, $\Delta V(y, B) - \Delta V(y, G) = \Delta\pi \, (\Delta\underline{u} - \Gamma) > 0$ at every y. At the Good-type cutoff $\Delta V(\tilde{y}_G, G) = 0$, so $\Delta V(\tilde{y}_G, B) > 0$: the Bad type strictly prefers cosigning there. $\Delta V(\cdot, B)$ is continuous, positive at $\tilde{y}_G$, negative for large y by Lemma 3, whose limit does not depend on the rate pair as long as the rates are bounded, and crosses zero at most once by (M4); its root therefore satisfies $\tilde{y}_B > \tilde{y}_G$. □

Proof of Lemma OA2 (Belief–rate fixed point). Self-map. $\Lambda_1(y; r) \in [0, 1]$ as a probability, so $\mu_{co} = \pi_B + \Delta\pi\Lambda_1 \in [\pi_B, \pi_G]$ and, by (OA5), $T_y(r) \in \mathcal{R}$ for every $r \in \mathcal{R}$: beliefs are bounded whatever parents do, so the posted rate cannot leave the interval.

Continuity. By (A1) u is C^2, so $\Delta V(y, \theta; r, r_{so})$ in (4) is continuous in r; F_η is a distribution function with a density, so $q_\theta(y, y_p; r) = F_\eta(I_\theta)$ is continuous in r; and since $q_\theta \in [0, 1]$, dominated

convergence carries continuity to $\bar{q}_\theta(y;r) = \int q_\theta \, dF(y_p \mid y)$. The denominator of (B.12) is $\lambda\bar{q}_G + (1-\lambda)\bar{q}_B$, strictly positive because F_η has full support (A3) and, being continuous on the compact set $\mathcal{R}$, bounded below by some $\underline{q} > 0$. Hence $\Lambda_1(y;\cdot)$ and therefore T_y are continuous on $\mathcal{R}$.

Existence. $T_y(\cdot) - \mathrm{id}$ is continuous on $\mathcal{R}$, weakly positive at $\min \mathcal{R}$ and weakly negative at $\max \mathcal{R}$ by the self-map property, so the intermediate value theorem gives an r with $T_y(r) = r$. That r is (OA1) evaluated at consistent beliefs.

Uniqueness and convergence. Differentiating (OA5),

$$T_y'(r) = -\frac{\Delta\pi}{\mu_{co}(r)^2}\,\frac{\partial \Lambda_1(y;r)}{\partial r}.$$

From (B.12), $\partial\Lambda_1/\partial r$ is a rational function of $(\bar{q}_G, \bar{q}_B, \bar{q}_G', \bar{q}_B')$ whose denominator is bounded below by $\underline{q}^2$, and $\bar{q}_\theta'(r) = \int f_\eta(I_\theta)\,(\partial I_\theta/\partial r)\,dF$ with $|\partial I_\theta/\partial r| = \beta\,\pi_\theta\,u_c\big(y - (1+r)L^*, h^*\big)L^*$ bounded on $\mathcal{R}$ by (A1). Hence $|\partial\Lambda_1/\partial r| \le C\bar{f}$ for a constant C depending only on primitives, and $|T_y'| \le C'\bar{f}$ with $C' = \Delta\pi\, C/\pi_B^2$. Under the dispersion condition (M2), $\bar{f}$ is small enough that $C'\bar{f} < 1$, so T_y is a contraction on the complete metric space $\mathcal{R}$; Banach's theorem then gives a unique fixed point and convergence of $r_{n+1} = T_y(r_n)$ from any $r_0 \in \mathcal{R}$. □

Proof of Proposition OA1 (Three-Region PBE). Existence and uniqueness of y_G^{pool}. Along the pooling schedule $r_{co}^{pool}(y) = 1/\bar{\pi}_1(y) - 1$, the map $y \mapsto \Delta V(y, G; r_{co}^{pool}(y), r_{so}(y))$ is continuous (by (A1) and continuity of $\bar{\pi}_1$). By (M4) it is strictly positive as $y \to y_F^+$ and strictly negative as $y \to \infty$, so a root exists; by (M4) it crosses zero only once, so the root y_G^{pool} defined in (OA3) is unique.

Region 1 ($y < y_G^{pool}$). Here $\Delta V(y, G) > 0$, and by Lemma 1, $\Delta V(y, B) > \Delta V(y, G) > 0$, so both types choose *co* when $o = 1$. The pool contains both types in proportions set by $\bar{q}_\theta(y)$, so Bayes' rule (B.12) gives $\mu_{co}(y) = \bar{\pi}_1(y)$ and the pooling rate (OA1); this is the rate used to define the region, so beliefs, pricing and sorting are mutually consistent. Parental screening is active with $q_G > q_B$ by Lemma 2.

Exit of Good types and the separating belief. For $y \ge y_G^{pool}$, Good types choose *so*. Any borrower observed choosing *co* is therefore a Bad type, so Bayes' rule pins $\mu_{co}(y) = \pi_B$ and the constant separating rate r_{co}^{sep} in (OA2); since $\bar{\pi}_1(y) > \pi_B$, $r_{co}^{sep} > r_{co}^{pool}(y)$.

Existence and uniqueness of y_B^{sep}. Under the survival condition, $\Delta V(y_G^{pool}, B; r_{co}^{sep}, r_{so}) > 0$, while by (M4) the same object is negative as $y \to \infty$; continuity and the single crossing of $\Delta V(\cdot, B)$ in (M4) deliver a unique root $y_B^{sep} > y_G^{pool}$, as in (OA4). For $y \in [y_G^{pool}, y_B^{sep})$ Bad types strictly prefer *co* at r_{co}^{sep} (Region 2); for $y \ge y_B^{sep}$ even Bad types prefer *so*, so no cosign pool forms (Region 3).

Off-path beliefs and equilibrium. Assign the worst-case belief $\mu_{co} = \pi_B$ to any out-of-equilibrium choice of *co* in Regions 2–3; this is admissible and makes deviation to the cosign menu unprofitable for Good types, so no profitable deviation exists. The cutoffs are uniquely determined by the two monotone crossings, and the belief and rate systems are then pinned by the cutoffs, giving a unique PBE with $y_F < y_G^{pool} < y_B^{sep} < \infty$ and the schedule (OA6). □

Proof of the Two-Region Case (Remark following Proposition OA1). Suppose the survival condition fails, i.e. $\Delta V(y_G^{pool}, B; r_{co}^{sep}, r_{so}) \leq 0$. By (M4), $\Delta V(\cdot, B; r_{co}^{sep}, r_{so})$ crosses zero only once, from above, so it is negative at every $y > y_G^{pool}$. Thus once Good types exit at y_G^{pool}, Bad types are unwilling to cosign at the separating rate they would face, so the cosign pool is empty for all $y \geq y_G^{pool}$: Region 2 has zero Lebesgue measure and $y_B^{sep} = y_G^{pool}$. The equilibrium collapses to two regions separated by the single cutoff y_G^{pool}: both types cosign below it (priced at $\bar{\pi}_1(y)$) and all borrowers go solo above it. Existence and uniqueness of y_G^{pool} follow exactly as in the proof of Proposition OA1. □

Proof of Proposition OA2 (Pool Quality Decomposition). Equations (OA9)–(OA10) are the mass-weighted average repayment probabilities of the two pools under the three-region sorting of Proposition OA1: the cosign pool is Region 1 (priced at $\bar{\pi}_1$) plus Region 2 (priced at π_B); the solo pool is the Region-2 block ($y_G^{pool} \leq y < y_B^{sep}$, priced at $\bar{\pi}_2$, since every Good type and only the *unoffered* Bad types sort there), plus the Region-3 block ($y \geq y_B^{sep}$, priced at the unconditional $\bar{\pi}$ because no cosign menu exists there), plus the low-income solo sub-pool ($y < y_G^{pool}$, priced at $\bar{\pi}_0$). Writing $B \equiv \omega_{so}^2 \int_{y_G^{pool}}^{y_B^{sep}} \bar{\pi} f_{so}^2 + \omega_{so}^3 \int_{y_B^{sep}}^{\infty} \bar{\pi} f_{so}^3$ for the high block evaluated at the benchmark and differencing,

$$\bar{\mu}_{so} - \bar{\mu}_{co} = B + \underbrace{\omega_{so}^2 \int_{y_G^{pool}}^{y_B^{sep}} (\bar{\pi}_2 - \bar{\pi}) f_{so}^2}_{=S} + \omega^{lo} \int_0^{y_G^{pool}} \bar{\pi}_0 f_{so}^{lo} - \omega_{co}^1 \int_0^{y_G^{pool}} \bar{\pi}_1 f_{co}^1 - \omega_{co}^2 \int_{y_G^{pool}}^{y_B^{sep}} \pi_B f_{co}^2. \quad \text{(OA16)}$$

On the three interior integrals add and subtract the unconditional benchmark $\bar{\pi}(y)$:

$$\omega^{lo} \int_0^{y_G^{pool}} \bar{\pi}_0 f_{so}^{lo} = \omega^{lo} \int_0^{y_G^{pool}} \bar{\pi} f_{so}^{lo} \underbrace{- \omega^{lo} \int_0^{y_G^{pool}} (\bar{\pi} - \bar{\pi}_0) f_{so}^{lo}}_{=D_2}, \quad \text{(OA17)}$$

$$-\omega_{co}^1 \int_0^{y_G^{pool}} \bar{\pi}_1 f_{co}^1 = -\omega_{co}^1 \int_0^{y_G^{pool}} \bar{\pi} f_{co}^1 \underbrace{- \omega_{co}^1 \int_0^{y_G^{pool}} (\bar{\pi}_1 - \bar{\pi}) f_{co}^1}_{=D_1}, \quad \text{(OA18)}$$

$$-\omega_{co}^2 \int_{y_G^{pool}}^{y_B^{sep}} \pi_B f_{co}^2 = -\omega_{co}^2 \int_{y_G^{pool}}^{y_B^{sep}} \bar{\pi} f_{co}^2 \underbrace{+ \omega_{co}^2 \int_{y_G^{pool}}^{y_B^{sep}} (\bar{\pi} - \pi_B) f_{co}^2}_{=A}. \quad \text{(OA19)}$$

The three retained $\bar{\pi}$ integrals combine into the reweighting term $R = \omega^{lo} \int_0^{y_G^{pool}} \bar{\pi} f_{so}^{lo} - \omega_{co}^1 \int_0^{y_G^{pool}} \bar{\pi} f_{co}^1 - \omega_{co}^2 \int_{y_G^{pool}}^{y_B^{sep}} \bar{\pi} f_{co}^2$. Collecting gives the identity $\bar{\mu}_{so} - \bar{\mu}_{co} = B + D_1 + D_2 + S + R + A$, establishing claim (i).[3]

Signs. $B > 0$ as $\bar{\pi} > 0$. $S > 0$ because $\Lambda_2(y) > \lambda(y)$ by (OA7): in Region 2 the Bad types

[3] A centres the Region-2 cosign block on the unconditional benchmark $\bar{\pi}$, as the other two interior integrals are centred. Centring instead on the pooling belief $\bar{\pi}_1(y)$ those borrowers would have carried had the Good types not left gives the strictly larger trap loss $\omega_{co}^2 \int_{y_G^{pool}}^{y_B^{sep}} (\bar{\pi}_1 - \pi_B) f_{co}^2 > A > 0$, the difference $\omega_{co}^2 \int_{y_G^{pool}}^{y_B^{sep}} (\bar{\pi}_1 - \bar{\pi}) f_{co}^2$ being a belief-level term that then moves into R. Neither centring changes the identity or the sign of the trap channel.

holding an offer exit to the cosign menu, so the solo pool they leave behind is strictly better than the unconditional benchmark. S and A are the two sides of one exit, and both widen the gap. By Lemma 2, $\bar{\pi}_1 > \bar{\pi} > \bar{\pi}_0$, so the integrands of D_1 and D_2 are positive and both terms are negative. R is *not* signed by the maintained primitives: it compares *mass-weighted* integrals of the same benchmark $\bar{\pi}$ under the two densities. Although higher-income parents cosigning more readily tilts the normalised cosign density f^1_{co} rightward relative to f^{lo}_{so} (which, with $\bar{\pi}$ increasing, would sign the comparison of the *normalised* integrals), the masses satisfy $\omega^{lo} \neq \omega^1_{co}$ in general, and no restriction among the maintained assumptions pins their ratio. The decomposition is an accounting identity; the sign of the overall gap is an empirical question, settled by the NSMO pool-quality evidence rather than by primitives. Finally $A > 0$ because the Region-2 cosign pool is priced at the worst case $\pi_B < \bar{\pi}(y)$: the trap holds those borrowers below the benchmark and so widens the gap. Claim (ii) is immediate from the identity: $\bar{\mu}_{co} < \bar{\mu}_{so} \iff B + D_1 + D_2 + S + R + A > 0$, for which $B + S + A > |D_1| + |D_2| + |R|$ is sufficient. Claim (iii) is (OA8) evaluated at a given income: where $\mu_{co}(y) < \mu_{so}(y)$, the pricing condition (B.17) gives $\partial r_k / \partial g = 1/\mu_k(y)$ and hence $\beta^{co}_{rate}(y) > \beta^{so}_{rate}(y)$ directly. The claim is pointwise and does not extend to the pool averages: Jensen's inequality bounds $\mathbb{E}[1/\mu_k \mid k]$ below by $1/\bar{\mu}_k$ for *each pool separately*, and two lower bounds do not order the two expectations. Since μ_{co} is degenerate at π_B over Region 2 while μ_{so} is not, the pools differ in the dispersion of μ and not only in its mean, so $\bar{\mu}_{co} < \bar{\mu}_{so}$ is consistent with $\mathbb{E}[1/\mu_{co} \mid co] < \mathbb{E}[1/\mu_{so} \mid so]$. □

Proof of Proposition 3 (Equal Rates, Smaller Rejection Response). Under (2) the rate on menu k is $1 + r_k = K/\bar{\pi}(y_{\min})$, so a cost shock dK moves it by $dr_k = dK/\bar{\pi}(y_{\min}) = (1 + r_k)\, dK$ at $K = 1$, (11): by the same amount on both menus where $y_p \geq y$, by $S(y, y_p)\, dK$ more on the cosigned menu where $y_p < y$. By (13), $dDTI_k = (L/Y_k)\, dr_k = DTI_k\, dK$, so $\sigma_k = DTI_k$: every ratio rises by the same proportion dK, on both menus and whatever the parent's file. The rejection rate is $p_k(K) = \mathbb{E}^k[H_k(DTI_k(K))]$; differentiating under the expectation, as in the proof of Proposition OA3 below, gives

$$\beta^k_{rej} = \mathbb{E}^k\left[H_k'(DTI_k)\, \sigma_k\right] = \mathbb{E}^k\left[H_k'(DTI_k)\, DTI_k\right].$$

(i) At $\tau = 0$ both menus face H. Since H is increasing and convex on the relevant range, $H' \geq 0$ is non-decreasing there, so $d \mapsto H'(d)\, d$ is non-decreasing for $d > 0$. If the cosigned ratios are first-order stochastically dominated by the solo ratios, the expectation of a non-decreasing function is weakly smaller under the cosigned distribution, so $\beta^{co}_{rej} \leq \beta^{so}_{rej}$; the reverse dominance reverses the inequality.

(ii) Write $\Phi(\tau) \equiv \mathbb{E}^{co}[H'(DTI_{co} - \tau\gamma(y_p))\, DTI_{co}] = \beta^{co}_{rej}$ and $\Phi_{so} \equiv \beta^{so}_{rej} > 0$, which is free of τ. Because $\gamma > 0$ and H is strictly convex, $\tau \mapsto H'(DTI_{co} - \tau\gamma)$ is strictly decreasing, so Φ is strictly decreasing, and it is continuous by dominated convergence (H' is bounded on the relevant range). If the cosigned ratios are higher, part (i) gives $\Phi(0) \geq \Phi_{so}$. If moreover $H'(d) \to 0$ as d falls below the cap, $\Phi(\tau) \to 0 < \Phi_{so}$ as $\tau \to \infty$. Hence there is a unique $\tau^* \geq 0$ with $\Phi(\tau^*) = \Phi_{so}$, and $\beta^{co}_{rej} < \beta^{so}_{rej}$ if and only if $\tau > \tau^*$. □

Proof of Proposition OA3 (Sensitivity Decoupling under the Menu-Reading Rule). Derivation of the rejection-pass-through formula (14). In this appendix the shock is the guarantee fee g, a rise in the lender's cost of a dollar lent, which the menu-reading rule passes to the rate on menu k at $1/\mu_k(y)$ (Online Appendix OA1); the ratio sensitivity is then $\sigma_k = L/(Y_k\,\mu_k(y)) = DTI_k$, as in (13). Write $p_k(g) = \mathbb{E}^k[H_k(DTI_k(g))]$. A marginal increase dg shifts every borrower's DTI upward by $\sigma_k\,dg$. Since H_k is differentiable with bounded derivative and σ_k is integrable (guaranteed by the loans applied for having a density with finite mean), differentiation under the expectation gives

$$\beta^k_{rej} = \frac{\partial p_k}{\partial g} = \mathbb{E}^k\big[H_k'(DTI_k)\,\sigma_k\big]\,, \tag{OA20}$$

which is (14). Under the sharp-cap special case $H_k = \mathbf{1}\{DTI_k > \overline{\text{dti}}\}$, H_k' is a Dirac mass at $\overline{\text{dti}}$ and the expression collapses to the classical threshold-density formula $\mathbb{E}^k[\sigma_k \mid DTI_k = \overline{\text{dti}}]\cdot f^k_{DTI}(\overline{\text{dti}}; g)$.

Decoupling. Dividing the two pools' expressions,

$$\beta^{co}_{rej} < \beta^{so}_{rej} \iff \frac{\mathbb{E}^{co}[H_{co}'(DTI_{co})\,\sigma_{co}]}{\mathbb{E}^{so}[H_{so}'(DTI_{so})\,\sigma_{so}]} < 1, \tag{OA21}$$

which is the leniency ratio (OA12) that (C2′) delivers. It remains to show that the restriction is not vacuous, i.e. that with a *common* enforcement hazard the ranking would run the other way. Each borrower's DTI sensitivity is $\sigma_k = L/(Y_k\mu_k(y))$. Under (C1), Proposition OA2 gives $\bar\mu_{co} < \bar\mu_{so}$, so the risk-pricing channel $1/\mu_{co} > 1/\mu_{so}$ raises σ_{co} relative to σ_{so}: with $H_{co} = H_{so}$ and comparable positioning of the two DTI distributions, $\mathbb{E}[\sigma_{co}] > \mathbb{E}[\sigma_{so}]$ pushes (OA21) *above* one and $\beta^{co}_{rej} > \beta^{so}_{rej}$.

Condition (C2′) overturns this through the enforcement margin, and under (12) it does so as a threshold in one primitive. Write

$$\Phi(\tau) \;\equiv\; \mathbb{E}^{co}\big[H'\big(DTI_{co} - \tau\gamma(y_p)\big)\sigma_{co}\big]\,, \tag{OA22}$$

so that $\beta^{co}_{rej} = \Phi(\tau)$ and $\beta^{so}_{rej} = \Phi_{so} \equiv \mathbb{E}^{so}[H'(DTI_{so})\sigma_{so}]$, the latter free of τ. Since $\gamma \geq 0$ and H is convex on the range of effective ratios, H' is non-decreasing there, so $\tau \mapsto H'(DTI_{co} - \tau\gamma)$ is non-increasing pointwise and strictly decreasing wherever $\gamma > 0$ and H is strictly convex. As $\sigma_{co} > 0$, Φ is continuous and strictly decreasing on $\{\tau : \gamma > 0\}$, so

$$\beta^{co}_{rej} < \beta^{so}_{rej} \iff \Phi(\tau) < \Phi_{so} \iff \tau > \tau^*,$$

with τ^* the unique root of $\Phi(\tau) = \Phi_{so}$ when one exists. At $\tau = 0$ the two hazards coincide, and the previous paragraph showed that (C1) then gives $\Phi(0) > \Phi_{so}$; hence $\tau^* > 0$ and the threshold is interior. If in addition $H'(d) \to 0$ as d falls below the cap, then $\Phi(\tau) \to 0 < \Phi_{so}$, so τ^* is finite and decoupling obtains for every large enough weight. Because Φ is strictly decreasing, the leniency ratio (OA12) is strictly decreasing in τ and crosses one exactly at τ^*, which is what makes the empirical slope comparison of Appendix OA5.9 an estimate of τ against τ^* rather than a restatement of the conclusion. Note that the sparsity route to (OA21) under the sharp cap (a small

cosigned threshold density $f^{co}_{DTI}(\overline{\text{dti}}; g)$; Appendix OA3) is *not* supported by the data, which show cosigned applications concentrated at and above the threshold; leniency, not distance, delivers the insulation. □

Proof of Proposition 5 (The Cost of the Split). (i) From (OA15), $\Delta V(y,B) - \Delta V(y,G) = \Delta\pi(\Delta\underline{u} - \Gamma)$, so

$$I_G - I_B = \beta\big[\Delta V(y,G) - \Delta V(y,B)\big] + \Delta\pi\,\Delta u_p(y_p) = \Delta\pi\big[\Delta u_p(y_p) - \beta(\Delta\underline{u} - \Gamma)\big], \tag{OA23}$$

positive by Assumption 1. Only the first bracketed term varies with y_p, and $\Delta u'_p(y_p) = u'(y_p) - u'(y_p - m) < 0$ by the strict concavity of (A1) — on the saturated region $\bar{p}(y_p) = 1$, so the parent's outlay is the whole payment m and is constant in y_p — so $I_G - I_B$ is strictly decreasing in y_p. Write $s \equiv -\Delta u'_p(y_p) > 0$; then $I'_\theta(y_p) = (1 - \pi_\theta)\, s > 0$, so both arguments rise with parental income, and $I'_G < I'_B$ because $\pi_G > \pi_B$.

Let $\varphi \equiv f_\eta / F_\eta$, strictly decreasing by the log-concavity of (A3). Differentiating $\log(q_G/q_B) = \log F_\eta(I_G) - \log F_\eta(I_B)$,

$$\frac{d}{dy_p} \log \frac{q_G}{q_B} = \varphi(I_G)\, I'_G - \varphi(I_B)\, I'_B \;<\; 0,$$

since $I_G > I_B$ gives $\varphi(I_G) < \varphi(I_B)$ while $0 < I'_G < I'_B$: both factors are smaller and all four quantities positive. The two channels reinforce rather than offset — the gap between the types' thresholds narrows, and both sit further into the upper tail of F_η where the screen discriminates less. Finally $\Lambda_1(y, y_p)$ in (B.24) is a strictly increasing transformation of q_G/q_B, hence strictly decreasing in y_p. Below saturation the outlay is $y_p - \underline{c}_p$ by (B.4), so $\Delta u_p = u(y_p) - u(\underline{c}_p)$ and $\Delta u'_p = u'(y_p) > 0$: both thresholds fall in y_p, $I'_\theta = -(1 - \pi_\theta)\, u'(y_p)$, and

$$\frac{d}{dy_p} \log \frac{q_G}{q_B} = u'(y_p)\big[(1 - \pi_B)\,\varphi(I_B) - (1 - \pi_G)\,\varphi(I_G)\big] \;>\; 0,$$

since $\varphi(I_B) > \varphi(I_G)$ and $1 - \pi_B > 1 - \pi_G$. The screen strengthens in y_p below saturation and weakens above it, the hump of the statement.

(ii) $\mu_{co}(y, y_p) = \pi_B + \Delta\pi\Lambda_1(y, y_p)$ inherits the strict monotonicity, so $\hat{r}_{co}(y, y_p) = 1/\mu_{co}(y, y_p) - 1$ is strictly increasing in y_p. By the law of total probability $\Lambda_1(y) = \mathbb{E}[\Lambda_1(y, y_p) \mid y, o = 1]$, so $\mu_{co}(y) = \mathbb{E}[\mu_{co}(y, y_p) \mid y, o = 1]$ lies strictly between the infimum and supremum of $\mu_{co}(y, \cdot)$ whenever the conditional distribution of y_p is non-degenerate. Continuity and strict monotonicity then give a unique $y^0_p(y)$ with $\mu_{co}(y, y_p) \gtrless \mu_{co}(y)$ as $y_p \lessgtr y^0_p(y)$, and hence $\hat{r}_{co}(y) \gtrless \hat{r}_{co}(y, y_p)$ on the same partition: cosigners below y^0_p pay more than the rule would charge them on the fuller information set, and those above pay less.

(iii) The cosigned rate enters the parent's problem only through $\Delta V(y, \theta; r_{co}, r_{so})$, and $\partial\Delta V/\partial r_{co} = -\pi_\theta\, u_c\big(y - (1 + r_{co})L^*, h^*\big)L^* < 0$ by (4). Since F_η is strictly increasing, $\partial q_\theta/\partial r_{co} < 0$. Consider the counterfactual in which the rule is permitted to read y_p and compare it with pooled pricing at the initial schedule. By *(ii)* pooled pricing raises the rate faced by every family with $y_p < y^0_p$ and

lowers it for every family with $y_p > y_p^0$, and the rate difference $\hat{r}_{co}(y) - \hat{r}_{co}(y, y_p)$ is itself strictly decreasing in y_p. The ratio of post- to pre-change participation, $q_\theta\big(y, y_p; \hat{r}_{co}(y)\big)/q_\theta\big(y, y_p; \hat{r}_{co}(y, y_p)\big)$, is therefore strictly increasing in y_p, crossing one at y_p^0. A density reweighted by a strictly increasing likelihood ratio dominates the original in the monotone-likelihood-ratio order and hence first-order stochastically, so the conditional distribution of y_p in the cosign pool shifts to the right. By *(i)* $\Lambda_1(y, \cdot)$ is strictly decreasing, so its expectation under the shifted distribution is strictly smaller: $\Lambda_1(y)$ under pooled pricing is strictly below its y_p-conditional counterpart. The statement is first-order, evaluated at the initial schedule; it does not require the belief–rate fixed point, which Proposition OA1 takes as given. □

Proof of Corollary 2 (Screen reversal). On the saturated region the outlay is m, fixed, so with u concave $\Delta u_p(y_p) = u(y_p) - u(y_p - m) \le m\, u'(y_p - m)$, which tends to 0 as $y_p \to \infty$ whenever $u'(c) \to 0$. Taking the limit in (OA23), $I_G - I_B \to -\beta\, \Delta\pi\, (\Delta \underline{u} - \Gamma)$, which is strictly negative at every income, since $\Delta \underline{u} > \Gamma$ there by Lemma 1. Since $I_G - I_B$ is continuous and strictly decreasing in y_p by Proposition 5(i), and strictly positive at the lower end of the support by Assumption 1, there is a unique $\bar{y}_p(y)$ at which it vanishes; above it $q_G < q_B$, the likelihood ratio falls below one, and $\Lambda_1(y, y_p) < \lambda(y)$. □

OA3 Threshold Sparsity under a Sharp Cap: A Lognormal Benchmark

Status of this appendix. Under the sharp-cap special case $H_j = \mathbf{1}\{DTI_j > \kappa\}$, the leniency condition (C2′) of Proposition OA3 reduces to a *threshold-sparsity* condition on the density of cosigned applications at the cap. This appendix develops that benchmark analytically. The direct empirical test (Appendix OA5.9) *rejects* sparsity — cosigned applications concentrate at and above the threshold — so the operative route to insulation is the enforcement-leniency channel of (C2′), and the analysis below is retained as the sharp-cap benchmark rather than as the maintained mechanism.

I verify the sparsity condition under a lognormal specification for the joint distribution of (L, y, y_p), showing that the extensive-margin insulation channel would dominate the intensive-margin quality channel for a sufficiently large parental income buffer.

Specification. Let $(\log L, \log y, \log y_p)$ be jointly normal, and approximate joint qualifying income $Y_{co} = y + y_p$ as lognormal, so that $\log Y_j$ is normal for both menus with $Y_{so} = y$.[4] With the ratio of Section 3.5, $DTI_j = (1 + r_j)L/Y_j$, hence

$$\log DTI_j = \log(1 + r_j) + \log L - \log Y_j \;\sim\; \mathcal{N}\big(m_j,\, s_j^2\big), \tag{OA24}$$

[4]The lognormal closure for the sum $y + y_p$ is the standard Fenton–Wilkinson approximation; it is exact in the limit where one of the two incomes dominates.

with $m_j = \log(1+r_j) + \mu_{\log L} - \mu_{\log Y_j}$ and $s_j^2 = \mathrm{Var}(\log L - \log Y_j)$. The lognormal density of DTI_j at the rejection threshold κ is

$$f^j_{DTI}(\kappa) = \frac{1}{\kappa s_j}\,\phi(d_j), \qquad d_j \equiv \frac{\log\kappa - m_j}{s_j}, \tag{OA25}$$

where ϕ is the standard normal density.

Closed form for the rejection pass-through. Because the sensitivity $\sigma_j = L/(Y_j\mu_j) = DTI_j/[(1+r_j)\mu_j]$ is proportional to DTI_j, the sensitivity *conditional on being at the margin* is the constant

$$\mathbb{E}^j[\sigma_j \mid DTI_j = \kappa] = \frac{\kappa}{(1+r_j)\,\mu_j}. \tag{OA26}$$

Substituting this and (OA25) into the rejection-pass-through formula (14) gives the closed form

$$\beta^j_{rej} = \mathbb{E}^j[\sigma_j \mid DTI_j = \kappa]\; f^j_{DTI}(\kappa) = \frac{\phi(d_j)}{(1+r_j)\,\mu_j\,s_j}. \tag{OA27}$$

Verification of (C2). Dividing the two pools' expressions in (OA27),

$$\frac{\beta^{co}_{rej}}{\beta^{so}_{rej}} = \underbrace{\frac{(1+r_{so})\,\mu_{so}\,s_{so}}{(1+r_{co})\,\mu_{co}\,s_{co}}}_{\text{intensive factor} = O(1)} \cdot \underbrace{\frac{\phi(d_{co})}{\phi(d_{so})}}_{\text{extensive factor}} = O(1)\cdot\exp\Big(-\tfrac{1}{2}\big(d^2_{co} - d^2_{so}\big)\Big). \tag{OA28}$$

The intensive factor is bounded: although $\mu_{co} < \mu_{so}$ raises $1/\mu_{co}$ (the quality channel of (C1), which on its own would push β^{co}_{rej} *up*), the ratio remains $O(1)$ in the parental buffer. The extensive factor is governed by the location shift. Since $\mu_{\log Y_{co}} = \mathbb{E}[\log(y+y_p)] > \mathbb{E}[\log y] = \mu_{\log Y_{so}}$, I have

$$m_{so} - m_{co} = \big(\mu_{\log Y_{co}} - \mu_{\log Y_{so}}\big) - \big(\log(1+r_{co}) - \log(1+r_{so})\big) \;>\; 0 \tag{OA29}$$

whenever the parental income buffer exceeds the rate wedge, i.e. $\mu_{\log Y_{co}} - \mu_{\log Y_{so}} > \log\frac{1+r_{co}}{1+r_{so}}$ — which holds for any non-trivial y_p because the rate wedge is second order. Hence $m_{co} < m_{so}$: the cosign DTI distribution is shifted left. With κ in the upper tail ($\log\kappa > m_j$, so $d_j > 0$) it follows that $d_{co} > d_{so} > 0$, and therefore

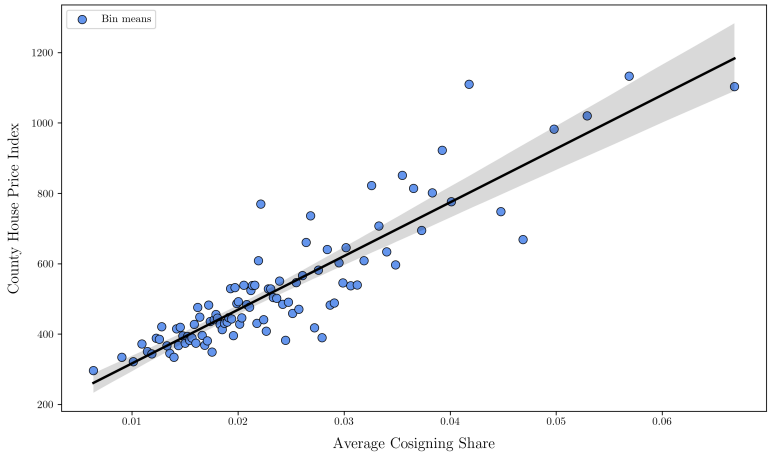
$$\frac{\phi(d_{co})}{\phi(d_{so})} = \exp\Big(-\tfrac{1}{2}\big(d^2_{co} - d^2_{so}\big)\Big) \;<\; 1, \tag{OA30}$$

decreasing in the buffer $m_{so} - m_{co}$. Because the Gaussian tail decays faster than any bounded factor, there exists a buffer threshold $\underline{\Delta}$ such that whenever $\mu_{\log Y_{co}} - \mu_{\log Y_{so}} > \underline{\Delta}$ the product (OA28) is strictly below one:

$$\beta^{co}_{rej} < \beta^{so}_{rej}. \tag{OA31}$$

Equivalently, in the sharp-cap threshold-density form of (C2′), the threshold-density ratio $f^{co}_{DTI}(\kappa)/f^{so}_{DTI}(\kappa) = \frac{s_{so}}{s_{co}}\,\phi(d_{co})/\phi(d_{so})$ falls below the sensitivity ratio $\mathbb{E}[\sigma_{so}]/\mathbb{E}[\sigma_{co}]$ once the parental buffer is large

FIGURE OA1. Cosigning prevalence and local house prices



County-level cosigning prevalence against the FHFA All-Transactions house price index, binned. Fact 1 of Section 4 states the gradient against price relative to income (Figure 2); this price-index version is the benchmark for the cross-county validation of Section OA4.5. Prevalence is the age-gap measure, so the gradient is in intergenerational cosigning rather than in multi-party borrowing.

FIGURE OA2. Cosigning and house prices relative to income, single years

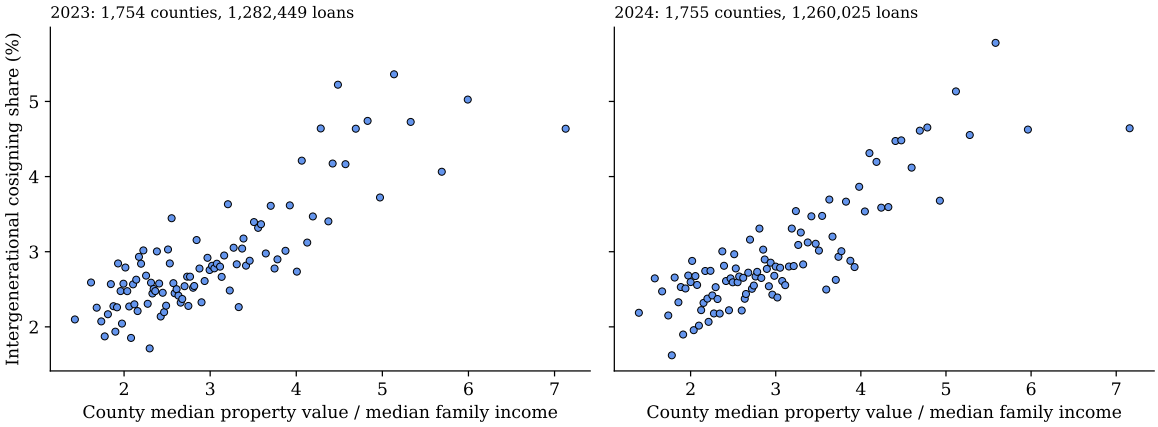


Figure 2 of Section 4 for a single year of originations at a time, 2023 (left) and 2024 (right), on the same county set: county intergenerational cosigning share against the county's median property value among purchase applications divided by its ACS median family income, in one hundred equal-count bins of counties weighted by loans. The 2024 income denominator uses the latest available ACS vintage where the 2024 release does not cover the county.

enough, which is exactly Condition (C2). This confirms that the insulation channel dominates the quality channel on the rejection margin under lognormality.

OA4 Data Construction

OA4.1 HMDA Sample Construction

The estimation sample is built from the HMDA Loan Application Register (LAR) snapshot files published by the CFPB for 2018–2024, read directly from the pipe-delimited annual archives.

Construction proceeds in two stages. A first pass (notebook `001_lar_preparation`) applies the structural product and scope filters and writes an intermediate file; the income, down-payment, and outcome filters follow (`002_lar_preparation_2`); and the conventional, fixed-rate restriction that defines the analysis sample is applied in the analysis notebooks (`003`–`007`). Table OA1 reports every filter in sequence, its effect on the row count, and is regenerated end-to-end from the public LAR by `empirics/sample_construction_funnel.py`.

TABLE OA1. HMDA sample construction. Each row applies one filter sequentially to the 2018–2024 LAR; "Remaining" is the count after the filter and "Dropped" the records it removes. Intermediate drops are order-dependent; the final count is not.

Filter step	Remaining	Dropped
RAW (all LAR rows read)	124,615,183	
Sex reported (drop Sex-Not-Available)	103,350,736	21,264,447
County & state non-missing	101,418,323	1,932,413
Home purchase (loan_purpose=1)	42,502,777	58,915,546
Principal residence (occupancy=1)	38,437,780	4,064,997
Single unit (total_units=1)	37,828,586	609,194
Consumer (business_purpose≠1)	37,785,492	43,094
Closed-end (open_end_loc=2)	36,563,979	1,221,513
First lien (lien_status=1)	35,686,317	877,662
Conventional+FHA (loan_type in 1,2)	31,437,797	4,248,520
Exclude territories (PR/GU/VI)	31,360,113	77,684
Forward mortgage (reverse≠1)	31,352,922	7,191
Applicant age <45	20,341,531	11,011,391
In an MSA/MD (derived_msa_md≠0)	20,341,513	18
Loan amount ≥ $20,000	20,263,206	78,307
Property value ≥ $25,000	16,745,236	3,517,970
Loan term ≥ 5 months	16,722,221	23,015
Income in (0, $10M]	16,443,150	279,071
Downpayment ≤ $500,000	16,249,160	193,990
Definitive outcome (action in 1,3,7)	14,944,588	1,304,572
Conventional first lien only	11,305,506	3,639,082
Fixed rate (intro_rate_period missing)	10,534,919	770,587

Product and structural filters. I keep first-lien, purchase-money mortgages on owner-occupied (principal-residence), single-unit properties: HMDA `loan_purpose` = 1, `occupancy_type` = 1, `total_units` = 1, and `lien_status` = 1. I exclude open-end lines of credit (`open_end_line_of_credit` = 2 retains closed-end loans), business- or commercial-purpose loans (`business_or_commercial_purpose` ≠ 1), and reverse mortgages (`reverse_mortgage` ≠ 1). The first stage retains loan types 1 and 2 (conventional and FHA); the analysis sample is then restricted to *conventional first liens* (`derived_loan_product_type` = "Conventional:First Lien"), which drops FHA. FHA loans are insured outside the GSE channel, are not priced through the risk-based loan-level price-adjustment grid, and are unaffected by the 2022 guarantee-fee reform, so they are inappropriate for both the pricing mechanism and the natural experiment. Finally, the analysis sample retains only *fixed-rate* loans (`intro_rate_period` missing,

which excludes adjustable-rate mortgages).

Scope filters. I restrict to the fifty states and the District of Columbia (dropping Puerto Rico, Guam, and the U.S. Virgin Islands) and to properties in a metropolitan area (`derived_msa_md` $\neq$ 0). Two scope choices are specific to this study. First, the sample is limited to a **primary applicant under age 45** (`applicant_age` $\in$ {$<$ 25, 25–34, 35–44}), the population for whom first-time purchase and parental support are relevant. Second, I keep applications whose reported applicant sex is male, female, *or joint* (two applicants of different sex), dropping only records where sex is not reported. Retaining the "joint" category is essential for the cosigning question: an intergenerational cosign in which parent and child are of opposite sex (a daughter and her father, a son and his mother) is coded "joint" in HMDA, and 47.2% of the intergenerational pairs the age-gap rule identifies are in fact of opposite sex (Section 4.4), so excluding the category would discard nearly half of them and bias the cosigning sample toward same-sex pairs. The cosigner is identified from the applicant–co-applicant age gap (Section 4.1), not from sex. The primary applicant's sex, used as a control, is taken from `applicant_sex` rather than the derived field, which is undefined for joint applications.

The age-gap rule, bin by bin. HMDA reports applicant and co-applicant age in bins, so the twenty-year gap of Section 4.1 is applied bin to bin. Table OA2 lists the admissible pairs. The rule is conservative in one direction: a genuine parent–child pair whose reported bins imply a gap of, say, eighteen years is never flagged, so the measure understates intergenerational cosigning rather than overstating it.

TABLE OA2. Age-bin pairs flagged as intergenerational cosigning. A pair is flagged when the reported bins imply an age gap of at least twenty years between the co-applicant and the primary applicant. Primary applicants aged 45 and over are outside the estimation sample.

Primary applicant age	Co-applicant age (cosigner)
$<$ 25	45–54, 55–64, 65–74, $>$ 74
25–34	55–64, 65–74, $>$ 74
35–44	65–74, $>$ 74

Range, quality, and outcome filters. I drop records with non-positive or implausibly large income (outside (0, \$10,000,000]; HMDA income is reported in thousands), implied down payments above \$500,000, loan amounts below \$20,000, property values below \$25,000, or loan terms under five months — predominantly data errors or non-residential loans that survive the product filters. I keep only applications with a definitive outcome: originated, denied, or pre-approval denied (`action_taken` $\in \{1, 3, 7\}$); applications approved but not accepted, withdrawn, incomplete, or purchased from another institution are excluded.

For each county, a dedicated Selenium-based scraper navigates the county clerk's web portal as websites change by county. The webscraped data always contains the official information on

mortgagors, mortgagees, loan amount, municipality and day of recording. Some website offer more detailed information such as property address and day of contract. Day of recording and day of contract are usually close but differ around holidays such as Christmas or New Year's. Whereas sometimes possible to filter directly for residential mortgages, many counties do not allow such targeted searches, so any mortgages with corporate identifiers (for example, LLC or INC) are dropped. All loan amounts below $20,000 are dropped as junior liens or home equity lines are sometimes included when scraped.

Full applicant and loan profile by application type. Table OA3 is the unabridged version of Table 4: the pooled means of every applicant and loan characteristic in the estimation sample, separately for solo, same-generation joint and intergenerationally cosigned applications. Section 4.4 reports the three rows that carry its argument; the remainder are here. Two rows need reading with care. Reported income is the *combined* gross income the underwriter relied on, so on a cosigned application it already includes the parent's earnings: that row measures the admission channel rather than the child's own earnings, and for the same reason the housing-payment-to-income row is deflated for cosigned and joint applications. The interest-rate and rejection rows are pooled means over the whole sample, and so answer a different question from Facts 2 and 3 of Section 4, which compare cosigned and solo loans within property-value-by-loan-amount cells and hold the size of the transaction fixed.

The cosigner's age profile. Figure OA3 plots graphically what Table OA4 reports in levels: the distribution of cosigner age, and how rejection and the contract rate vary with it *within* each primary-applicant age bin. The gradient is reported only here: an older guarantor contributes less qualifying income and less credit enhancement, and the market prices the difference, but no later section cuts on guarantor age. The 45–54 row contains only applicants under 25, because the twenty-year age-gap rule admits that pairing for no older applicant. The marginal distribution therefore mixes cosigner age with applicant age and the rows are not comparable across bins; the gradient has to be read within a primary-applicant bin.

Where cosigned loans go. Table OA5 gives the purchaser of record in the year of origination, by application type. It is the evidence behind the claim of Section 4.4 that cosigned lending sits further inside the loan-level price-adjustment grid than ordinary lending does: cosigned loans are more likely to be sold to Fannie Mae or Freddie Mac and less likely to be retained on balance sheet.

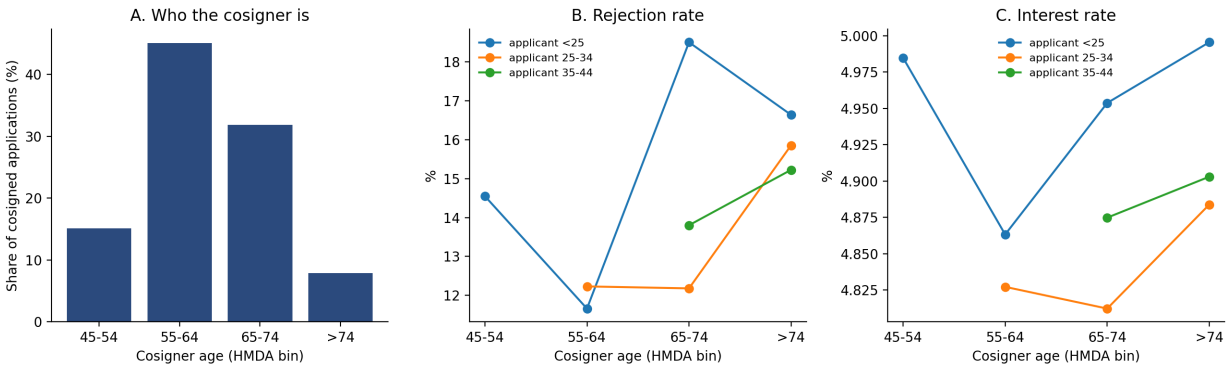
Why applications are denied. The descriptive facts report whether an application is denied; HMDA also records why. Table OA6 gives the decomposition by application type, and it separates the two margins Section 2 distinguishes. Cosigned denials are the most debt-to-income–coded of the three columns and the least coded for insufficient cash, which is the pattern the underwriting rules imply: joint income is what a cosigner supplies, so it is the constraint a cosigned file is still pressing against when it fails, while the down payment is the one margin no cosigner reaches. The

Table OA3. Applicant and loan profile by application type, full

	Mean by application type			Difference	
	Solo (1)	Joint (2)	Cosigned (3)	Cos.−Solo (4)	Cos.−Joint (5)
Reported income (combined, $000s)	94.2 (108.5)	143.1 (112.4)	172.6 (225.4)	78.4*** (0.4)	29.5*** (0.4)
Property value ($000s)	332.9 (240.2)	443.5 (286.8)	355.7 (225.9)	22.9*** (0.4)	−87.8*** (0.4)
Loan amount ($000s)	280.4 (198.3)	367.9 (226.1)	290.1 (175.9)	9.6*** (0.3)	−77.9*** (0.3)
Down payment ($000s)	52.4 (85.5)	75.6 (89.3)	65.7 (79.7)	13.3*** (0.2)	−9.9*** (0.2)
Loan-to-value (%)	87.19 (42.09)	85.65 (14.69)	84.25 (15.11)	−2.94*** (0.03)	−1.40*** (0.03)
Combined LTV, reported (%)	86.93 (12.03)	85.29 (11.78)	83.67 (13.12)	−3.26*** (0.03)	−1.62*** (0.03)
Share with CLTV > 95 (%)	20.30 (40.22)	11.60 (32.02)	5.42 (22.64)	−14.88*** (0.05)	−6.18*** (0.05)
Share with DTI ≥ 43% (%)	32.35 (46.78)	24.30 (42.89)	39.74 (48.94)	7.39*** (0.09)	15.44*** (0.09)
Share with DTI ≥ 45% (%)	23.28 (42.26)	17.06 (37.61)	30.15 (45.89)	6.87*** (0.09)	13.09*** (0.09)
Housing payment / income (%)	21.07 (33.69)	17.98 (31.88)	14.89 (28.92)	−6.19*** (0.06)	−3.09*** (0.06)
Interest rate (%)	4.618 (1.606)	4.633 (1.603)	4.869 (1.631)	0.251*** (0.003)	0.236*** (0.003)
Rate spread over APOR (pp)	0.449 (0.979)	0.403 (22.514)	0.504 (0.856)	0.055*** (0.002)	0.101*** (0.011)
Discount points ($)	2,633 (3,370)	3,224 (4,665)	2,975 (3,559)	342*** (11)	−249*** (11)
Share 30-year term (%)	93.19 (25.19)	92.61 (26.16)	94.11 (23.55)	0.92*** (0.05)	1.50*** (0.05)
Rejection rate (%)	13.30 (33.96)	9.57 (29.42)	13.18 (33.83)	−0.12* (0.06)	3.61*** (0.06)
Share female primary applicant (%)	38.64 (48.69)	34.15 (47.42)	49.82 (50.00)	11.18*** (0.09)	15.68*** (0.09)
Share White (%)	71.63 (45.08)	78.52 (41.07)	74.24 (43.73)	2.61*** (0.08)	−4.28*** (0.08)
Share Black (%)	7.99 (27.11)	3.90 (19.35)	5.34 (22.47)	−2.65*** (0.04)	1.44*** (0.04)
Share Asian (%)	10.44 (30.58)	8.59 (28.02)	10.02 (30.03)	−0.42*** (0.06)	1.43*** (0.06)
Share race not provided (%)	8.81 (28.34)	8.01 (27.14)	9.25 (28.97)	0.44*** (0.05)	1.24*** (0.05)
Share Hispanic or Latino (%)	14.34 (35.05)	11.42 (31.80)	14.69 (35.40)	0.34*** (0.07)	3.27*** (0.07)
Share above the conforming limit (%)	2.64 (16.02)	6.06 (23.86)	1.41 (11.78)	−1.23*** (0.02)	−4.65*** (0.02)
County house price index	749.2 (427.5)	800.1 (462.4)	866.3 (477.2)	117.1*** (0.9)	66.2*** (0.9)
Applications	5,837,560	4,403,026	294,333		

Notes: HMDA LAR 2018–2024, estimation sample of this appendix; the three application types are defined in Section 4.1. Columns (4) and (5) are differences in means, Welch t-test, though with millions of applications per cell the magnitudes and not the stars carry the content. Rate, spread, points, term and payment-to-income are measured on originations, every other row on applications; combined LTV is winsorized to $(1, 150]$.

Figure OA3. The cosigner's age and the terms the family gets



Notes: Panel A is the distribution of cosigner age; Panels B and C plot rejection and the mean contract rate against cosigner age *within* each primary-applicant age bin. The age-gap rule makes the series different lengths by construction — a 45–54 cosigner appears only for an applicant under 25.

Table OA4. Who the cosigner is, and how outcomes vary with cosigner age

Cosigner age (HMDA bin)	Share of cosigned (%)	Rejection rate (%)	Interest rate (%)	Property value ($000s)	Combined LTV (%)	Share DTI ≥43 (%)
45-54	15.2	14.55	4.985	300.4	86.01	32.36
55-64	45.1	12.10	4.835	368.9	83.90	34.15
65-74	31.9	13.45	4.854	405.3	81.66	41.10
¿74	7.9	15.63	4.906	375.9	82.79	41.84

Notes: Cosigned applications only, by the co-applicant's reported HMDA age bin; column 1 is the share of all cosigned applications. Rejection is measured on applications; interest rate, property value, combined LTV and the DTI share on originations.

Table OA5. Where cosigned loans go: purchaser in the year of origination

Purchaser in year of origination	Solo	Joint	Cosigned
Not sold in calendar year	17.48	19.18	14.25
Fannie Mae	29.62	28.33	32.41
Ginnie Mae	0.01	0.01	0.01
Freddie Mac	23.48	22.76	26.31
Farmer Mac	0.00	0.00	0.00
Private securitizer	0.87	0.89	0.71
Commercial bank / thrift	10.52	10.95	9.54
Affiliate institution	1.44	1.60	1.59
Other purchaser	5.10	4.45	3.44
Credit union / mortgage co.	10.99	11.35	11.31
Life insurance company	0.49	0.47	0.43
Originations	5,060,885	3,981,507	255,540

Notes: Column percentages of originations, by application type. HMDA records a purchaser only for loans sold in the same calendar year, so "not sold in calendar year" pools loans genuinely retained with loans sold the following January; the comparison across columns is unaffected so long as that timing is common to the three application types.

collateral and cash codes are reported separately but are substitutes for one event: an appraisal below the contract price is a failure of the security and, for that reason, leaves the buyer short of funds to close, and nothing in the reporting rules dictates which of the two the lender records. The table therefore also reports their union, which is invariant to that choice. It changes nothing: the cosigned deficit on the equity margin is −3.6 percentage points measured on the cash code alone and −3.6 measured on the union, because the collateral code is flat across the three columns. Two cautions apply to reading the credit-history row as evidence on the pricing axis. The shares are conditional on denial, so a column that is heavier on one reason is mechanically lighter on the rest; and HMDA carries no credit score, so the table cannot hold the applicant's own score fixed. The weakest-link claim of Appendix OA4.9 rests on the Selling Guide, not on this table.

TABLE OA6. Reasons reported for denied applications, by application type

Reason reported for the denial	Solo	Joint	Cosigned	All
Debt-to-income ratio	38.91	30.69	41.43	36.19
Credit history	39.79	43.01	38.55	40.85
Collateral	11.81	10.03	11.77	11.20
Insufficient cash (down payment, closing costs)	13.39	12.90	9.77	13.11
Employment history	4.07	5.50	4.43	4.57
Mortgage insurance denied	0.20	0.21	0.23	0.20
Collateral *or* insufficient cash	24.68	22.46	21.13	23.81
of which both cited	0.53	0.48	0.41	0.51
Any recorded reason	81.36	78.98	83.90	80.63
Recorded reasons per denial	1.08	1.02	1.06	1.06
Denied applications	776,675	421,519	38,793	1,236,987
Denial rate (% of applications)	13.30	9.57	13.18	11.74

Notes: Percent of *denied* applications flagged with each reason, on the sample of Table OA3. A denial is HMDA `action_taken` 3 (application denied) or 7 (preapproval request denied), the two actions for which a reason must be reported. Columns do not sum to 100: a lender may report several reasons for one denial, and a denial may carry none of the reasons listed. Lenders may report up to four reasons and the sample construction reads the first three; of the ten HMDA reason codes it carries the six shown, omitting unverifiable information, incomplete application, and other. "Any recorded reason" is the share of denials with at least one of the six, and so bounds how much the two omissions remove. Neither omission disturbs the comparison across columns unless it sorts by application type. "Collateral *or* insufficient cash" is the union of those two codes, not their sum. The co-citation row is reported for transparency rather than as a test of whether lenders treat the two codes as interchangeable: at 1.06 recorded reasons per denial, denials citing any two reasons are rare enough that the overlap is uninformative about coding practice.

Prevalence under both filing directions. Table OA7 gives, year by year and age bin by age bin, the two measures the filing-order paragraph of Appendix OA4.7 compares in the pooled sample: the published measure, which requires the child to be the primary applicant, and the measure that also counts pairs filed with the parent first. The gap widens over the sample period and is widest among the youngest buyers, so the published series understates both the level of cosigning and its growth.

TABLE OA7. Cosigning prevalence under both filing directions

	Applications	Cosigning rate (%)		Δ (pp)
		Child filed first	Both directions	
Panel A: by age of the young party				
Under 25	932,799	8.89	12.10	3.21
25–34	5,332,525	2.81	3.53	0.71
35–44	4,269,595	1.44	1.91	0.47
Panel B: by application year				
2018	1,392,389	2.43	3.11	0.68
2019	1,453,679	2.32	3.05	0.73
2020	1,760,139	2.45	3.17	0.72
2021	1,973,341	2.73	3.54	0.81
2022	1,542,093	3.20	4.19	0.99
2023	1,201,065	3.32	4.42	1.10
2024	1,212,213	3.34	4.44	1.10

Notes: "Child filed first" is the published measure, the age-gap rule of Section 4.1 on the estimation sample. "Both directions" adds, to numerator and denominator alike, the applications in which the older party was filed first, which the estimation sample excludes. In Panel A the "young party" is the primary applicant under the first measure and the co-applicant under the second.

Geographic reference: town to census-tract mapping. Because HMDA records the census tract of the property while some deed records report only the municipality, I construct a crosswalk between New York jurisdictions and 2020 census tracts using the TIGER (FHFA Tracts & Blocks and Places/County Subdivisions layers). For each county, spatial intersection of the county subdivision layer with the census tract layer recording all tracts that fall within each town boundary.

Lender identification and LEI matching. The lender name as it appears in the deed may differ from the institution's legal name in HMDA. To link deed records to the HMDA LAR lenders, each unique mortgagee string is matched to a Legal Entity Identifier (LEI) via the GLEIF API, using two search strategies: (i) an exact `legalName` field lookup, and (ii) a full-text search across GLEIF's name index (covering legal names, trade styles, and prior names). MERS (Mortgage Electronic Registration Systems) entries and government agency entries (HUD, USDA Rural Housing, state housing authorities) are flagged as ineligible and excluded from the LEI-match pipeline. Unmatched lenders that appear frequently are resolved manually.

Merging deed records to HMDA. The merge procedure links deed records to originated HMDA loans using a hierarchical key: records are first matched on (activity year, LEI, loan amount, town) simultaneously. Mortgage amounts are rounded to the nearest $5,000 as HMDA is also rounded and the inconsistent denomination of fees associated with the documentation of the deed. Only exact matches are considered. This comes at an obvious caveat that occasionally, the same

loan amount is requested in a given town, at a given bank in a given year. These deeds are dropped.

OA4.2 NSMO: Provenance and Analysis Samples

The NSMO supplies the default wedge, the typed default ratio (M_5), the pre-cure wedge (M_6), the parental-coverage benchmark and the cosign income gradient, and three different analysis samples appear in the text. This subsection states the file, the coverage, and how those samples relate.

The survey. The National Survey of Mortgage Originations (Federal Housing Finance Agency and Consumer Financial Protection Bureau, 2026) samples borrowers from the National Mortgage Database, a stratified random sample of closed-end first-lien residential mortgages, and surveys them shortly after origination. Loan performance is appended from the NMDB rather than self-reported, which is what makes the delinquency variables administrative and pre-cure (Appendix OA5.11). I use the Public Use File, which covers 62,359 respondents across survey waves 1 to 46 and origination years 2013 to 2024. The PUF is disclosure-protected: several variables, the origination credit score among them, are released binned rather than continuous, which is why the score enters every specification below as a categorical rather than a linear control.

Table OA8. Construction of the NSMO analysis samples

Step	**Respondents**	**% of file**
(1) NSMO PUF respondents, all waves	62,359	100.0
(2) Conventional (loan_type = 1)	48,129	77.2
(3) + first-time buyer (first_mort_r = 1)	8,249	13.2
(4) + cosigning question answered (x20a)	8,249	13.2
(5) + fixed rate, single-family detached, primary residence	4,663	7.5
(6) + origination credit score present	4,663	7.5

Notes: From `nsmo_v70_1324_puf.csv`, the NSMO Public Use File, survey waves 1–46 and origination years 2013–2024; survey weights (`analysis_weight`) applied to all reported shares. Step (3) is the universe the paper calls conventional × first-time buyer; step (5) adds the product filters used for the default and pricing work. The survey and its disclosure protection are described in the text above.

Three samples, and why they differ. Table OA8 gives the funnel; the three samples the text reports sit at different points on it, and the differences are substantive rather than incidental. The cosign income gradient uses conventional first-time buyers (8,249 respondents), which is the paper's stated universe. The rate-by-FICO schedule adds the product filters and restricts to solo borrowers (4,280), since it is the solo pricing schedule the model calibrates against. The default-wedge controls ladder, by contrast, runs on the 42,631 fixed-rate, single-family, primary-residence loans with complete controls in Table OA28, with *no* conventional-loan and *no* first-time-buyer restriction.

That last point is worth stating plainly rather than leaving to be inferred from a table note. The wedge is therefore measured on a broader population than the gradient, than the paper's

stated universe, and than the model object M_6 targets. Widening the population should if anything attenuate a wedge driven by first-time buyers, since it adds repeat buyers with longer credit histories, so the direction works against the paper rather than for it. But the reader should know the populations differ, and any comparison between the wedge and the model moment inherits that difference.

Data availability. The HMDA Loan Application Register is public and is obtained from the Consumer Financial Protection Bureau. The NSMO Public Use File is public and is obtained from the Federal Housing Finance Agency. The Fannie Mae Single-Family Loan Performance Data are obtained from Fannie Mae after free registration. The PSID is public and is obtained from the University of Michigan. The deed records are collected from New York county clerk land-record portals that provide free public access to recorded instruments; the scraping and cleaning code is part of the replication package, and because redistribution rights for the recorded instruments themselves rest with the individual county clerks, the package ships the code and the derived county-level aggregates rather than the raw scraped records. All code required to reproduce every table and figure is included; every number in the text is emitted by one of the scripts listed in `writing/stats.tex` rather than typed by hand.

OA4.3 Fannie Mae Loan Performance and the Link to HMDA

Facts 3 and 4 follow loans after origination in the Fannie Mae Single-Family Loan Performance Data (Fannie Mae, 2026b). Fannie Mae publishes one file per acquisition quarter holding every monthly report on the fixed-rate, fully documented single-family loans it acquires: the origination terms, both borrowers' credit scores, and each month's delinquency status, forbearance, modification and deferral flags, and the reason the loan ends. I keep purchase loans on one-unit principal residences. The files used here cover acquisitions from 2018Q1 to 2024Q4.

The link to HMDA. Fannie Mae records no borrower ages, so the parent–child flag comes from HMDA. I match each Fannie Mae loan to the HMDA originations of the same year on state, county (through the loan's three-digit ZIP code), exact note rate, loan amount in \$10,000 bands, term, combined loan-to-value ratio and debt-to-income band within one step, and the presence of a co-borrower, and keep only loans with exactly one HMDA candidate. Of the 6.85 million one-unit Fannie Mae loans originated in 2018–2024, 4.56 million are matched, between 55% and 84% by origination year. Rerunning the match with every Fannie Mae rate shifted by an eighth of a point puts false matches at 0.3–1.4% of matches. Unique matches are harder in dense cells, so first-time buyers, high loan-to-value loans and the 2020–21 vintages match less often, which is why every comparison is made within the score band or the price cell. The analysis sample is 2,295,805 matched first-time purchases: 71,819 cosigned, 929,696 other two-borrower and 1,294,290 solo-borrower loans.

Definitions. Delinquency is Fannie Mae's month-end status. Its lowest category covers current loans and loans less than 30 days late together, so a payment made late within the month is not observed. Months in forbearance count neither as delinquency nor as exposure, and neither do months of arrears carried out of forbearance until the loan is next current. A forced exit is a foreclosure or deed-in-lieu, a short sale, a third-party sale or a sale of the loan as non-performing. The last rows of Table OA9 follow each loan's first 30-day delinquency for twelve months, excluding delinquencies with less than twelve months of follow-up, with forbearance in the window, or ended by a repurchase.

Results. Table OA9 reports the full conditioning ladder. The 30-day gap turns from positive to slightly negative once the score band is held fixed. The gaps in serious delinquency, forced exits and recovery keep their sign and most of their size as leverage, origination quarter, debt-to-income and state are added. At the same score, 0.7% of cosigned loans that fall 30 days behind end in a forced exit, against 2.4% of solo-borrower loans, and a year after a first 30-day delinquency 75.7% are current without a servicer workout, against 66.4%. The NSMO ladder of Table OA28 records a raw delinquency gap of similar size in percentage points but cannot tell a parent from a spouse; Fact 4 uses these records for that reason.

The note rate, and Fact 3 in other data. In the linked loans, cosigned loans pay +0.23 percentage points more than solo-borrower loans. Holding the score band or the loan-to-value ratio fixed leaves the gap about where it is; holding the whole price cell fixed, the origination quarter together with the score band and the loan-to-value ratio, cuts it to +0.08. Two other sources give the same reading. In the National Survey of Mortgage Originations (NSMO; Federal Housing Finance Agency and Consumer Financial Protection Bureau, 2026), which records the score, the cosigned premium is +0.01 pp once FICO and the combined loan-to-value ratio are held fixed. HMDA records no score, and there cosigned loans price +0.251 percentage points above solo-borrower loans, 4.869% against 4.618%; the gap survives property-value, loan-amount, year, state and county cells (Table OA18).

OA4.4 Deed Sample: Construction, Validation, and Provenance

The HMDA sample is documented by the twenty-two step funnel of Table OA1. The deed records, which supply the augmented prevalence measure of Section 4.2 and the cosign-share moment M_3, had no equivalent. This appendix supplies one, adds the first validation of the $>$ 2-signatory rule, and corrects the provenance of the intensive-margin estimates. Everything below is emitted by scripts under `empirics/utils/` (`deed_construction_funnel.py`, `deed_signatory_audit.py`, `deed_intensive_margin.py`); no number here is typed by hand. Table OA10 is that funnel, from raw scrape to matched sample.

TABLE OA9. Delinquency and Its Resolution: Cosigned vs. Solo Loans (Fannie Mae)

		Difference to solo (pp), conditioning ladder				
Outcome	**Solo**	**Raw**	**Score**	**+LTV**	**Price cell**	**+Full**
Ever 30+ days past due						
Cosigned (parent–child)	11.54%	+2.32***	−0.68***	−0.36**	−0.24	−0.35**
		(0.20)	(0.14)	(0.15)	(0.15)	(0.16)
Other two-borrower		−3.48***	−5.33***	−5.23***	−5.14***	−4.68***
		(0.12)	(0.11)	(0.11)	(0.11)	(0.13)
Ever 90+ days past due						
Cosigned (parent–child)	2.59%	−0.31***	−1.23***	−1.00***	−1.00***	−1.03***
		(0.07)	(0.07)	(0.07)	(0.06)	(0.06)
Other two-borrower		−1.18***	−1.74***	−1.68***	−1.70***	−1.58***
		(0.06)	(0.06)	(0.05)	(0.05)	(0.05)
Forced exit, loans ever 30+ days past due						
Cosigned (parent–child)	2.43%	−1.48***	−1.68***	−1.32***	−1.36***	−1.28***
		(0.15)	(0.16)	(0.14)	(0.14)	(0.14)
Other two-borrower		−1.09***	−1.29***	−1.26***	−1.33***	−1.02***
		(0.11)	(0.12)	(0.12)	(0.12)	(0.09)
Current without a workout, 12 months after the first 30-day delinquency						
Cosigned (parent–child)	66.36%	+6.27***	+9.34***	+7.47***	+7.58***	+7.83***
		(0.58)	(0.56)	(0.54)	(0.54)	(0.50)
Other two-borrower		+2.19***	+5.21***	+5.03***	+4.92***	+5.33***
		(0.35)	(0.30)	(0.27)	(0.27)	(0.29)

Notes: Linear-probability estimates on Fannie Mae Single-Family Loan Performance loans matched one-to-one to HMDA: first-time buyers, purchase, principal residence, one unit, originated 2018–2024 (N = 2,295,805 loans: 71,819 cosigned, 929,696 other two-borrower, 1,294,290 solo). Cosigned is the HMDA age-gap flag for a parent–child pair of borrowers. Months in forbearance, and arrears carried out of forbearance until the loan is next current, count neither as delinquency nor as exposure. The forced-exit rows use the 234,299 loans ever 30 days past due (forced exit: foreclosure, short sale, deed-in-lieu, third-party or non-performing note sale); the last rows use the 176,191 first 30-day delinquencies with 12 months of follow-up and no forbearance or repurchase in that window. **Solo** is the solo mean; the other columns are differences to solo in percentage points. Ladder: **Score** fixed effects for the lower of the borrowers' credit scores in nine bands (the score the GSE grid prices); +**LTV** adds LTV bands; **Price cell** origination quarter × score band × LTV band; +**Full** adds DTI-band and state fixed effects, log loan amount and the high-balance flag (and the calendar quarter of the delinquency in the last rows). Standard errors clustered by state. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

TABLE OA10. Construction of the web-scraped deed sample

	Step	Records	Counties
(1)	Instruments scraped	4,307,527	40
(2)	After cleaning (corporate borrowers, loans below $20,000)	1,035,671	27
(3)	Lender resolved to a GLEIF entity identifier	773,005	27
(4)	Matched to an originated HMDA loan	36,958	27
	of which flagged by the deed rule (> 2 signatories)	1,286	
	of which flagged by the HMDA age-gap rule	1,304	
	flagged by both	178	
	flagged by either (the augmented measure)	**2,412**	

Notes: Step (2) to step (4) is a 3.6% match rate: the scrape is every recorded mortgage instrument, the HMDA sample only conventional fixed-rate first-lien purchase loans to owner-occupiers under 45 (Table OA1). The county column counts merged *files*, one of which bundles the five New York City boroughs. The two flags overlap on 7.4% of flagged loans (Jaccard).

Coverage, and a correction. The scrape reaches 40 county directories, of which 27 clear the cleaning stage and 27 produce a merged file. One of those bundles the five New York City boroughs, so the sample covers 31 counties rather than the thirty-four previously reported in Section 4.2; that figure was not reproducible from the pipeline and has been corrected. Fourteen counties were scraped but never cleared cleaning, in most cases because the portal returned partial records. The step (2) to step (4) match rate of 3.6% is not a measure of merge failure: the scraped universe is every recorded mortgage instrument, while the HMDA sample is restricted to conventional, fixed-rate, first-lien purchase loans to owner-occupiers under 45, so most scraped instruments have no eligible counterpart. Table OA11 names every county in the sample and reports the instruments scraped in each recording year, so the coverage behind the prevalence measure can be read directly rather than inferred. Coverage is not uniform: portals differ in how far back their free full-text access reaches, and the resulting county–year pattern, not any sampling design, is what determines which markets the deed measure observes.

Validating the > 2-signatory rule. The rule cannot, on its own, separate a child and a parent from a couple and a sibling, three co-investors, or a trust. Borrower names survive only in the raw scrapes, as the `OR Party` field, so the checks below run there and are joined back to the merged sample on county and instrument identifier; they reach 53% of flagged loans, three merged files having been written without an identifier column.

Two mechanical contaminants are bounded and small. County clerks record aliases as separate party rows, which could inflate a two-party instrument into a "cosigned" three-party one; collapsing alias rows moves only 0.7% of instruments below three, so the practice adds rows to instruments that already have three distinct people rather than manufacturing them. Non-person parties (limited companies, trusts, estates) appear on 11.2% of all 3+ instruments but only 2.4% of those that reach the merged sample, the corporate filter and the purchase-loan restriction having removed most of them.

Table OA11. Web-scraped deed coverage by county and year

County	2018	2019	2020	2021	2022	2023	2024	**Total**	**Merged**
Albany	13.1	15.2	21.2	29.8	20.7	13.0	12.9	126.0	1,848
Broome	11.3	11.1	11.5	15.0	12.7	9.3	9.5	80.4	1,324
Cayuga	3.7	3.8	4.7	5.3	4.6	3.4	3.5	29.0	501
Chenango	1.0	0.9	0.9	1.3	1.0	0.8	0.8	6.7	134
Columbia	3.2	3.1	4.2	4.9	4.0	2.4	2.3	24.1	354
Cortland	2.1	2.0	2.3	3.1	2.5	1.8	1.8	15.6	278
Delaware	2.1	2.0	2.7	3.4	2.5	1.9	2.0	16.6	258
Dutchess	254.4	580.6	367.4	228.3	100.7	58.2	45.3	1,634.8	888
Erie	48.9	48.7	55.3	67.3	49.6	39.2	38.0	347.1	1,019
Essex	1.9	1.8	2.4	2.1	2.2	1.5	1.6	13.5	81
Greene	2.2	2.2	2.7	3.6	3.2	2.0	1.7	17.6	447
Herkimer	2.4	2.2	2.2	2.9	2.8	1.9	2.2	16.6	358
Kings	156.7	89.1	105.7	137.4	110.5	57.0	65.0	721.4	–
Livingston	–	1.5	1.7	2.1	1.7	1.4	1.3	9.8	131
Monroe	14.4	14.1	15.1	18.7	16.4	13.7	6.7	99.1	1,705
Nassau	138.7	57.8	47.6	–	–	–	–	244.1	2,741
Oneida	8.0	8.7	9.6	11.4	10.4	7.5	8.2	63.7	1,529
Onondaga	38.5	39.7	46.8	53.7	41.9	34.0	33.7	288.4	8,237
Ontario	3.1	3.4	4.3	4.9	3.6	2.6	2.7	24.6	1,064
Otsego	2.5	2.4	2.6	3.5	2.9	2.0	2.2	18.1	295
Putnam	4.9	4.4	6.6	10.3	6.5	4.0	3.7	40.5	1,448
Rensslaer	6.4	6.7	9.0	12.3	10.2	7.2	7.4	59.1	928
Rockland	6.1	6.3	8.6	12.3	7.7	5.1	5.1	51.2	3,307
Schoharie	1.3	1.3	1.4	2.0	1.8	0.8	1.1	9.6	187
Steuben	11.0	11.2	11.8	13.4	11.5	8.4	8.6	76.0	713
Tompkins	2.0	2.0	2.3	2.7	2.1	1.5	1.4	14.0	575
Ulster	7.8	7.6	9.7	13.4	10.2	6.5	6.4	61.6	1,857
Total	747.7	929.8	760.5	664.9	444.0	287.1	275.2	4,109.3	36,958

Notes: Instruments scraped, in thousands, by county and recording year, read off the raw scrape filenames; the *Merged* column is the count matched to an originated HMDA loan and is in units, not thousands. "–" means no file for that county-year, which reflects portal coverage rather than an absence of recorded instruments. One merged file bundles the five New York City boroughs and is reported under *Nyc*. Counties scraped but not carried through cleaning, in most cases because the portal returned partial records: Chautauqua, Madison, Niagara, Orange, Oswego, Saratoga, Schenectady, Seneca, Warren, Westchester.

The third check fails, and the failure is worth reporting because it bounds what the measure can be asked to do. Surname structure looks like a natural way to separate families from co-investors, but it cannot: a daughter who has taken her husband's name, her husband, and her father produce three distinct surnames, and so do three unrelated parties. Testing this directly, the surname distribution among deed-flagged loans that the HMDA age-gap rule independently confirms as intergenerational is statistically indistinguishable from the distribution among those it does not confirm ($\chi^2_2 = 1.82$, $p = 0.40$). Surname structure therefore carries no information about the parent–child question and is not reported as though it did. What remains is the overlap between the two rules, 9.4% of deed-flagged loans, and that is itself confounded: with three or more signatories HMDA reports two, and the age gap is computed on whichever two. The deed measure is accordingly used as a measure of multi-party borrowing that HMDA's two-borrower cap conceals. Whether it carries intergenerational content of its own is a separate question, and the next paragraph tests it rather than settling it by argument.

A positive test the deed rule can pass. The surname check fails in one direction only: it cannot confirm the family reading, and it equally cannot refute it. A test that can confirm it uses the other instrument on the same loan. HMDA reports at most two borrowers and their age bands, so on deed-flagged loans that the age-gap rule does *not* flag, HMDA's own second borrower is independent evidence about who is signing. If those loans are intergenerational they should report a second borrower more often, and an older one, than loans neither rule flags. The test is one-sided by construction: with three signatories and two HMDA slots the guarantor may sit outside the HMDA record entirely, so a skew is evidence for the family reading while its absence would not have been evidence against it.

Table OA12 runs it on the 1,108 deed-only loans against the solo group, excluding the 178 loans both rules flag so that nothing in the age rows can be mechanical. Deed-only loans report a second borrower on 65.9% of applications against 44.9% of solo ones, an adjusted difference of 18.71 points (2.35) once county and year fixed effects and the applicant controls are in place. That settles the first half: the rule is finding multi-party transactions rather than recording noise. On the age margin the reported second borrower is at least forty-five on 9.4% of deed-only loans against 5.2% of solo ones, an adjusted 3.74 points (1.72, $p = 0.029$). The two sharper margins move the same way without reaching significance on 27 county clusters: at least fifty-five, 1.50 points ($p = 0.130$), and a mean age gap 0.47 years wider ($p = 0.143$).

The age-gap rule therefore still carries most of the intergenerational content of the augmented measure, but it no longer carries all of it. On the one margin HMDA permits us to observe, loans the age rule never touched are drawn from an older second-borrower distribution at the five percent level, which is the direct evidence the surname test could not supply.

An external benchmark. Both of the paper's prevalence measures are constructed here, so it is worth asking whether an outside source puts cosigning in the same range. The National Survey of Mortgage Originations asks borrowers directly, and on the paper's universe (conventional, first-time

TABLE OA12. What HMDA reports about the second borrower on deed-flagged loans

	Solo	Deed only	Deed (all)	HMDA age-gap
Loans	34,546	1,108	1,286	1,304
Reports a co-applicant (%)	44.9	65.9	70.6	100.0
Conditional on a reported co-applicant:				
Co-applicant ≥ 45 (%)	5.2	9.4	27.2	100.0
Co-applicant ≥ 55 (%)	0.7	2.3	19.2	87.7
Mean age gap (years)	0.57	1.28	7.52	33.22
Deed only minus solo, county and year fixed effects and applicant controls:				
	Estimate	(cluster)	(HC1)	N
reports a co-applicant (pp)	$+18.71^{***}$	(2.35)	[1.35]	35,652
co-applicant ≥ 45 (pp)	$+3.74^{**}$	(1.72)	[1.11]	16,217
co-applicant ≥ 55 (pp)	+1.50	(0.99)	[0.59]	16,217
age gap (years)	+0.47	(0.32)	[0.27]	16,217

Notes: The 36,958 loans of the merged deed × HMDA sample. *Deed only* is deed-flagged and *not* age-gap flagged, so its age columns cannot be mechanical; the 178 loans both rules flag sit in *Deed (all)* and in *HMDA age-gap*. Ages are HMDA bands evaluated at midpoints, the open bands at the midpoint of a ten-year interval on the closed side, which understates the top gap. Loans whose co-applicant age is coded not-provided count as reporting a co-applicant but drop out of the three age rows. The bottom block regresses each outcome on the deed flag over the age-gap-unflagged sample, with county and year fixed effects, log income, and applicant age band, race and sex. Standard errors clustered by county in parentheses, over 27 clusters, which is few enough that the heteroskedasticity-robust alternative is reported in brackets alongside rather than suppressed. Stars follow the clustered p-value. ***, **, *: 1, 5, 10%.

buyers, 8,249 respondents, survey-weighted) the answer depends entirely on which question is used. Asked whether they had to add a co-signer *or co-borrower* to qualify, 9.81% say yes — but that counts a spouse added to the note, which the intergenerational measures here deliberately exclude, so it is not a comparable figure and is reported only to forestall its being quoted as one. Restricting to respondents who also report that nobody else lives in the home, which isolates a cosigner who does not occupy the property, gives 4.20%.

That is the comparable number, and it falls between the two measures built here: above the HMDA age-gap rate on the matched sample and below the augmented rate. The ordering is what one would expect. The survey measure is stricter than mere presence, since it requires the cosigner to have been *needed* to qualify, but it catches non-occupant cosigners that HMDA's two-borrower cap and twenty-year age-gap rule both miss. An independent source using a different instrument on a different sample therefore brackets the paper's headline rather than contradicting it. It does not validate the deed rule specifically, and it inherits the usual limits of a self-reported survey measure.

Who the deed records add. The surname test above cannot say whether the deed rule finds parents. A different comparison can say something almost as useful: whether the people the deed rule finds *look like* the people the age-gap rule finds. Every descriptive fact in Section 4 is computed on the age-gap measure alone, so the roughly half of the augmented prevalence that the deed records contribute has never been described. Table OA13 describes it, profiling 1,108 loans flagged by the deed rule and *not* by the age-gap rule against 1,304 flagged by the age-gap rule and against solo

borrowers on the matched sample.

TABLE OA13. Who the deed records add: profile by cosign measure

	Solo	HMDA age-gap	Deed only	Union
Observations	34,546	1,304	1,108	2,412
Income ($000s)	130.00	210.31	147.86	181.63
Property value ($)	385,539	436,557	443,854	439,909
Loan amount ($)	315,069	343,972	355,217	349,138
Implied equity ($)	70,470	92,584	88,637	90,771
Interest rate (pp)	4.61	4.81	4.88	4.84
Combined LTV	84.97	81.46	82.51	81.94
Share with DTI $\geq$ 43%	24.25	40.74	32.16	36.79

Notes: The merged deed×HMDA sample. *Deed only* are loans flagged by the > 2-signatory rule and not by the age-gap rule. A deed is recorded only when a loan closes, so the sample carries no rejected applications and rejection variables are omitted. Levels are local to New York and to a loan-amount-matched sample; the object of interest is the comparison across columns.

Read the table in two blocks. On everything that does not depend on who is listed on the application — property value, loan amount, implied equity, the interest rate and combined loan-to-value — the deed-only group differs from solo in the same direction as the age-gap group and by a similar amount, the ratio of the two gaps running between 0.70 and 1.39. On the two variables that *do* depend on the applicant list, the groups diverge sharply: the age-gap group reports income $80k above solo where the deed-only group reports only $18k, and the same pattern appears in the share above a 43% debt-to-income ratio.

That divergence is the point rather than a problem, and its direction is a check. HMDA reports the *combined* income of the parties on the note and computes debt-to-income from it. A cosigner caught by the age-gap rule is on the application, so the parent's income sits inside the reported figure; a third signatory caught only by the deed rule is not on the application — which is exactly why HMDA misses them — so the reported income excludes it. Were the deed rule instead picking up couples, siblings or co-investors already visible to HMDA, the income gap would resemble the age-gap group's. It does not. This is the one positive piece of evidence that the deed rule finds genuinely off-application parties, and it is weaker than a relationship test would be, but it is not nothing.

Do the sample filters select on the outcome? Two filters in the HMDA construction funnel are applied before the definitive-outcome filter and could in principle remove denials differentially, which would put every rejection result in the paper on a sample selected on its own outcome. Replaying the funnel over the raw register and tabulating outcomes at each of the two, the answer is not the one the ordering suggests. The property-value floor removes 3,517,970 records of which 61.1% are denials against 11.7% of those it keeps, an enormous imbalance: HMDA property value is frequently missing on denied applications, where no appraisal was obtained. The sex-reported

filter, by contrast, removes 21,264,447 records whose denial share, 22.7%, is close to the 21.0% it retains.

Levels are not what matters here. Fact 2, the debt-to-income leniency evidence and the rejection difference-in-discontinuities are all cosigned-versus-solo *contrasts*, and a filter that removes denials from both arms alike shifts the level without disturbing the contrast. Splitting each filter by cosign status reverses the ranking. The property-value floor removes denials at +47.1 points among cosigned applications against +49.5 among solo, a difference-in-differences of -2.4 points against a level effect near fifty: it is almost purely a level effect, and the contrasts survive it. The sex filter, innocuous in levels, carries a differential of +4.4 points. I report that as a caveat rather than a finding, because cosign status is itself constructed from the co-applicant age field, and records with sex unreported are records with weaker applicant reporting generally, so the flag is measured with more error in precisely the set being dropped.

What the deed measure can and cannot be used for. One further limitation governs how the deed records are used in the rest of the paper, and it concerns geography rather than levels. Across the 26 counties with more than a hundred matched loans, the deed-based cosign rate ranges from 0.38% to 12.85%, with a mass-weighted coefficient of variation of 0.85 against 0.51 for the HMDA age-gap rate — a variance ratio of 2.8. Thirty-fold variation in the same underlying behaviour invites the question of whether the deed rule is measuring cosigning or measuring how a county clerk fills out a form.

Section 4 supplies the benchmark for answering it. Fact 1 documents that cosigning rises where houses are expensive relative to local incomes, and the HMDA measure, built on a uniform federal reporting standard, carries the price gradient strongly: the county-level correlation between the HMDA cosign rate and the local house-price index is +0.90. The deed measure does not. Its correlation with the same index is +0.31, while its correlation with the mean number of parties recorded per instrument — computed over *all* instruments, including the one- and two-party majority the > 2 rule never flags, so a property of recording practice rather than of cosigning — is +0.52. The two measures correlate with each other at only +0.34 across counties.

The deed rate therefore tracks recording practice more closely than it tracks the economic driver of cosigning, and the reverse holds for the HMDA rate. I read this as a limit on scope rather than as grounds for discarding the deed data: the records remain the only source that sees the multi-party arrangements HMDA's two-borrower cap conceals, which is what the augmented *level* is built from, but their cross-county variation is not interpretable as variation in family behaviour. Every geographic result in the main text — Fact 1, the county map, and the tract-income gradient — accordingly rests on the HMDA measure alone, and the extrapolation from New York to the nation inherits whatever recording heterogeneity the scraped counties carry. With 26 counties none of these correlations is precisely estimated; the exercise bounds the concern rather than settling it.

Provenance of the intensive-margin estimates. The estimates previously reported for this sample could not be retained. They were generated on 28 October 2025, against merged county

files last rebuilt on 25 June 2026, and were drawn from the New York HMDA sample estimated *without* the deed records, with heteroskedasticity-robust rather than county-clustered standard errors. Table OA14 re-estimates the specification on the current merge, separately for the three cosign measures, since they are distinct objects.

TABLE OA14. Intensive-margin outcomes on the current deed-matched sample

	Log property value	**Log loan amount**	**Interest rate (pp)**
HMDA age-gap	−0.2133***	−0.2456***	+0.0902***
	(0.0329)	(0.0352)	(0.0164)
implied equity gap	+0.0323		
deed (¿2 signatories)	−0.0269	−0.0414	+0.1088***
	(0.0206)	(0.0262)	(0.0243)
implied equity gap	+0.0145		
augmented (union)	−0.1207***	−0.1446***	+0.1061***
	(0.0264)	(0.0300)	(0.0182)
implied equity gap	+0.0239		
County FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Demographics	Yes	Yes	Yes
Observations	36,956	36,956	36,889

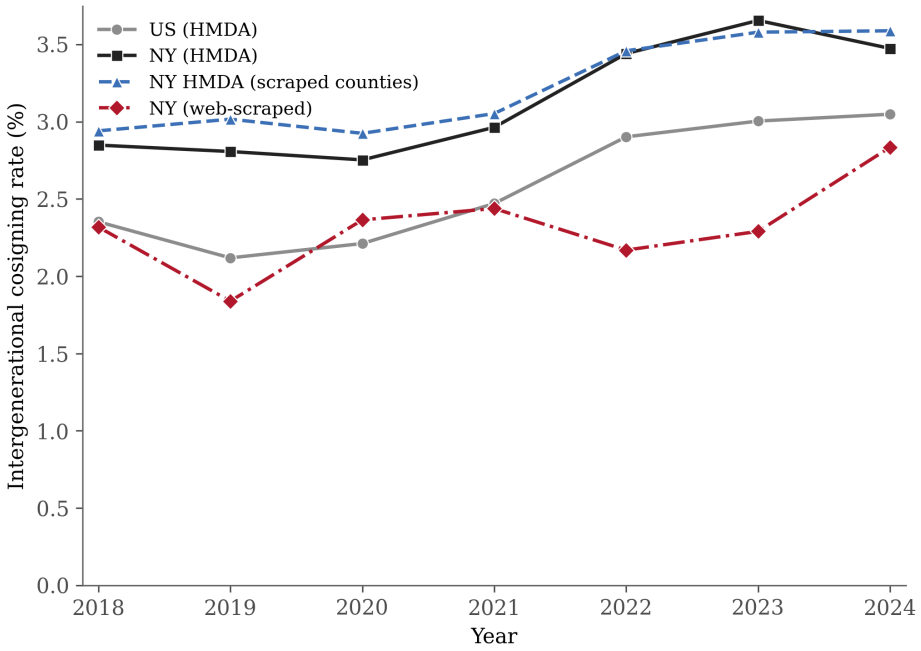
Notes: OLS on the merged deed×HMDA sample, re-estimated on the current merge (Appendix OA4.4). Demographics are applicant age bin, race, sex and log income; standard errors clustered by county. The *implied equity gap* is the property-value coefficient less the loan-amount coefficient, the share of the property-value difference not financed by a larger loan. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Two things follow. The conditional rate premium is unchanged in substance: the augmented measure gives +0.1061 percentage points with a county-clustered standard error of 0.0182, against the +0.1045 previously reported. The property-value and loan-amount coefficients, by contrast, change sign. That is a difference in sample rather than a reversal of the equity gap of Section 4.4: the deed-matched sample is matched *on loan amount*, which reweights the loan-size distribution mechanically, and it compares cosigned loans against all non-cosigned rather than against solo borrowers. What the body actually claims is about the difference between the two coefficients, since that difference is the part of the property-value gap not financed by a larger loan, and it is stable: +0.0239 log points here against +0.0212 in the earlier estimates. The guarantee still arrives with a down payment attached.

OA4.5 Representativeness of the Web-Scraped Sample

Figure OA4 sets the scraped counties against the full New York HMDA series and that series against the national one. It is parallel movement rather than a common level that the argument needs, for the reason the note gives.

FIGURE OA4. Intergenerational Cosigning Rates: US, New York, and Web-Scraped Sample



Notes: The level gap between the HMDA and deed series is definitional: HMDA identifies two-party cosigning by age difference, deeds capture additional signatories. Parallel trends, not levels, establish representativeness.

How the augmented column is constructed. The *Augmented* column of Table 3 adds a single national uplift of +3.00 percentage points to the HMDA age-gap baseline. I measure the uplift on the 36,958-loan New York deed-matched sample (Table OA10) as the share of loans flagged by *either* the deed rule or the age-gap rule, 6.53%, less the share flagged by the age-gap rule alone, 3.53%. It is applied as one national scalar rather than county by county because, as Appendix OA4.4 shows, the deed measure's cross-county variation substantially reflects recording practice rather than family behaviour. The two columns of Table 3 therefore share no denominator — the baseline is a national count, the uplift a New York extrapolation — so the augmented column is an estimate, and no difference column is reported.

OA4.6 The Descriptive Facts on the Augmented Measure

The facts of Section 4 use the HMDA age-gap rule. This subsection repeats three of them on the deed-matched New York sample with the augmented measure, which counts a loan as cosigned when either the deed rule or the age-gap rule flags it.

Fact 1 across census tracts. Figure OA5 replaces the county with a census tract in a given year. The ratio is the tract's median property value among purchase applications over its median family income, and no geographic fixed effects are absorbed, so the comparison runs across markets as in Figure 2. The cosigned share rises from 4.1% in the fifth of tract-years with the lowest ratio, which averages 1.5, to 8.6% in the highest fifth, where it averages 5.9. Each rule on its own rises

with the ratio too, except that the deed rule falls back in the highest fifth, which is mostly New York City tracts.

FIGURE OA5. Cosigning and House Prices Relative to Income: New York Census Tracts



Notes: Deed-matched HMDA sample, New York 2018–2024: 31,446 loans in 6,628 tract-years with at least ten purchase applications in the estimation sample behind the tract's median property value. Share of loans cosigned under the augmented measure, the deed rule or the age-gap rule, against the tract-year's median property value divided by the tract's median family income in the 2019–2023 American Community Survey, in 50 equal-count bins of tract-years weighted by loans. HMDA reports 2010 tracts until 2021; their income is the land-area-weighted mean over the 2020 tracts they overlap. Logarithmic horizontal axis.

Facts 3 and 4 on deed-linked loans. The deed records carry no credit score and say nothing about repayment, so Facts 3 and 4 need the Fannie Mae loans of Appendix OA4.3. I rerun that appendix's link for New York, which reproduces its matches, and join the deed-matched sample to the same HMDA records on the fields both carry, keeping one-to-one links. Of the deed records, 28,510 link, and 6,933 of the Fannie Mae loans they reach are first-time purchases. Table OA15 reports both facts on these loans. Within a price cell the cosigned-minus-solo gap in the note rate is +0.08 percentage points, with a standard error of 0.02, against +0.08 on all linked loans. At the same credit score, cosigned loans fall 30 days behind less often than solo loans, 9.0% against 12.7%, and reach 90 days on 1.5% against 2.7%, where all linked loans give 1.4% against 2.6%. That last difference rests on 12 serious delinquencies among cosigned loans and has a standard error of 0.7 points, so it agrees with Fact 4 without establishing it on its own.

OA4.7 Who Cosigns: Composition, Geography, and Checks on the Age-Gap Rule

Section 4.4 reports the three features of the cosigning population that Sections 5 and 6 use. This subsection collects the remainder: who the applicants are and how old the guarantor is, where cosigning happens, the evidence that the market's thinness is not a supply-side phenomenon, and two checks on the age-gap rule that identifies intergenerational pairs.

TABLE OA15. Facts 3 and 4 on deed-linked New York loans

	New York, deed-linked			All linked
	Cosigned	**Solo**	**Difference**	**Difference**
Note rate (%), same price cell	4.70	4.62	+0.08***	+0.08
			(0.02)	
Ever 30+ days past due (%), same score	9.0	12.7	−3.7***	−0.6
			(1.0)	
Ever 90+ days past due (%), same score	1.5	2.7	−1.2*	−1.2
			(0.7)	
Loans	551	3,648		

Notes: Fannie Mae Single-Family Loan Performance Data linked to HMDA and to the New York deed records (Appendix OA4.4): first-time purchases of one-unit principal residences originated 2018–2024 in the scraped counties. **Cosigned** is the augmented measure, flagged by the deed rule or the HMDA age-gap rule; **Solo** loans have one borrower and neither flag; other two-borrower loans enter the regressions as a separate group. Note rate: within a price cell, origination quarter by lower-score band by loan-to-value band. Delinquency: share of loans ever 30 or 90 days past due, within the lower-score band; months in forbearance, and arrears carried out of forbearance until the loan is next current, do not count. The cosigned level is the solo level plus the difference. Standard errors in parentheses, clustered by three-digit ZIP code. **All linked**: the differences of Facts 3 and 4 on every linked first-time purchase (Section 4). Of the cosigned loans, 12 reach 90 days past due. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Thinness is not a supply-side phenomenon. One candidate explanation for a market this thin is that few lenders write the product; they do. Of 3,883 lenders originating in the sample, the 2,348 with at least 100 originations are almost all in the business: only 2.9% never originate a cosigned loan, the median such lender's cosigning rate is 2.37% and the 90th percentile is 4.38%. Cosigned originations are barely more concentrated than lending as a whole (Herfindahl 128 against 98; top-ten share 27.7% against 24.7%), the cosigning rate is flat across lender size deciles, and depositories and independent mortgage banks write the product at nearly the same rate (2.35% and 2.86%). Nor do lenders keep the paper: cosigned loans are *more* likely than solo loans to be sold to Fannie Mae or Freddie Mac in the year of origination (58.7% against 53.1%) and less likely to be retained (14.3% against 17.5%, Table OA5), so cosigned lending sits further inside the loan-level price-adjustment grid than ordinary lending does, not outside it.

Applicant composition: more women, fewer Black applicants. Women are 49.8% of cosigned primary applicants against 38.6% of solo ones, a gap of +11.2 percentage points (s.e. 0.094). The racial pattern runs the other way and is larger in relative terms: Black applicants are 5.3% of cosigned applications against 8.0% of solo ones, so the family-guarantee channel is used least by the group with the widest homeownership gap and the thinnest parental balance sheets — the supply-side constraint of Section 8 appearing directly in the cross-section.

Fathers cosign more often than mothers. HMDA reports the sex of both parties, so the pair itself can be typed. Among the 280,649 intergenerational pairs in which both parties report male or female, the cosigner is male — a father, in the modal case — in 58.4% of pairs and

female in 41.6%. The four cells are son–father 30.6%, daughter–father 27.8%, daughter–mother 22.2% and son–mother 19.4%. The two sides of the pair are tilted in opposite directions: sons and daughters receive a cosign in almost exactly equal numbers, which already makes daughters over-represented relative to their 38.6% share of solo applicants, while on the supply side fathers outnumber mothers by roughly three to two. 47.2% of pairs are of opposite sex, which is what makes retaining HMDA's "Joint" category indispensable (Section 4.1). Because `co_applicant_sex` is not retained by the estimation pipeline, these shares are computed in a separate pass over the raw Loan Application Register (`empirics/utils/cosign_direction_and_sex.py`).

Validating the age-gap rule. The cosigning indicator infers a family relationship from an age difference, so it is worth asking whether the pairs it selects look like families. They do. Among applications where both parties report a race, 96.8% of intergenerational pairs share the primary applicant's race, against 93.1% of same-generation joint applications. The age-gap rule selects pairs that are *more* racially concordant than married couples in the same data, which is what parent–child pairs should look like and not what unrelated co-investors would look like.

What filing order hides. HMDA's "primary applicant" is a filing convention, not an economic role, and the estimation sample keeps only applications whose primary applicant is under 45. An intergenerational pair recorded with the *older* party first is therefore dropped outright. Replaying the sample construction without the age restriction turns up 94,061 such applications against the 294,333 the age-gap rule keeps, so 24.2% of two-party intergenerational pairs are filed parent-first. Counting both directions raises the two-party cosigning rate from 2.79% to 3.65%, a difference of 0.86 percentage points; Table OA7 gives the two measures year by year and age bin by age bin. The gap is widest exactly where cosigning matters most — 0.86 points on average but over three points among buyers under 25 — and it widens over the sample period, so the published series understates both the level of cosigning and its growth. Read that as an upper bound on what filing order hides rather than as a correction: HMDA does not say which party is buying, so a parent-first pair may be a child's purchase recorded the other way round or a parent's own purchase with an adult child as co-borrower, and the two are indistinguishable in these fields. The sign is not in doubt in either case — the headline rate is a floor — which is a further reason to carry the deed records of Section 4.2, where every party to the instrument is named and filing order carries no information.

Where cosigning happens. Figure OA6 maps the rate across the 2,388 counties with at least 200 applications, together 98.8% of the sample. Dispersion is wide: the ninetieth-percentile county cosigns at 3.86% against 1.38% at the tenth, a ratio of 2.8 around a median of 2.29%. The geography is not the one a story about dense coastal metros would predict. Rates run highest through the mountain West and in amenity counties — Gunnison, Summit and Routt in Colorado, Gallatin in Montana, Mono in California, Dare in North Carolina, Monroe in Florida — and lowest across the interior Midwest and Plains. What those counties share is a high ratio of house prices to

local earnings rather than a high price level as such, the county-level counterpart of the tract-level gradient in Table OA16 below. Counties with fewer than 200 applications are left grey rather than plotted, since a rate on a thin denominator is mostly noise and a choropleth would give it the same visual weight as a large county; 2,388 counties clear the threshold, covering 98.8% of the estimation sample.

Table OA16 reports that tract-level gradient itself, on a measure of neighbourhood income the parent's earnings cannot inflate: HMDA's applicant income cannot locate cosigning in the income distribution, because on a cosigned application that field already contains the parent's earnings. Measured instead by the tract's income position relative to its own metropolitan median, the rate is 3.14% in the lowest-income tracts, peaks at 3.50% in moderate-income ones, and falls across the remaining bins to 2.19% in the highest. Table OA17 asks the prior question, how many young adults have a parent able to help at all. The either-channel row is the headline because a cosign needs either earnings to add to the file or a balance sheet to stand behind the loan, and which channel binds differs across families. The second both-channel row proxies the parent's debt-to-income headroom by whether they own outright, a stock test standing in for a flow constraint that discards every parent carrying a mortgage they could comfortably service. Every share conditions on the parent being alive and interviewed, so young adults whose parents have died or left the survey, whose capacity is zero, sit outside the denominator: each figure is an upper bound on the population share with a capable parent.

TABLE OA16. Cosigning by the neighbourhood's income position

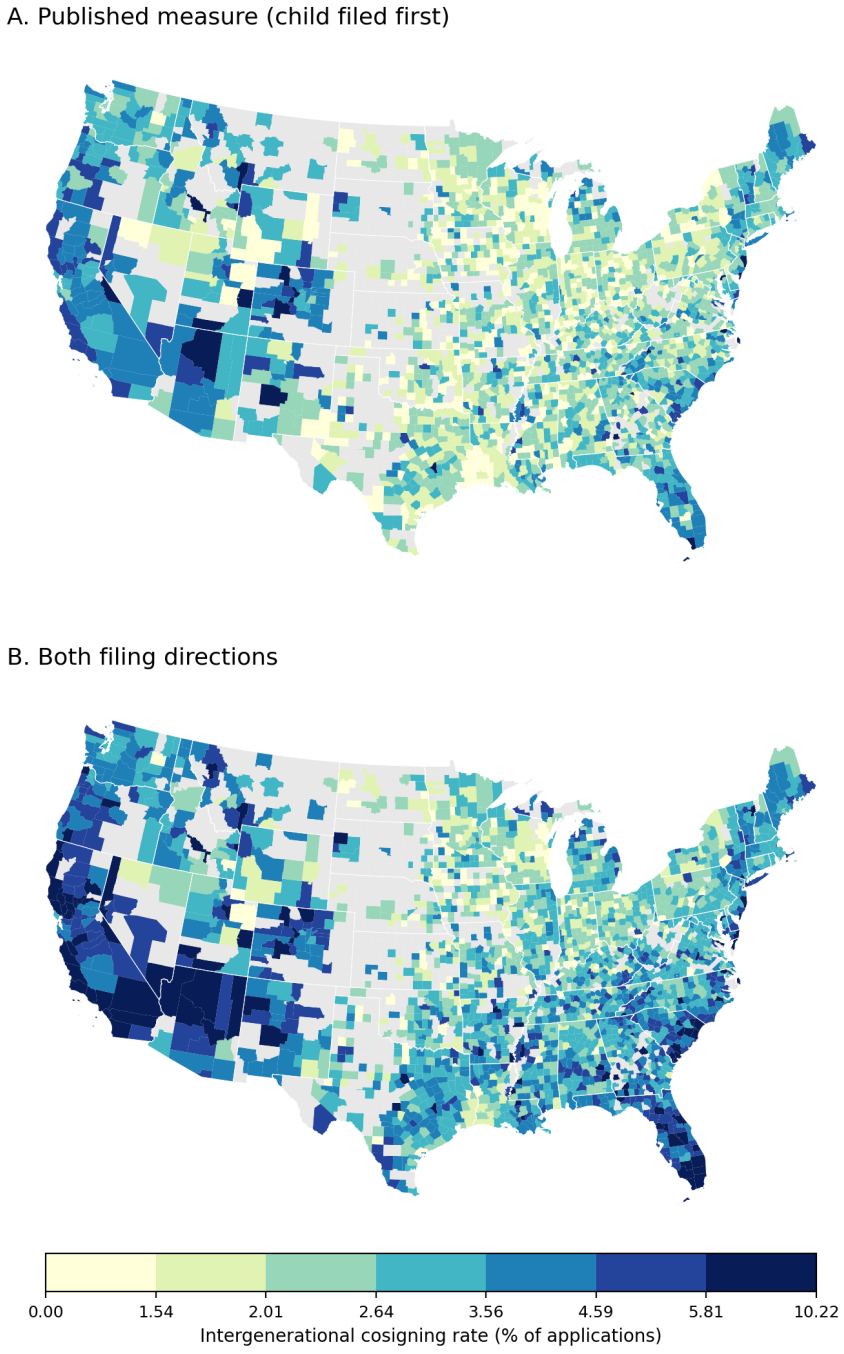
Tract income relative to metro median	Applications	Cosigning rate (%)		Δ (pp)
		Child filed first	Both directions	
Low (<50% of metro median)	402,098	3.14	4.42	1.28
Moderate (50–80)	1,546,873	3.50	4.70	1.21
Middle (80–100)	2,136,092	3.06	4.06	0.99
Middle-upper (100–120)	2,366,502	2.76	3.57	0.81
Upper (120–150)	2,339,769	2.51	3.22	0.71
High (≥150)	1,743,585	2.19	2.73	0.54

Notes: Applications grouped by the HMDA field `tract_to_msa_income_percentage`, the tract's median family income as a percentage of its metropolitan median, cut at the standard low/moderate/middle/upper points. Being a property of the tract, it is not inflated by the parent's earnings entering a cosigned file. Columns as in Table OA7.

OA4.8 Descriptive Facts: Stability Across Stratification

Table OA18 reports the four descriptive facts of Section 4 across cell definitions: its first column is the property-value by loan-amount cells the body states, and the rest add year, state or county. The three facts separate. The interest-rate premium of Fact 3 is the most stable, positive at every level and largest under county cells. The property-value and loan-amount gaps of Section 4.4 attenuate

FIGURE OA6. Intergenerational cosigning by county



Notes: Cosigning rate as a percentage of applications, by county, pooled over 2018–2024. Panel A is the published child-filed-first measure, Panel B adds parent-filed-first applications (Table OA7); the two share one colour scale, so Panel B is uniformly darker and the geographic pattern should be read within each panel. Counties under 200 applications are grey; Alaska and Hawaii are in the data but off the map.

TABLE OA17. Who could cosign: parental capacity among young adults

	Share of young adults (%)	
Parental capacity measure	All aged 25–44	Renters only
Parent owns their home	77.9	70.5
Parent owns outright (no mortgage)	36.0	31.0
Parent housing equity $\geq$ \$50k	68.6	60.5
Parent housing equity $\geq$ \$100k	58.1	50.9
Parent net worth $\geq$ \$100k	67.3	58.8
Parent net worth $\geq$ \$250k	49.9	42.8
Parent income $\geq$ half the child's	74.7	81.6
Parent income $\geq$ the child's	51.1	65.1
Either channel: income $\geq$ half the child's *or* net worth $\geq$ \$100k	88.5	88.4
Either channel, strict: income $\geq$ the child's *or* net worth $\geq$ \$250k	72.1	74.8
Both channels: equity $\geq$ \$50k and income $\geq$ half	53.4	51.7
Both, with DTI headroom: outright owner, equity $\geq$ \$100k, income $\geq$ half	21.2	19.6
Person-years	9,382	5,032

Notes: PSID matched parent–child file, 2019 and 2021 waves, weighted by the child's individual weight; young adults aged 25–44 linked to an observed parent. Housing equity is PSID imputed wealth including home equity less imputed wealth excluding it. The headline measure is the *either-channel* row; the two *both-channel* rows are reported for comparison only.

once geography is absorbed, which is the same composition effect the fixed-effects estimates in Table OA14 pick up, though their difference, the equity gap, keeps its sign. The rejection difference of Fact 2 is the one that does not survive: it is −0.06 points under year cells, so it should not be read as evidence that cosigned files are rationed more tightly, only that they are not rationed less.

TABLE OA18. Cosigning, credit access, and housing demand: stability across stratification

	Additional stratification			
Cosigned − solo	none	year	state	county
Rejection rate (pp)	+1.26	-0.06	+0.21	+1.20
Interest rate (pp)	+0.169	+0.112	+0.257	+0.308
Property value (\$000s)	+21.1	+16.2	-0.7	-1.4
Loan amount (\$000s)	+7.4	+3.6	-8.8	-5.2

Notes: The "none" column is the property-value by loan-amount cells reported in the text of Section 4; the rest add year, state or county to the cell definition. Cosigned minus solo-borrower means, excluding same-generation joint applications.

OA4.9 Cosigning Mechanics: Additional Underwriting Detail

Section 2 summarizes how income pooling and weakest-link pricing move the admission and price margins in opposite directions. This subsection adds the underwriting mechanics behind that summary.

A primary borrower with income y and a cosigning parent with income y_{parent} qualifies on the basis of joint income $Y_{co} = y + y_{parent}$, which can clear the DTI threshold even when y alone cannot; for loans processed through Desktop Underwriter (DU), the automated underwriting track, the DTI is calculated on the combined income and liabilities of all borrowers. The representative credit score used for GSE pricing is determined by taking the middle bureau score for each borrower and then, for loans with multiple borrowers, selecting "the lowest applicable score from the group as the representative credit score for the loan." That representative score is used "for pricing purposes (i.e., assessing LLPAs) on all loans" (Fannie Mae, 2026a). The *admission* side of this rule softened during the sample period while the pricing side did not: since September 2021, Desktop Underwriter applies the 620 eligibility floor to the *average median* credit score across borrowers, whereas loan-level pricing continues to use the lowest representative score (Fannie Mae, 2021). This mid-sample change widened the admission–pricing wedge and leaves the weakest-link pricing rule intact throughout 2018–2024.

For manually underwritten loans, the guide constrains the occupying borrower directly: "[u]sing only the income of the occupying borrower(s) to calculate the DTI ratio, the maximum allowable DTI ratio is 43%." No occupant-only ceiling applies to loans underwritten through Desktop Underwriter. DU "analyzes the risk factors in the loan casefile for all borrowers" and evaluates the blended ratio within its overall risk assessment (Fannie Mae, 2020). A cosigner therefore operates as a compensating factor in automated underwriting: debt loads that would trip the threshold for a solo applicant can be approved when a non-occupant co-borrower backs the loan. The main text sets the manual route aside: the loan records do not say which route a file took, and the model admits on joint income throughout, so the occupant-only ceiling is treated as not binding.

The overlay is a ceiling that is slack for most cosigned loans rather than one that is irrelevant. Cosigned first-time buyers are materially less levered than solo ones — mean CLTV 83.7 against 86.9, and 5.4% above a 95% CLTV against 20.3% (Section 4) — and part of that gap is the overlay truncating the upper tail mechanically rather than selection on family equity. What is true is that the cap sits well above where the cosigned distribution is concentrated, so the estimated model sets the two limits equal. The approximation errs in a known direction: relative to the rulebook it gives cosigned borrowers *more* equity room than they in fact have, so the deposit relief priced in Section 8 is a conservative estimate of what the equity constraint costs a cosigning family.

OA4.10 International Comparison of Family Guarantee Regimes

The sharp segmentation between access and pricing that characterizes the U.S. cosign market is not universal. Sorting major mortgage markets by the mechanism they use shows that take-up is substantially higher wherever a family guarantee that reaches the price or the deposit is available, and lowest in the two markets where none is.

Private resolution: pooled pricing and collateral substitution. Most private markets resolve the admission-pricing wedge that the U.S. leaves open. Norway offers two instruments and

they divide the work between them. A *medlåntaker* (co-borrower) is assessed jointly with the buyer on income and debt-servicing capacity and adds nothing to the security, so it reaches the affordability test and stops there. A *realkausjon* is a third-party pledge of real property, typically the parents' home, and it is what satisfies the equity requirement: the pledge counts as *tilleggssikkerhet* inside the regulatory *belåningsgrad* calculation, so it relaxes the 85% loan-to-value cap directly (Smarte Penger, 2026; Finanstilsynet, 2024). A Norwegian family blocked by the deposit therefore has an instrument built for it; a U.S. family has none. Canada (17%) absorbs high-LTV risk into a national insurance pool run by the CMHC, with premiums largely invariant to credit score above an eligibility threshold, socializing the intensive-margin penalty for a risky young borrower (Government of Canada, 2024). Australia (10%) goes further: family guarantors pledge actual home equity as secondary collateral to avoid Lenders Mortgage Insurance, improving *both* margins through a direct collateral enhancement (Housing Australia, 2026; Aussie Home Loans, 2026).

The United Kingdom is the exception among the private regimes, and the instructive one. *Joint borrower sole proprietor* (JBSP) arrangements, on the order of 7% of borrowers, pool the parent's income for the affordability assessment and stop there. The co-borrower signs the mortgage but is not named on the title deeds and pledges no property of their own. The lender's security therefore remains the subject property alone, and the loan-to-value ratio, with it the deposit, is exactly what it would have been (Accord Mortgages, 2026). Nor does the parent's credit standing reach the rate. U.K. mortgages are priced off loan-to-value bands and product terms rather than a borrower-level risk grid, so a JBSP loan carries essentially the pricing of the equivalent standard residential product at the same LTV (Duck, 2024). The parent's income clears the door and reaches nothing else.

Two further details sharpen the parallel with the United States. Where the supporting parent still carries a mortgage of their own, several U.K. lenders cap the JBSP loan-to-value *below* the standard maximum. That is the same direction as the U.S. non-occupant overlay described above: on both sides of the Atlantic, bringing a parent onto the note can tighten the loan-to-value test rather than relax it. And the exception is beginning to appear. A small number of lenders now write JBSP above the ordinary ceiling, up to 100% of purchase price with family support (Mortgage Solutions, 2026). Such a product is close to the instrument Section 8 finds should exist, and its emergence at the edge of the market rather than at its centre is what the mechanism here would predict.

The United States and the United Kingdom therefore reach the same outcome by different institutional routes: in the U.S. the pricing rule explicitly discards the cosigner's credit file, while in the U.K. there is no borrower-level risk price for it to enter. They are the two thinnest markets in the table.

Public resolution: the state as cosigner. Across much of Southern Europe and the Nordics the private guarantee is subsidized or replaced by a public one. State guarantees back roughly 24% of first-time-buyer loans in Portugal (The Portugal Post, 2026; Banco de Portugal, 2026, 2025;

Banco Santander, 2026) and 20% in Italy (the *Consap* fund, for buyers under 36) (Finizio, 2026). Spain's ICO *aval* guarantees up to 20% of the price for young buyers, though take-up remains low (about 15% of young buyers, barely 1% of originations) (López, 2025; Regidor, 2025), and Finland's state-backed "ASP" system covers roughly 19% of first-time-buyer volume by pairing subsidized savings with sovereign guarantees (State Treasury of Finland, 2026). In the Iberian markets the public scheme also operates *in combination with* the family: the traditional instrument there is the parental guarantee (the Portuguese *fiadores* and the Spanish *aval*, typically provided by a parent), which long predates the state programs and continues alongside them. The quoted shares therefore describe parent-combination regimes in which the state supplements rather than replaces the family guarantor. In these bank-centric systems the state's presence is a binary condition for access that circumvents the continuous risk-based pricing friction seen in the U.S.

Comparability, sources and caveats. The shares in Table 2 are orders of magnitude rather than comparable statistics. Definitions are not harmonized across countries, and the sources are of mixed type: official statistics for Canada, Norway and Finland, national press for Italy, Portugal and Spain, and industry or lender material for Australia and the United Kingdom. Three rows carry qualifications of their own, marked in the table. Norway's two entries are both taken from the 2021 survey, the last vintage that publishes a level for each instrument on the same population and denominator: 18.5% of new amortising loans to first-time buyers had an older co-borrower outside the household, and 13.4% a guarantor or additional security in a dwelling owned by someone else (Finanstilsynet, 2021). The two overlap slightly and cannot be summed, since 1.6% of first-time buyers had both. Both series have moved since, and the co-borrowing one is volatile: it fell to 8% by the 2024 survey before rising to 13% in 2025 (Finanstilsynet, 2025), while the guarantor series has continued to decline without a published level. Finland's 19% is a share of first-time-buyer loan *volume* rather than of borrowers. Spain's ICO *aval* dates only from 2024, so no period average exists: the 15% is take-up among eligible young buyers, and barely 1% of all originations. And no official U.K. series on JBSP take-up could be located, so the 7% is a trade-press estimate (Duck, 2024) and should be read as an order of magnitude; the *mechanism* entry for the United Kingdom is sourced to lender criteria (Accord Mortgages, 2026) and is not subject to that caveat. The U.S. row is the augmented HMDA-plus-deed measure of Section 4, computed on purchase applications to primary applicants under 45; HMDA carries no first-time-buyer flag, so age proxies the first-time margin, and the U.S. row is therefore not on exactly the population the peer rows describe.

Table 2, reported in Section 2, makes the point that a family guarantee which clears the access constraint and improves nothing else is close to an Anglo-American peculiarity. In seven of the nine markets some instrument that clears the access constraint, whether a creditworthy relative or a public guarantor, also reaches the price or the deposit. Norway supplies both kinds, and the contrast there is instructive: its co-borrowing reaches neither margin, exactly as U.S. cosigning does, but a family stopped by the deposit can pledge the parents' home instead. Only in the United States and the United Kingdom does the family instrument reach neither margin with no alternative

that does. The two regimes get there differently, and the distinction is worth keeping: the U.S. weakest-link rule is the only one that *explicitly* pools income for admission and then discards the cosigner's credit file in the price. U.K. JBSP simply has no borrower-level risk price for a cosigner to enter. The mechanism is American; the outcome is shared. That wedge is the institutional foundation of what follows, and it is what makes the access value of cosigning separable from its pricing value.

A second regularity runs through the table, and it bears directly on the reform Section 8 evaluates: almost everywhere except the United States, a family or public guarantee is allowed to stand in for the *deposit*. In Norway the standard parental instrument is the *realkausjon*, a lien on the parents' own property, used precisely when the child cannot meet the equity requirement of the lending regulation (Finanstilsynet, 2024). An Australian family pledge works the same way, securing the loan against the parents' home so the effective loan-to-value ratio falls to the level at which lenders' mortgage insurance no longer applies (Housing Australia, 2026; Aussie Home Loans, 2026). The public schemes are built on the same margin rather than on price: Italy's *Consap* fund guarantees the tranche of the loan above the ordinary 80% loan-to-value ceiling, which is what makes a 100% mortgage possible for a buyer under 36 (Finizio, 2026). Spain's ICO *aval* guarantees up to 20% of the purchase price (López, 2025), and Portugal's scheme and Finland's ASP pair a sovereign guarantee with the same objective (The Portugal Post, 2026; State Treasury of Finland, 2026). Even in the United Kingdom, where the guarantee instrument itself reaches neither margin, the deposit is addressed by a separate product rather than by JBSP. The Barclays Family Springboard has the helper post ten percent of the price into a linked account, so that the buyer needs no deposit at all (Duck, 2024).

The United States fails on both margins at once, and it is not quite alone, which is the useful part. A U.S. cosigner's income clears the income test and is then discarded in the rate, and their balance sheet never touches the loan-to-value test, because the cosigner is liable for the debt rather than pledging collateral. A U.K. JBSP borrower ends up in the same position by a different route. The two regimes whose family instrument reaches neither the price nor the deposit are also the two thinnest markets in the table, while every market above ten percent lets a guarantee substitute for equity or absorbs the high-leverage risk publicly. Nine countries with very different housing and credit institutions identify nothing, so this is a pattern and not evidence. But it is the same pattern the estimated model produces from the inside in Section 8: pricing the guarantee is not what buys access, and letting it stand in for the deposit is.

OA5 Natural Experiment: Further Robustness and Validity

OA5.1 Mapping the Estimates to the Model

Each natural-experiment estimate is the empirical image of a structural object. The chain runs through the pricing condition. Under the grid (2) a common fee passes through equally to both menus; when the rate also reads the menu (B.17) (Online Appendix OA1), it passes through as

$1/\mu_k$, so the differential $\beta_3^{co} - \beta_3^{so} = \Delta g^{reform}(1/\mu^{co} - 1/\mu^{so})$ is positive if and only if $\mu^{co} < \mu^{so}$ (OA35) — and, because μ^k is near one for any GSE-eligible pool, its magnitude is capped at 0.03–0.12 basis points (§5.4), below any detectable range. The reform's rate estimate is therefore mapped to the model as a *bound* — no economically large repricing of the guarantee — rather than as a measurement of the pool-quality gap $\bar{\mu}_{so} - \bar{\mu}_{co}$, whose existence is established by the NSMO wedge and whose size the structural estimation identifies.

Calibrating μ^k from the NSMO. Write expected repayment per dollar promised as $\mu^k = 1 - p^k \cdot \text{LGD}$, where p^k is the default rate of pool k and LGD is loss given default, the share of the balance a lender loses when a loan terminates in foreclosure. The NSMO wedge of Online Appendix Table OA28 gives $p^{so} = 4.58\%$ and $p^{co} = 4.58\% + +1.47 = 6.05\%$ on the 30/60-day rung. Two conversions bracket the object the pricing rule needs. Taking early delinquency at face value, with every spell terminating in default and LGD = 0.40, gives $\mu^{so} = 0.982$ against $\mu^{co} = 0.976$; allowing that most early delinquency cures, so that 30% of spells terminate in default, and taking LGD = 0.25, gives $\mu^{so} = 0.997$ against $\mu^{co} = 0.995$.[5]

The rule is the right order of magnitude on levels. Fed these μ^k, the pricing rule implies a level cosign premium $r_{co} - r_{so} = (1+\rho+g)(1/\mu^{co} - 1/\mu^{so})$ of 12 to 64 basis points, where $1+\rho+g$ is the lender's gross cost of a dollar lent, its cost of funds plus the guarantee fee, which the paper normalizes to one: a rule that priced the measured risk would charge tens of basis points. Within the price cell the data show a cosigned rate only +0.08 points above the solo rate (Fact 3), and the pricing grid in the model charges none. The model's own counterpart is larger: before the guarantee, the lender's book rule of Section 8 would charge the cosigned pool 134.7 basis points (Figure 9). What matters here is that the multiplicative rule, calibrated externally, produces a premium of the size the *risk* warrants, and that none of it is charged.

The reduced form thus returns what the model predicts at its resolution — no detectable repricing of the guarantee on either margin, with the insulating sign on rationing. The *magnitude* of the pool-quality gap $\bar{\mu}_{so} - \bar{\mu}_{co}$ is what the structural estimation of Section 7 identifies. (That gap is the whole of (OA11); A names one component of it, the Region-2 trap loss, and the two should not be conflated.)

OA5.2 The Year-FE Primary: The Rate Margin's Robustness Ladder

Table OA20 re-estimates the year-FE rate differential of Table C.1 with one design choice changed per row and everything else held fixed. The differential is a null in every row: across the variants and the four windows it spans −0.11–+0.06 pp, and no cell's p-value falls below 0.66. The two choices that moved the retired estimate most do not move this one: the covariate convention, worth a third of a percentage point there (Appendix OA5.12), is worth -0.09 against -0.06 at h = \$20k here, and

[5] The loss-given-default band is the one Section 8 uses, from Qi and Yang (2009); the cure fraction is asserted rather than measured. Neither choice matters for the conclusion drawn here, since both conversions put the implied premium an order of magnitude above what lenders charge.

TABLE OA19. Mapping the natural-experiment estimates to the model.

Empirical object	Model counterpart	Sign
Rate differential $\beta_3^{co} - \beta_3^{so}$: null, bounded small	Prop. OA2 via (OA35): mechanism-implied differential 0.03–0.12bp, below any detectable range	≈ 0
Rejection differential $\beta_3^{co} - \beta_3^{so} < 0$	Prop. 3(ii): leniency $\tau > \tau^*$; joint income $Y_{co} = y + y_p$ in part (i)	$-$
No economically large repricing of the guarantee on either margin	The pricing rule reads the child's file, not the guarantee	bound
NSMO default wedge $\hat{\gamma} > 0$	Premise $\bar{\mu}_{co} < \bar{\mu}_{so}$ (trap term A)	$+$
Wedge concentrated in the GSE-priced channel	(2) binds via the risk-based g-fee/LLPA grid	$+$

closing the post window at end-2022 — so the January-2023 LLPA re-grid never enters — returns -0.11 pp, because each year carries its own intercept either way. The rejection margin's full-sample counterpart, computed in the same pipeline as the denial variants (`rejection_variants.csv`), is -0.053–-0.024 across the wide grid, negative at 9 of 9 windows (smallest $p = 0.26$): the insulating sign is not an artifact of the above-AMI restriction.

TABLE OA20. The Year-FE Primary Specification: Robustness Ladder (Rate Margin)

	Cosigned − solo differential, by bandwidth ($000s)			
Design variant	**18**	**20**	**22**	**24**
Year-FE primary (headline, Table C.1)	−0.01 (0.23)	−0.06 (0.23)	−0.06 (0.23)	−0.05 (0.23)
Donut: drop the $5k bracket straddling the limit	−0.03 (0.25)	−0.09 (0.25)	−0.06 (0.25)	−0.01 (0.25)
+ quadratic in the running variable	−0.10 (0.32)	−0.06 (0.23)	−0.09 (0.23)	−0.07 (0.23)
Triangular kernel	−0.09 (0.24)	−0.07 (0.23)	−0.07 (0.23)	−0.07 (0.23)
Lender FE only (no covariates)	−0.06 (0.22)	−0.09 (0.22)	−0.10 (0.22)	−0.09 (0.21)
Full sample (waiver-exposed)	+0.06 (0.22)	+0.03 (0.22)	+0.03 (0.22)	+0.03 (0.22)
Post window closed end-2022	−0.07 (0.32)	−0.11 (0.33)	−0.10 (0.32)	−0.10 (0.32)
Solo reweighted to cosigned LTV mix	+0.02 (0.23)	−0.03 (0.24)	−0.02 (0.23)	−0.02 (0.23)

Notes: Each cell is the cosigned-minus-solo difference-in-discontinuities in the note rate under the year-FE primary (15) with only the named design choice changed, lender-clustered standard errors in parentheses; the headline row is the above-AMI estimate of Table C.1 (N at h = $20k: 23,985 / 2,586). Each row is defined in the paragraph above. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

OA5.3 The Four Pricing Cells and the Coefficient Mapping

The pricing rule of Section 3.1 normalizes the lender's cost of a dollar lent to one. Here the cost is written out, as the cost of funds ρ_t and the guarantee fee $g_{s,t}$, because the experiment moves both, and the rule is read by cosigning group $k \in \{so, co\}$ as under the menu-reading rule

of Online Appendix OA1, $1 + R^k = (1 + \rho_t + g_{s,t})/\mu^k$; the posted rule of the paper is the case $\mu^{so} = \mu^{co} = \bar{\pi}$. This gives an interest rate in each of the four cells defined by side of the conforming limit ($s \in \{-, +\}$, below or above) and reform timing ($t \in \{Pre, Post\}$):

$$1 + R^k_{s,t} = \frac{1 + \rho_t + g_{s,t}}{\mu^k_{s,t}}, \qquad \begin{aligned} \rho_t &= \rho + \Delta\rho\, \mathbf{1}\{t = Post\}, \\ g_{s,t} &= g_0 + \Delta g^{HB} \mathbf{1}\{s = +\} + \Delta g^{reform} \mathbf{1}\{s = +,\, t = Post\}, \end{aligned} \tag{OA32}$$

where ρ is the lender's cost of funds, $\Delta\rho$ a common macro funding shock between periods, g_0 the baseline guarantee fee, Δg^{HB} the pre-existing high-balance fee premium for loans above the threshold, Δg^{reform} the 2022 fee increase, and $\mu^k_{s,t} \in (0, 1]$ the average repayment quality of pool k in that cell. Keeping the multiplicative structure, rather than log-linearizing it away, is what allows the design to speak to heterogeneous pass-through.

Differencing the four cells recovers every coefficient of the Diff-in-Disc regression $E[Y^k \mid D, Post] = \alpha^k + \beta^k_1 D + \beta^k_2 Post + \beta^k_3 (D \times Post)$ in primitives. Under time-invariant composition, $\mu^k_{s,Post} = \mu^k_{s,Pre} \equiv \mu^k_s$, the above- and below-threshold changes are

$$R^k_{+,Post} - R^k_{+,Pre} = \frac{\Delta\rho + \Delta g^{reform}}{\mu^k_+}, \qquad R^k_{-,Post} - R^k_{-,Pre} = \frac{\Delta\rho}{\mu^k_-}, \tag{OA33}$$

and under local continuity, $\mu^k_+ \approx \mu^k_- \equiv \mu^k$, the common macro term cancels in their difference, leaving

$$\beta^k_3 = \frac{\Delta g^{reform}}{\mu^k}. \tag{OA34}$$

The reform fee is scaled by the *inverse pool quality* of group k. Whenever cosigned pools carry lower average quality ($\mu^{co} < \mu^{so}$), an ordering Fact 4 (§4.5) establishes for the borrowers' own repayment before any parental cover, the same reform fee passes through to a larger rate increase for cosigners:

$$\beta^{co}_3 - \beta^{so}_3 = \Delta g^{reform} \left(\frac{1}{\mu^{co}} - \frac{1}{\mu^{so}} \right) > 0. \tag{OA35}$$

The differential is Sensitivity Decoupling on the intensive margin: heterogeneous pass-through of a *common* cost shock, generated by the level difference in pool quality rather than by any differential change in composition.

Additivity. The *additivity* condition of Picchetti et al. (2024) is mild here: the 2022 shock is a flat increment layered on the standing high-balance premium — cost components add in the numerator of (OA32) while pool quality scales within group — so nothing makes the reform scale the pre-existing discontinuity.

Exposure. High-balance status switches deterministically at the limit, so assignment is sharp. The fee is not. Fannie Mae and Freddie Mac charge it when they buy the loan, so a high-balance loan kept on balance sheet or sold privately never pays it. If the agencies buy a share s^k of pool

k, (OA34) becomes $\beta_3^k = s^k \, \Delta g^{reform}/\mu^k$, and the β_3^k of Section 5 are intention-to-treat effects of high-balance status rather than effects of paying the fee. The cosigned pool is the more exposed one: the agencies buy 58.7% of cosigned originations in the year of origination against 53.1% of solo ones (Table OA5, all originations rather than the high-balance window alone). On the admission margin the gap runs against the result, since the more exposed pool is the one whose denials show no detectable response. On the rate margin it runs with (OA35), so a rate differential would have to be netted of the exposure gap before it could be read as pool quality; with no rate response detected in either pool, the point does not bind. The share s^k can itself respond — a lender facing the fee can keep the loan instead — and the estimates include that response.

Written out state by state, the four cells are

$$1 + R^k_{-,Pre} = \frac{1 + \rho + g_0}{\mu^k_{-,Pre}} \tag{OA36}$$

$$1 + R^k_{-,Post} = \frac{1 + \rho + \Delta\rho + g_0}{\mu^k_{-,Post}} \tag{OA37}$$

$$1 + R^k_{+,Pre} = \frac{1 + \rho + g_0 + \Delta g^{HB}}{\mu^k_{+,Pre}} \tag{OA38}$$

$$1 + R^k_{+,Post} = \frac{1 + \rho + \Delta\rho + g_0 + \Delta g^{HB} + \Delta g^{reform}}{\mu^k_{+,Post}} \tag{OA39}$$

and differencing them gives the three remaining regression coefficients in full:

$$\alpha^k = \frac{1 + \rho + g_0}{\mu^k_{-,Pre}} - 1, \tag{OA40}$$

$$\beta_1^k = R^k_{+,Pre} - R^k_{-,Pre} = \frac{1 + \rho + g_0 + \Delta g^{HB}}{\mu^k_{+,Pre}} - \frac{1 + \rho + g_0}{\mu^k_{-,Pre}}, \tag{OA41}$$

$$\beta_2^k = R^k_{-,Post} - R^k_{-,Pre} = \frac{1 + \rho + \Delta\rho + g_0}{\mu^k_{-,Post}} - \frac{1 + \rho + g_0}{\mu^k_{-,Pre}}. \tag{OA42}$$

The intercept is the baseline (Below, Pre) rate; β_1^k is the pre-existing jump at the threshold, carrying both the high-balance fee and any cross-cutoff difference in pool quality; β_2^k carries the common macro funding shock together with any quality drift below the threshold.

What each identifying assumption buys. Local continuity does no work on β_1^k: the cross-cutoff composition in the pre-reform jump — the sorting that motivates a difference-in-discontinuities rather than a plain RD — is differenced out by *time-invariance*, not assumed to be zero. Smoothness is invoked only to cancel the macro residual $\Delta\rho(1/\mu^k_+ - 1/\mu^k_-)$ in the per-group β_3^k, so β_1^k keeps its composition term intact.

The differential needs less than local continuity. Written without it, the macro term in (OA35) enters only as $\Delta\rho\big[(1/\mu^{co}_+ - 1/\mu^{co}_-) - (1/\mu^{so}_+ - 1/\mu^{so}_-)\big]$, which vanishes when the cross-cutoff

sorting gap is *similar across cosign groups* — a weaker condition than full smoothness, and one the Difference-in-McCrary of Appendix C.3 supports, since bunching is stable and of similar magnitude in both pools.

The denial margin. The algebra above is for the interest rate, in which fees and the funding shock add. The headline outcome is a denial citing debt-to-income, a threshold outcome, and there time-invariance has to hold for more than pool composition. The pre-reform jump in denials compares the share of applicants past the cap just above and just below the limit, and each share depends on how many applicants sit near the cap. A rise in mortgage rates moves that mass, so the old jump — the high-balance fee's effect and the sorting together — can change between periods with composition held fixed. What the denial estimates need is therefore that the pre-existing jump in denials would have stayed the same without the reform. The yearly event study of Figure 6 supports this over 2018–2021, when mortgage rates fell, but no pre-reform year saw a rise like the three hundred basis points of 2022–2024, so for that rise the condition is assumed rather than tested.

OA5.4 Selection on Observables: Balancing Tests

Even if the number of bunchers is constant, the *type* of borrowers sorting across the limit could have changed. I test for jumps in the composition of *pre-determined* borrower characteristics at the threshold using the same DiDisc specification, replacing the dependent variable with each characteristic. I restrict attention to characteristics not co-determined with the loan-amount choice — applicant income, age and gender — since loan size, property value, and LTV are jointly chosen with the running variable, so a discontinuity in them cannot separate selection from a behavioral response. As shown in Table OA21, the cosigned-minus-solo DiDisc is statistically insignificant for every characteristic ($p > 0.10$): the composition of borrowers at the cutoff did not shift differentially with the reform.

Table OA21. Composition Stability — Difference-in-Discontinuities on Pre-Determined Characteristics

Characteristic	**Non-cosigned $\hat{\beta}_3$**	**Cosigned $\hat{\beta}_3$**	**DiDisc $\hat{\Delta}$ (cos − solo)**
Log income	-0.0533^{**} (0.0243)	+0.0163 (0.0760)	+0.0851 (0.0773)
Applicant age (ord.)	−0.0163 (0.0239)	−0.0204 (0.1055)	−0.0228 (0.1118)
Female applicant	+0.0208 (0.0175)	+0.0921 (0.0877)	+0.0514 (0.0893)

Notes: DiDisc on pre-determined characteristics not co-determined with loan size. An insignificant cosigned-minus-solo differential ($p > 0.10$) indicates no differential composition shift at the cutoff. SE clustered by lender.

Crucially, I test for *Sorting into Contracts*: if the reform caused safer borrowers to switch from solo to cosigned status to mitigate the fee hike, I would see a jump in the probability of cosigning at the threshold. I estimate the difference-in-discontinuities with an indicator for a cosigned application as the outcome, on the sample and specification of the denial result: applications above

120% of area median income within ±$67.1k of the limit, year effects with year-specific slopes, lender fixed effects and standard errors clustered by lender. The controls omit the co-applicant's age, which defines the indicator, and income, which includes the co-applicant's. The estimate is $\hat{\beta}_3 = -0.26$ percentage points (standard error 1.05, $p = 0.81$), against a pre-reform cosigning share of 8 to 11% near the limit. There is no detectable change, although the 95% interval does not exclude a shift of about two percentage points in either direction. The estimate is insignificant at ±$50k and ±$90k as well; only on the full sample at the widest window does the cosigned share fall (-1.19 pp, $p = 0.015$), which runs against switching into cosigning rather than toward it. The same condition covers exits from the sample. A reform that moved applicants above the limit to an FHA loan, to a jumbo loan, or out of the market would thin the density just above the limit, the same signature as heavier bunching, and the Difference-in-McCrary of Appendix C.3 finds no change in that density's jump in either pool. That test is more than a balance check. The model of Section 3 predicts that a fee increase drives Bad-type parents out of the cosign pool faster than Good-type parents, because the obligation the fee raises enters the cosigning propensity with weight $(1 - \pi_\theta)$. A composition response would therefore have surfaced precisely here, and finding none is a direct test of the time-invariant-composition condition on which (OA34) rests rather than a demonstration that the sample happens to be balanced.

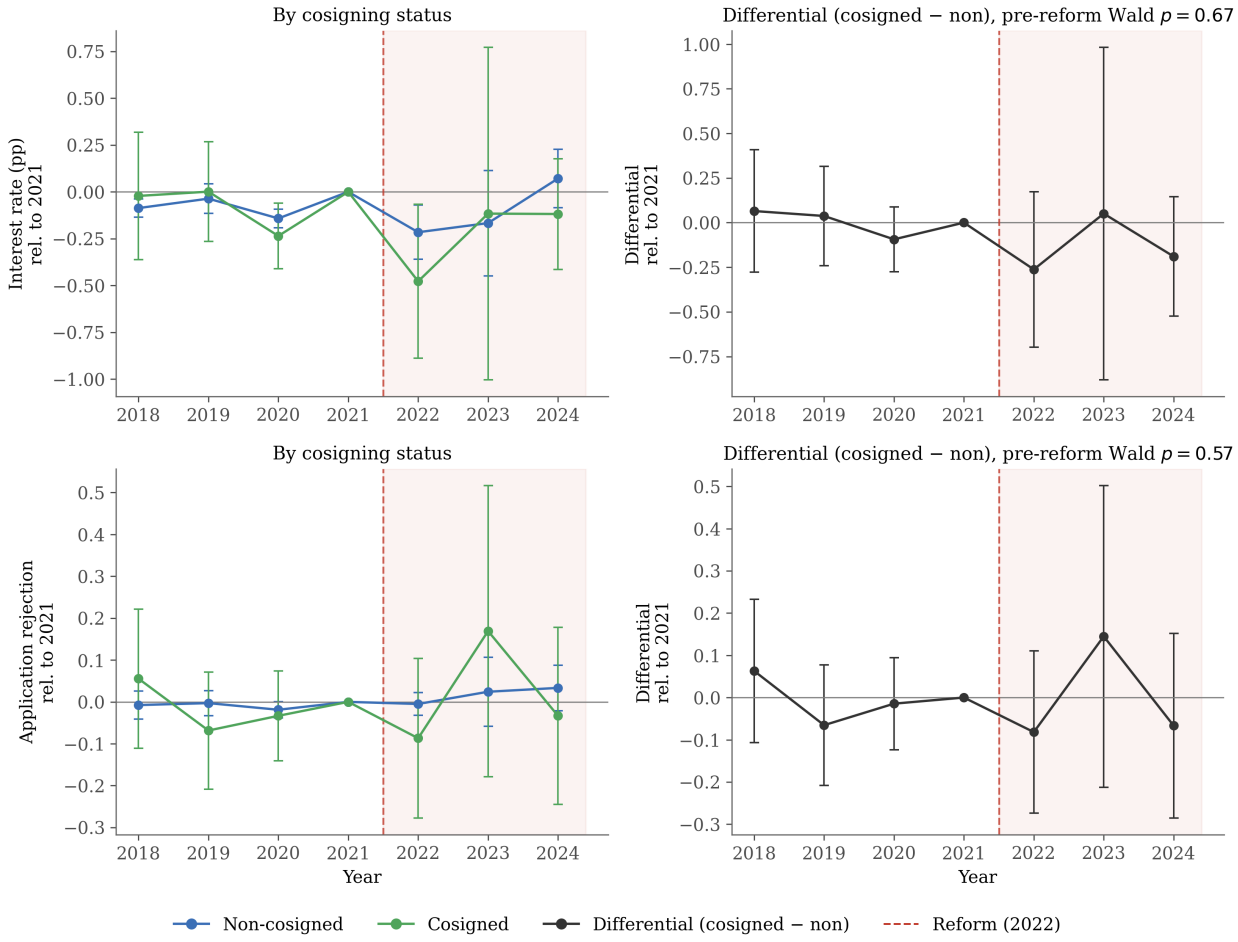
OA5.5 Parallel Pre-Trends: The Event Study

A second threat is that cosigned and solo borrowers might have been on diverging trajectories before the reform, so that the post-reform divergence reflects pre-existing dynamics rather than the cost shock. I assess this with an event study: I re-estimate the discontinuity at the limit separately for each year, normalized to the 2021 baseline, for each cosigning group.[6] Figure OA7 plots the resulting year-by-year series (left column) and the cosigned-minus-solo differential (right column). The pre-reform estimates are flat and statistically indistinguishable from zero for both groups, and the differential exhibits no pre-trend: joint Wald tests on the pre-reform years fail to reject equality of trends across groups for both outcomes (reported in the panel titles). These tests are precisely the validity check Picchetti et al. (2024) propose for the difference-in-discontinuities design, a stacked regression-discontinuity estimate across pre-treatment periods Wald-tested for a constant discontinuity, which formalizes the time-invariance of the confounding discontinuity that identification requires. The dynamic response emerges only from 2022 onward, consistent with a causal effect of the g-fee shock rather than a pre-existing divergence. The wide 2023 intervals reflect that year's thin above-limit cell and are not evidence of a reversal, and the underlying estimates, including the same event study at every headline bandwidth on the above-AMI sample, are in

[6] One mid-sample institutional change touches the admission margin: since September 2021, Desktop Underwriter applies the 620 eligibility floor to the *average median* credit score across borrowers, while loan-level pricing continues to use the lowest representative score (Appendix OA4.9). The change is common to both sides of the conforming limit, so it differences out of the discontinuity; its only design implication is that 2021 is not a perfectly clean reference year on the admission margin. Re-referencing the event study to 2020 leaves the conclusion unchanged: the joint pre-reform Wald test fails to reject parallel trends on both margins ($p = 0.46$ for the interest rate, $p = 0.23$ for rejection).

`event_study_grid.csv`.

FIGURE OA7. Event-study difference-in-discontinuities, by outcome



Each point is the estimated discontinuity in the outcome at the conforming limit in a given year, normalized to 2021; interest rate on top (percentage points), application rejection below (probability). Left column, the two groups separately; right column, their difference, with the joint pre-reform Wald p-value in the title. Bars are 95% intervals at the RD-optimal bandwidth; shading marks the post-reform period.

OA5.6 Institutional Specificity: Placebo Cutoff Tests

A potential concern is that cosigners and non-cosigners may simply have different "local trends" in interest rate sensitivities or rejection probabilities that would have diverged regardless of the CLL reform. To rule out this possibility, I conduct a placebo experiment by shifting the institutional threshold to an arbitrary point in the distribution where no cost shock occurred.

Rather than rely on a single false threshold, I re-estimate my main specification across a grid of placebo cutoffs, set at fractions 0.70 through 1.30 of the actual Conforming Loan Limit (excluding the true cutoff). At each placebo the running variable is re-centered on the false threshold and observations within ±$5,000 of the true CLL are removed, so the placebo cannot pick up the genuine discontinuity. Table OA22 reports the cosigned-minus-solo differential at every placebo. It

is insignificant at the 5% level at all ten cutoffs for both outcomes (0/10); two rejection placebos (0.90 and 1.15 of the limit) clear 10%. Away from the institutional threshold, then, the two borrower pools respond to loan-size variation similarly enough that no placebo differential is resolved.

TABLE OA22. Placebo Cutoffs — Difference-in-Discontinuities at False Thresholds

Placebo cutoff	**DiDisc $\hat{\Delta}$ (cosigned − solo)**	
(× CLL)	**Interest rate (pp)**	**Rejection (prob.)**
0.70	+0.1916 (0.1544)	+0.0202 (0.0400)
0.75	+0.3106 (0.2323)	−0.0826 (0.0454)
0.80	+0.0044 (0.1812)	−0.0287 (0.0390)
0.85	−0.0380 (0.1756)	−0.0311 (0.0406)
0.90	+0.3064 (0.2287)	+0.0756 (0.0400)
1.10	+0.1455 (0.4399)	+0.0779 (0.0649)
1.15	+0.1063 (0.4836)	+0.1051 (0.0608)
1.20	+0.3955 (0.4310)	+0.0734 (0.0888)
1.25	+0.1496 (0.4144)	+0.0268 (0.0848)
1.30	+0.2062 (0.4083)	−0.0858 (0.1173)

Notes: Cosigned-minus-solo DiDisc at false cutoffs (fractions of the CLL), with cutoffs snapped to the $5,000 reporting-heap grid and a per-placebo donut; observations within ±$5,000 of the true limit removed. Differential insignificant at every placebo for both margins.

Under the retired pooled-intercept specification the true cutoff returned a positive rate differential and a negative rejection differential, whereas at every placebo threshold both differentials are indistinguishable from zero. The test is one-sided in what it can establish, however, and the placebo standard errors say how far: at the upper placebos they run 0.41–0.48, larger than the headline point estimate itself, so at those cutoffs the design could not have rejected a placebo differential equal to the true one. Insignificance at ten placebos is therefore consistent with, but not proof of, a zero placebo effect, and two of the lower placebos (0.75 and 0.90 of the limit) return point estimates near +0.31. Randomisation inference puts the headline differential at the $p = 0.074$ percentile of the placebo distribution — suggestive, and short of conventional significance. That test runs over a finer grid of 26 false cutoffs than the ten tabulated here, since a randomisation distribution needs more draws than a table needs rows. The placebo grid is best read as failing to find the generic loan-size divergence that would explain the headline away, rather than as ruling it out.

OA5.7 Sample: the Above-AMI Primary Specification against the Full Sample

The primary specification restricts to applications with income above 120% of the area median. The reason is a second fee change moving in the opposite direction: from late 2022 the first-time-buyer upfront-fee waiver zeroes the high-balance surcharge for qualifying borrowers, and the threshold is 100% of area median income in general but 120% *in a high-cost area*. This sample is high-cost by construction — a high-balance loan exists only where the conforming limit is raised — so the operative cut is the 120% one, and a restriction at the area median would leave waiver-

eligible applicants in a specification that claims to be waiver-immune. HMDA income on a multi-borrower application is the blended figure, so a cosigned application is mechanically less likely to fall below any income threshold than an otherwise identical solo one, and the waiver therefore lands disproportionately on the *solo* pool: a full-sample differential compares a fee increase with a fee increase net of an exemption granted to one side. Above 120% of area median income no applicant in this sample is eligible and the shock is common by construction.

Under the retired pooled-intercept specification the same design on the full sample gives a rate differential of +0.62 to +0.71 pp at $h \leq$ \$24k against +0.63–+0.74 on the above-AMI sample, and a rejection differential of −0.03 to −0.05 against −0.054–−0.012: the two samples agree on sign and magnitude under that specification. They disagree about the composition of the rate differential. On the full sample the solo discontinuity is flat at every bandwidth (−0.19 to +0.06 pp), so the differential looks like a cosigned-only response; on the primary specification the solo jump at $h =$ \$20k is +0.21 pp against a cosigned +0.88, so it is a gap between two positive responses. The waiver is the natural explanation — it suppresses the below-median part of the solo pool's fee increase — and an earlier draft preferred the above-AMI decomposition on that ground. Under the year-FE primary specification the question dissolves: both samples return a null differential (full sample +0.03–+0.06 pp across $h =$ \$18–24k, Table OA20), and the above-AMI restriction survives as the design's waiver-immunity requirement rather than as a choice between decompositions.

OA5.8 A Second Falsification: the Untreated Jumbo Boundary

The placebo-cutoff grid of Appendix OA5.6 moves the threshold within the high-cost sample. A sharper test moves the *sample* instead, holding the threshold institutional. In non-high-cost counties the baseline conforming limit is a genuine boundary — loans above it are jumbo and GSE-ineligible — but the 2022 high-balance fee did not apply there. A differential at that cutoff cannot be pass-through of the reform, so a null is what the design requires.

TABLE OA23. Falsification: the Jumbo Boundary in Non-High-Cost Counties (Untreated)

h (\$000s)	Non-cosigned $\hat{\beta}_3$	Cosigned $\hat{\beta}_3$	DiDisc $\hat{\Delta}$ (cosigned − solo)	N_{non}	N_{cos}
18	+0.9731*** (0.1161)	+1.3006* (0.7770)	+0.3274 (0.7700)	66,780	2,380
20	+0.9506*** (0.1189)	+1.2664 (0.7867)	+0.3158 (0.7800)	67,727	2,396
22	+0.7910*** (0.1200)	+1.1999 (0.7839)	+0.4089 (0.7746)	72,786	2,555
24	+0.7545*** (0.1201)	+1.1570 (0.7819)	+0.4025 (0.7719)	73,046	2,564
26	+0.7606*** (0.1036)	+1.4386** (0.5990)	+0.6780 (0.5982)	80,493	2,849
28	+0.7884*** (0.1030)	+1.5408*** (0.5882)	+0.7525 (0.5877)	83,945	2,973

Notes: Identical design at the baseline conforming limit in NON-high-cost counties, where loans above the cutoff are jumbo (GSE-ineligible) and the 2022 high-balance fee did not apply. A null differential here rules out that the retired differential reflects generic cosigned-vs-solo divergence at loan-size thresholds. 2022-only post window. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table OA23 does not deliver a clean null. The cosigned-minus-solo differential runs +0.32 to +0.75 pp across the grid, insignificant at every bandwidth but of the same sign and a larger

magnitude than the +0.19 to +0.21 pp the treated cutoff returns when its post window is closed before the January-2023 LLPA re-grid.

OA5.9 The DTI Margin: Stretch and Leniency

Two facts about the debt-to-income margin, documented directly from the underwriter-reported DTI in HMDA (computed on *joint* income for multi-borrower applications), pin down the mechanism behind the rejection insulation of Proposition OA3. First, cosigned applications are *not* sparse near the underwriting threshold — they concentrate at and above the 43–45% range (Table OA24; Figure OA8): families stretch on joint income rather than using it to create slack, consistent with the leverage evidence from the NSMO. Second, the rejection hazard behaves asymmetrically through that range, and the asymmetry is on the passage *through* the cap rather than on the approach to it. From the 43–45 band to 46–49 the solo rejection hazard rises +0.46–+0.68 pp (largest $p = 0.0077$) while the cosigned hazard does not move at all; the cosigned-minus-solo difference is −0.88–−0.64 pp and is significant at 5% in all three specifications (largest $p = 0.045$), including on pre-reform years alone, which is the right test of leniency as a standing institutional fact rather than a reform artefact (Panel B of Table OA24). Over the *approach* to the cap the differential runs the other way, cosigned applications closing on the threshold faster (+0.39 pp without controls), so the endpoint contrast from 36–42 to 46–49 nets the two segments against one another and returns -0.25 pp, indistinguishable from zero (Panel C). Both are reported, and the through-cap contrast is the one the insulation mechanism is about. The institutional basis is the two-track underwriting rule of Section 2: an occupant-only 43% ceiling applies on the manual track only, while the automated track evaluates the blended ratio of all borrowers within an overall risk assessment (Fannie Mae, 2020). Threshold *leniency*, not threshold sparsity, is what insulates the cosigned rejection margin — and it implies the fee shock's predicted rejection differential is an order of magnitude below this design's detectable range, so the near-zero rejection difference-in-discontinuities is read as a bounded null rather than a precise zero.

The margin the leniency story predicts. If the insulation works through the debt-to-income cap specifically, the reform's rejection response should concentrate on denials that cite debt-to-income as a reason, rather than on rejection in general. Table OA27 runs the retired pooled-intercept design on that outcome (2022-only window). The cosigned-minus-solo differential is negative at every bandwidth and shrinks monotonically as the window widens (−0.044, −0.036, −0.029, −0.015), the sign the leniency mechanism predicts, but it is insignificant throughout (standard errors 0.022–0.047). Under the year-FE primary specification on the pooled window the solo leg carries conventional stars: Table OA25 puts the solo response at +0.96–+1.95 percentage points, significant at 5% at every window and confirmed by a score wild cluster bootstrap ($p \leq 0.022$), with every cosigned estimate at or near zero. The placebo standard is applied in two batteries, and their disagreement is itself informative. The far battery — 20 cutoffs displaced ±$70–160k, estimated at the wide windows — ranks the real cutoff's t second of twenty-one ($p_{RI} = 0.095$), and its

TABLE OA24. DTI Density and Rejection by Cosign Status — Threshold Leniency, Not Sparsity

Panel A: shares and raw rejection rates by DTI zone

DTI zone	**Solo share**	**Cosigned share**	**Solo rejection**	**Cosigned rejection**
36–42	47.1%	40.0%	4.40%	3.82%
43–45 (cap)	24.3%	24.1%	4.93%	4.73%
46–49	28.6%	35.8%	5.59%	4.75%

Panel B: passing through *the cap — the step from* 43–45 *to* 46–49 *(pp)*

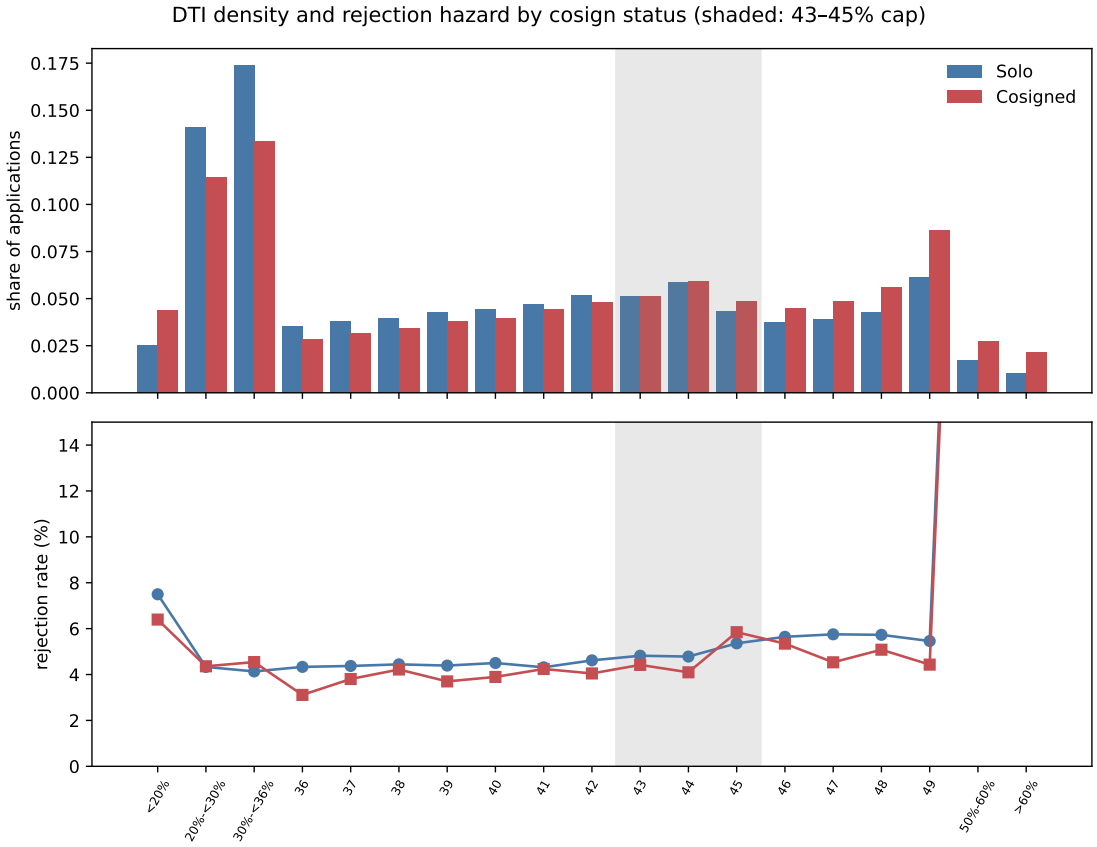
	Solo step	**Cosigned step**	**Cosigned − solo**
No controls	$+0.66^{***}$	$+0.02$	-0.64^{**}
	(0.17)		(0.32)
Year, county, lender FE + controls	$+0.46^{***}$	-0.21	-0.67^{**}
	(0.17)		(0.32)
Pre-reform years only (2018–21)	$+0.68^{***}$	-0.20	-0.88^{**}
	(0.17)		(0.41)

Panel C: the endpoint contrast — 36–42 *to* 46–49 *(pp)*

	Solo step	**Cosigned step**	**Cosigned − solo**
No controls	$+1.19^{***}$	$+0.94$	-0.25
	(0.17)		(0.27)
Year, county, lender FE + controls	$+0.97^{***}$	$+0.54$	-0.43
	(0.21)		(0.28)
Pre-reform years only (2018–21)	$+1.26^{***}$	$+1.00$	-0.26
	(0.22)		(0.34)

Notes: Conventional first-lien applications with underwriter-reported DTI in the integer brackets 36–49, spousal-joint co-applicants dropped; N = 544,622 solo / 37,917 cosigned. HMDA's DTI is computed on *joint* income. Panels B and C are the *same* linear probability model of rejection on three DTI zones interacted with cosign status, lender fixed effects absorbed, errors clustered by lender; they differ only in the omitted zone, so the fit is identical and only the reported contrast changes. Panel B omits the cap zone and therefore reports the passage *through* the threshold, which is the margin the insulation claim is about; Panel C omits 36–42 and pools that passage with the approach to the cap, over which the differential runs the other way (cosigned applications approach the cap *faster*: +0.39 pp without controls), which is why the endpoint contrast is closer to zero. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

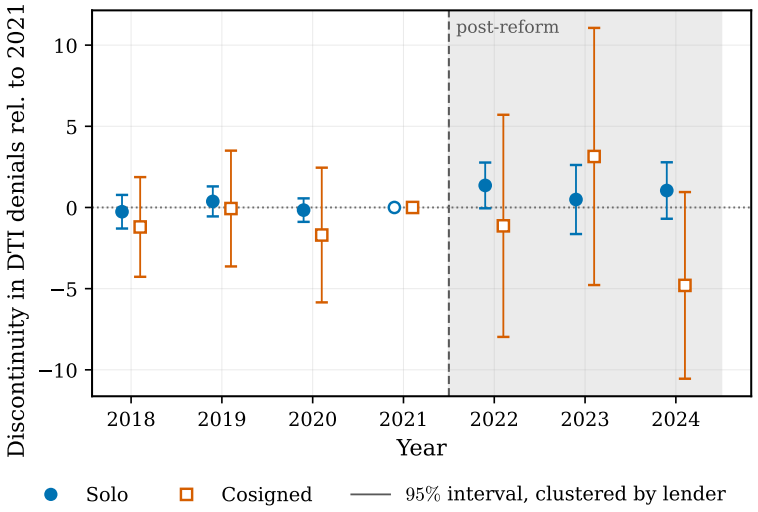
FIGURE OA8. DTI density and rejection rate, by cosign status



DTI density (top) and rejection rate (bottom) by cosign status across the underwriter-reported DTI distribution; the shaded band marks the 43–45% underwriting threshold.

window-stability extension finds 2 of 12 eligible placebos reproducing the real estimate's full profile ($p_{stab} = 0.23$): at distant thresholds this design does generate window-stable pseudo-responses. But distant is the operative word. The above-side far placebos sit in ceiling-selected counties against real area limits, and the far territory's composition differs from the cutoff's own. The near battery closes the gap: 12 placebo thresholds at ±\$30–80k — the estimate's own neighbourhood, in \$10k steps that preserve the reporting-heap geometry, each estimated at the same narrow windows as the real cutoff, valid because no treatment discontinuity exists at any of them (below the limit nothing is treated, above it everything is) — produces a placebo distribution exactly as dispersed as the standard errors claim, and *no* placebo reaches the real estimate's $|t|$ at either bandwidth: the real cutoff ranks first of thirteen at both ($p_{near} = 0.077$), where the local response is +2.39 pp ($t = 2.50$), roughly double the wide-window estimate. The noise floor grows with distance from the cutoff; at the cutoff itself the response stands clear of it. The solo response is therefore reported as a detection, localised to the threshold's neighbourhood, with the far battery disclosed as the bound on how far that claim travels. A *positive* cosigned response would have counted against the leniency reading, and none appears. Panel B of Figure 6 in the body draws the near bat-

FIGURE OA9. Year-by-year discontinuity in DTI-cited denials, by pool



The discontinuity in denials citing debt-to-income at the conforming limit, estimated year by year relative to 2021 for each pool on the above-AMI sample at $h = \$67.1$k, with year-specific slopes on each side, controls and lender effects; percentage points with 95% intervals clustered by lender. Hollow markers at 2021 are the normalisation; shading marks the post-reform years. The solo series is Panel A of Figure 6; no cosigned estimate is distinguishable from zero in any year.

tery; the year-by-year discontinuity for both pools is Figure OA9 below, where the cosigned series is flat before and after the reform (joint pre-reform Wald on the cosigned-minus-solo differential $p = 0.90$; solo alone $p = 0.76$). The year-by-year fit uses the headline window, \$67.1k, so that the pre-trend test runs on the sample the estimate is made on; that window is the MSE benchmark for the pooled jump, not for a single year's, and since each year's jump is estimated on a fraction of the observations its own optimum would be wider (the $n^{-1/5}$ scaling puts it near \$90k) and its interval narrower. Re-estimating at $h = \$90$k changes nothing of substance: the solo pre-reform years stay flat (joint Wald $p = 0.84$), and 2022 is +1.46 pp (s.e. 0.58) against +1.36 pp (s.e. 0.72) at the headline window. Underlying estimates: `rejection_variants.csv`, `denial_wcb_ri.csv`, `denial_ri_stability.csv`, `denial_ri_near.csv`.

The two grids behind Figure 5 are Table OA25 for the denial margin and Table OA26 for rejection, both below.

TABLE OA25. The Reform and DTI-Cited Denials

	Panel A: above-AMI (primary)			Panel B: full sample		
h (**$000s**)	Solo $\hat{\beta}_3$	Cosigned $\hat{\beta}_3$	DiDisc $\hat{\Delta}$	Solo $\hat{\beta}_3$	Cosigned $\hat{\beta}_3$	DiDisc $\hat{\Delta}$
40	$+1.95^{***}$	-0.71	-2.66	$+0.91$	-0.77	-1.68
	(0.59)	(3.31)		(0.58)	(3.69)	
50	$+1.72^{***}$	-1.35	-3.07	$+1.27^{**}$	-0.95	-2.22
	(0.58)	(2.84)		(0.52)	(3.31)	
60	$+1.30^{**}$	-1.62	-2.92	$+1.24^{***}$	-2.61	-3.85
	(0.54)	(2.70)		(0.47)	(3.08)	
67.1†	$+1.11^{**}$	-0.11	-1.22	$+0.84^{*}$	-1.84	-2.68
	(0.47)	(2.58)		(0.48)	(3.08)	
70	$+1.08^{**}$	$+0.67$	-0.40	$+0.79^{*}$	-1.08	-1.87
	(0.47)	(2.53)		(0.47)	(3.03)	
76.6†	$+0.96^{**}$	-0.14	-1.11	$+0.75$	-1.82	-2.57
	(0.43)	(2.08)		(0.46)	(2.33)	
80	$+0.96^{**}$	-0.09	-1.05	$+0.70$	-1.86	-2.55
	(0.42)	(2.00)		(0.46)	(2.27)	
90	$+1.05^{***}$	-0.16	-1.21	$+0.83^{*}$	-1.48	-2.31
	(0.40)	(1.98)		(0.42)	(2.16)	

Notes: Outcome: application denied with debt-to-income cited as a reason (HMDA), in percentage points. Specification, controls and clustering as in the year-FE primary (15). Bootstrap and placebo batteries: Online Appendix OA5.9. †: this outcome's MSE-optimal bandwidths, solo then cosigned; the solo optimum is the estimate Section 5 leads with (on the heaped support, Online Appendix OA5.12). Panel A samples: 35,490–92,643 solo, 3,839–10,341 cosigned; Panel B: 76,206–213,082 and 5,147–14,433. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

TABLE OA26. Pass-Through to Application Rejection — Year-FE Difference-in-Discontinuities (Primary Specification)

h (**$000s**)	**Solo** $\hat{\beta}_3$	**Cosigned** $\hat{\beta}_3$	**DiDisc** $\hat{\Delta}$	p_{Δ}^{WCB}	N_{non}	N_{cos}
30	$+0.0142$ (0.0115)	-0.0423 (0.0661)	-0.0565 (0.0691)	0.47	35,490	3,839
40	$+0.0165$ (0.0112)	-0.0303 (0.0618)	-0.0469 (0.0633)	0.50	45,527	4,928
50	$+0.0180^{*}$ (0.0094)	-0.0152 (0.0529)	-0.0332 (0.0557)	0.59	55,779	6,028
60	$+0.0167^{*}$ (0.0088)	-0.0275 (0.0511)	-0.0442 (0.0548)	0.45	65,700	7,142
70	$+0.0106$ (0.0084)	-0.0027 (0.0480)	-0.0133 (0.0512)	0.81	75,257	8,264
77.2†	$+0.0117$ (0.0078)	-0.0275 (0.0464)	-0.0393 (0.0485)	0.46	81,930	9,032
80	$+0.0119$ (0.0077)	-0.0284 (0.0453)	-0.0404 (0.0472)	0.42	83,994	9,269
83.2†	$+0.0118$ (0.0078)	-0.0252 (0.0450)	-0.0369 (0.0467)	0.46	86,390	9,555
90	$+0.0145^{**}$ (0.0073)	-0.0152 (0.0404)	-0.0297 (0.0422)	0.52	92,643	10,341

Notes: Specification (15) on the above-AMI sample; all of calendar 2022 is coded post, HMDA reporting no origination month. Covariates and lender fixed effects as in Appendix OA5.15.1; standard errors clustered by lender, p_{Δ}^{WCB} the differential's score wild cluster bootstrap p-value. Outcome is application rejection in probability units. † covariate-adjusted MSE-optimal bandwidths, benchmarks only. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

TABLE OA27. Differential Response of DTI-Reason Denials — Difference-in-Discontinuities

h (\$000s)	Non-cosigned $\hat{\beta}_3$	Cosigned $\hat{\beta}_3$	DiDisc $\hat{\Delta}$ (cosigned − solo)	N_{non}	N_{cos}
62	+0.0123* (0.0067)	−0.0317 (0.0440)	−0.0440 (0.0473)	137,848	9,030
82	+0.0070 (0.0059)	−0.0288 (0.0307)	−0.0357 (0.0322)	179,269	11,803
102	+0.0066 (0.0049)	−0.0227 (0.0219)	−0.0293 (0.0230)	221,094	14,507
123	+0.0063 (0.0044)	−0.0084 (0.0220)	−0.0147 (0.0223)	264,454	17,361

Notes: Outcome = application denied *with debt-to-income cited as a denial reason* (HMDA). Threshold leniency predicts the cosigned insulation concentrates on this margin: the DTI cap binds solo applicants but not cosigned ones (Selling Guide B2-2-04). 2022-only post window; controls and clustering as in the retired pooled-intercept tables. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

OA5.10 The Mechanical Bound on the DTI-Denial Response

Section 5 reads the solo DTI-denial response of +1.11 pp against the largest response the payment arithmetic could deliver. This subsection works that bound in full, on the estimation sample. Every input is measured on solo applications just above the limit within the $h = \$67.1$k window, above-AMI, in the post-reform years 2022–24, so the bound is sized where the estimate lives rather than on national distributions.

From fee to payment. The fee is tiered by loan-to-value ratio: 75 basis points of the loan amount above 75% and 50 basis points at 60–75% (the untreated tier at or below 60% is excluded from the sample, Online Appendix OA5.14). In the cell, 83% of loans sit in the upper tier and 17% in the lower, an average upfront fee of 71 basis points. The enterprises annualise upfront fees by a present-value multiplier of four (Federal Housing Finance Agency, 2025, fn. 7), so the fee is 0.18 pp of note rate, against the 0.19 pp of the top tier alone that the body uses as an upper bound. At the cell's mean 2022–24 note rate of 5.73%, a 30-year payment per dollar borrowed is $m/(1-(1+m)^{-360})$ with m the monthly rate; raising the rate by 0.18 pp raises that payment by 1.9%.

Who crosses the cap. If the mortgage is the borrower's only debt, the debt-to-income ratio scales with the payment: a ratio r becomes $r(1+g)$ with $g = 1.9\%$, and it crosses the 50% cap when $r \geq 50/(1+g) = 49.05$. HMDA reports ratios between 36 and 49 as whole numbers without stating whether the exact ratio is truncated or rounded into the bin. Taking ratios as uniform within a bin, the share of the reported-49 bin that crosses is $50 - 49.05$, that is 95%, under truncation (the bin is $[49, 50)$), and $49.5 - 49.05$, that is 45%, under rounding (the bin is $[48.5, 49.5)$). Nothing in the 48 bin crosses under truncation; under rounding only its top sliver does, which I ignore.

The bound. In 2022–24, 5.3% of solo applications just above the limit reported a ratio of 49, up from 2.5% before the reform as interest rates rose. That is the observed post-reform share, not a counterfactual: a 1.9% payment increase moves ratios of 49.05–49.99 out of the bin and ratios of 48.1–48.9 into it, so the fee leaves the bin's size roughly unchanged and no imputation is needed.

Multiplying, the fee could have pushed between $5.3 \times 45\% = 2.4$ pp and $5.3 \times 95\% = 5.0$ pp of solo applications across the cap. The estimate of +1.11 pp sits inside that range: a share of between 0.22 and 0.47 of the applicants the fee pushed over the cap were denied for it. Other debts make the payment a smaller part of the ratio, so fewer cross; the bound is therefore an upper bound and the denied share a lower bound.

The cosigned comparison. The same arithmetic on cosigned applications just above the limit gives a reported-49 share of 8.9% in 2022–24 (6.2% before the reform), 1.69 times the solo share, and a mechanical bound of 4.0 to 8.4 pp. The fee's push therefore reaches a larger part of the cosigned pool than of the solo pool, and the cosigned denial response is -0.11 pp (s.e. 2.58): no response the data can detect where the arithmetic predicts a larger one than for solo applicants. That is the leniency of Section 3, the hazard shifted out by the guarantor, though the cosigned standard error is too wide to size it.

OA5.11 Direct Evidence on the Mechanism: The Default Wedge

The body's pass-through results bound the market's repricing of the guarantee at economically small, so the premise doing the work in the mechanism is not a priced gap but a real one. I verify it directly — that cosigned pools *are* riskier, $\bar{\mu}_{co} < \bar{\mu}_{so}$ — using external default data. This step is what distinguishes genuine adverse selection from a benign alternative in which the two pools share the same mean risk but differ only in variance: only the former predicts a positive conditional default differential.

I use the National Survey of Mortgage Originations (NSMO) Public Use File, which contains a direct question on whether the borrower *had to add a co-signer to qualify*, together with loan performance and underwriting variables. The primary outcome is whether the loan is *ever 30/60-day delinquent*. Terminal outcomes (foreclosure, charge-off) understate borrower risk because the cosigner's role is precisely to cure missed payments, so the loan performs even when the underlying borrower is risky; early delinquency is the rung a later cure does not erase. I therefore read the early-delinquency wedge as the primary estimate and the 90+-day and terminal rungs as a severity ladder. I estimate $\text{default}_i = \gamma\,\text{cosign}_i + X_i'\beta$ on pooled cohorts with survey weights, where $\gamma > 0$ conditional on observables is the adverse-selection wedge.

Measurement: the delinquency flags are administrative and pre-cure. The performance outcomes in the NSMO Public Use File are not survey self-reports: they are servicer-reported credit-record variables appended from the National Mortgage Database — a quarterly panel of point-in-time severity codes (`perf_status`) running from "30 to 59 days past due" through foreclosure and charge-off.[7] The ever-30/60-day indicator scans this panel, so a delinquency spell is retained permanently even when the loan is subsequently brought current. This is precisely the pre-cure

[7] FHFA, *Appendix C: National Survey of Mortgage Originations Public Use File Codebook and Unweighted Tabulations*, NMDB Technical Report 2, v7.0 (May 2026), pp. C-146–C-148, https://www.fhfa.gov/data/nsmo.

object the mechanism requires: a cosigned borrower who misses payments and is then rescued by the family still appears delinquent at the early rungs, while the terminal codes record only the failures the cure did not absorb. Two measurement caveats both work against finding a wedge rather than for it. Because the archives are quarterly, a short 30–59-day spell that begins and cures between archive dates is missed entirely — and cosigned spells, cured faster by construction, are missed more often than solo spells — so the early-delinquency wedge is, if anything, attenuated (90+-day spells necessarily span an archive date and cannot be missed). And a family rescue that arrives before day 30 never registers at all, so the measured wedge understates the cosigned pool's underlying missed-payment propensity. (Pandemic-era forbearance, 2020–21, additionally suppressed reported delinquency for both pools.)

Two estimators carry the result, and the difference between their headline numbers is attribution, not disagreement. The estimator of record, which calibrates the model mapping of Appendix OA5.1, is the Firth logit, the bias-reduced choice for a sample with 124 cosigned delinquency events: its within-cohort average marginal effect is +1.47 pp ($p< 0.001$). Table OA28 reports the companion linear-probability ladder on the same sample, whose within-cohort wedge is +1.36 percentage points (significant at the 5% level), a 30% increase over the 4.58% baseline rate: under either estimator, cosigned borrowers are substantially more likely to fall behind. Conditioning on observable credit risk (FICO and CLTV) attenuates the wedge to +0.56pp, with a standard error of 0.56 — a point estimate of the magnitude the mechanism would predict, and one this sample cannot separate from zero.

What this sample could have detected. The decomposition rests on 124 cosigned delinquency events, and it is worth stating what that buys rather than leaving the reader to infer it from a standard error. At s.e. = 0.56pp, a two-sided 5% test has 80% power against a true residual wedge of about 1.6pp and essentially none against residuals below 1pp. The residual the mechanism plausibly implies is in the latter range. A null here is therefore close to uninformative: the honest statement is that the private-information component is not identified in these data, in either direction, and no conclusion — including the conclusion that it is absent — should be drawn from the insignificance. The estimate is robust in magnitude to a logit specification (average marginal effect +0.71pp) and to controlling for subordinate financing, but robustness of an imprecise estimate is not precision.

What the NSMO does establish, at conventional levels and on the outcome the mechanism cares about, is the raw wedge: cosigned pools reach early delinquency substantially more often than solo pools, so $\bar{\mu}_{co} < \bar{\mu}_{so}$. That is what the pricing argument requires. It also shows that most of the measured gap is *observable* risk — lower-FICO, higher-CLTV borrowers admitted through the joint-income door. This is sufficient for the paper's mechanism and worth being explicit about: selection on observables through an admission rule that reads joint income, priced by a rule that reads the weakest single file, produces mispricing without any hidden information. A private-information component would reinforce that account; establishing one requires either a larger sample (a pooled vintage stack, or the restricted-use National Mortgage Database) or a design rather than a control

set, and I do not claim it here.

TABLE OA28. The Adverse-Selection Wedge in Default — Cosigners vs. Solo Borrowers (NSMO)

		Cosign coefficient $\hat{\gamma}$ (pp), controls ladder			
Outcome	**Baseline**	**+Cohort**	**+FICO/CLTV**	**+Full**	**Rel. (Cohort)**
Ever 30/60-day delinquent	4.58%	+1.36**	+0.56	+0.47	+30%
		(0.59)	(0.56)	(0.56)	
Ever 90+ days delinquent	1.78%	+0.51	+0.22	+0.18	+29%
		(0.35)	(0.35)	(0.35)	
Terminal (FC / charge-off / BK)	0.28%	+0.08	+0.03	+0.02	+29%
		(0.11)	(0.11)	(0.11)	

Notes: Linear-probability estimates from the NSMO Public Use File, cohorts pooled with survey weights on a constant sample (N = 42,631; 2,651 cosigned), cosigning being the survey item "had to add a co-signer to qualify". $\hat{\gamma}$ is in percentage points against column 1's baseline; the last column is $\hat{\gamma}_{+\text{Cohort}}$ over that baseline. Ladder: +**Cohort** wave and metro-LMI fixed effects, +**FICO/CLTV** observable credit risk, +**Full** DTI, PTI and income. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

OA5.12 Bias-Aware Inference on a Heaped Running Variable

The conforming loan limit is bunched, and lender originations heap on the $5,000 HMDA reporting brackets, so the running variable is discrete and an MSE-optimal regression-discontinuity bandwidth in the sense of Calonico et al. (2014) is not well grounded. Even an inference-optimal (coverage-error-rate-minimizing) bandwidth (Calonico et al., 2020) presumes a continuous running variable; with a discrete, heaped support conventional local-polynomial confidence intervals undercover regardless of the window, and one response is bias-aware inference rather than a finer bandwidth (Kolesár and Rothe, 2018). I therefore report bias-aware ("honest") inference in the sense of Armstrong and Kolesár (2018) alongside the conventional estimates: the conventional standard error is augmented with a worst-case bound on the local smoothing bias over the class of conditional mean functions whose second derivative is bounded by M, and the estimate and its honest interval are traced across a grid of bandwidths. Table OA29 reports the result. The interest-rate differential is +0.41 to +0.44 pp at the narrow windows the discrete support allows, attenuating toward +0.15 as the window widens, and the rejection differential is a tight near-zero. The honest intervals are wide and include zero — they deliberately price in worst-case curvature on a heaped support.

Why this was an alternative and not the headline. The bias-aware route imposes a second choice that the conventional one does not, and in the opposite direction from the usual reading. `RDHonest` accepts no covariates, so the honest pipeline residualises the *outcome* on the control set before the discontinuity is estimated and then runs the discontinuity on the raw running variable — it never touches the running variable or the treatment. Frisch–Waugh equates that to a covariate-adjusted local regression only if both sides are residualised, so the two are different estimands rather than one estimand with and without a bias correction, and they coincide only under local orthogonality of the controls, which fails here. The paper's covariate convention is the standard

covariate-adjusted local regression (Calonico et al., 2019), which is what the estimates in Section 5 report. Bias-awareness itself is cheap: holding the estimator fixed it costs on the order of 0.01pp. The distance between +0.67 and +0.24 is almost entirely the covariate convention (≈ 0.33 of it), then the kernel (≈ 0.09), with the residualisation frame worth 0.01–0.03 and county-versus-state fixed effects and loan term nothing at all. Two further prices are worth naming. The honest pipeline pools the residualisation across cosigned and solo applications, because a cosigned post-reform cell of a few thousand observations cannot support a lender-effect vector estimated on both pools separately; a formal test rejects that pooling on the numeric controls alone. And the bounded-misspecification-error estimator of Kolesár and Rothe (2018), the natural discrete-support alternative, is computed on every call in this pipeline and used by none of them: its bias bound on this design is roughly ±12 pp on the rate margin and larger than 100 pp on rejection, which is not an inference anyone can report.

TABLE OA29. Bias-Aware (Honest) Inference for the Difference-in-Discontinuities

Bandwidth h ($000s)	**DiDisc $\hat{\Delta}$ (cosigned − solo)**	**Honest 95% CI**
Panel A: Interest rate (pp)		
20	+0.413	[−0.143, +0.968]
25	+0.436	[−0.110, +0.982]
30	+0.176	[−0.265, +0.616]
40	+0.137	[−0.242, +0.516]
50	+0.172	[−0.168, +0.512]
Panel B: Application rejection (prob.)		
50	−0.039	[−0.147, +0.069]
65	−0.043	[−0.141, +0.054]
82	−0.044	[−0.135, +0.048]
100	−0.046	[−0.129, +0.037]

Notes: Bias-aware (honest) inference (Armstrong–Kolesár) on the heaped running variable: the differential with worst-case curvature bias bounded by M, traced across bandwidths. Honest intervals deliberately price in worst-case curvature and are wide.

What the bandwidth grid in that table is not. Each row fixes the worst-case bias at Bmax and sets the curvature bound accordingly, $M = 2\,\text{Bmax}/h^2$. The bound therefore falls with the square of the window, and each row assumes a *different* smoothness class rather than re-estimating one object under one class at several windows. Read the grid as a menu — what the differential looks like if one is willing to assume this much curvature over this much of the support — and not as a robustness check, which is what a constant-M grid would be.

Two further things this table cannot show on its own. First, the bias-aware point estimate is insensitive to the curvature bound itself: varying Bmax widens the interval and leaves the point unchanged. What moves it is the sample and the data window. The +0.41 reported here is the full sample; the +0.28 from the primary-spec pass and the +0.24 from the common-window grid of Table OA36 are the above-AMI sample at successively wider windows. Only the step from +0.28 to

+0.24 is a window effect holding the sample fixed; the larger step is sample and window together, and I do not attribute it to either alone. Table OA30 sets every estimator of this single differential side by side, with the interval that belongs to each, and names the retired headline.

TABLE OA30. Reconciling Every Estimator of the Interest-Rate Differential at $h = \$20$k

Estimator	**What it is**	$\hat{\Delta}$	**95%**
Conventional, above-AMI*	Controls in the local regression (CCFT), donut, lender-clustered SE	+0.6669	[+0.026,
Conventional, full sample	Same estimator, no above-AMI restriction	+0.6651	[+0.052,
Bias-aware, rd_win=$60k	Residualise-first, Armstrong–Kolesár conservative bound	+0.2760	[−0.318,
Bias-aware, common window	Residualise-first, rd_win=$125k, both margins on one grid	+0.2408	[−0.351,
Bias-aware, appendix grid	Residualise-first, separate pipeline, full sample	+0.4130	[−0.143,

Notes: Five estimators of the same cosigned-minus-solo rate differential at $h = \$20$k, differing in covariate convention, sample and data windo row is the retired pooled-intercept headline (Appendix OA5.15.1; the year-FE primary is Table C.1). The 0.43pp gap between the conventiona inference, which costs about 0.01pp; the decomposition is in the paragraph above.

The admission margin. The same machinery applied to DTI-cited denials, the outcome Section 5 leads with, adapts the single-cutoff procedure of Kolesár and Rothe (2018) to a difference of two jumps. Each pool–period cell is fitted separately at a common bandwidth and curvature bound; the response is the post-reform cell less the pre-reform cell; the standard error clusters the per-lender scores across the two cells; and the worst-case bias of the difference is bounded by the sum of the two cells' bounds, which is conservative because it lets the two periods curve in opposite directions. The outcome is residualised on the control set and on lender and year effects, the closest the procedure comes to the year-effect specification (15), which it cannot reproduce: it takes no covariates, no year-specific slopes and a triangular kernel, so the honest point estimate at the solo MSE-optimal bandwidth, +1.56 pp (s.e. 0.50), is not the +1.11 pp of the body. Residualising on lender effects alone gives +1.43 pp, so the year effects are not what separates them.

The curvature bound has to be chosen, not estimated; no data-driven choice can be honest (Armstrong and Kolesár, 2018). Kolesár and Rothe (2018) recommend choosing it from what is plausible in the application, reporting a range, and reading the bound through the heuristic that it lets the conditional mean depart from a straight line by at most $ML^2/8$ over any interval of length L on one side of the cutoff. The bound the body quotes, Bmax = 1 pp, lets the denial rate stray 0.25 pp from a line over any \$67.1k interval, a quarter of its pre-reform level of 0.98%; the rate in fact moves by about 0.2 pp across the whole window before the reform. Table OA31 reports the honest intervals across the bandwidth grid at that bound and at half of it. The solo interval excludes zero at all 9 windows under both; no cosigned interval excludes zero under any bound, including the conventional one.

Tracing the bound at the optimal bandwidth gives the breakdown value. The solo interval excludes zero until the denial rate is allowed to stray from a straight line by 0.52 pp over a \$67.1k interval (an extrapolation error of 2.06 pp at the window's edge), half its pre-reform level and more than twice the 0.2 pp by which the rate actually moves across the window before the reform.

TABLE OA31. Bias-Aware (Honest) Inference for the DTI-Denial Response

	Solo				Cosigned	
h **($000s)**	Estimate	Conventional	Honest, Bmax = 0.5pp	Honest, Bmax = 1pp	Estimate	Honest, Bmax = 1pp
30	+1.90 (0.71)	[+0.52, +3.29]	[+0.46, +3.35]	[+0.30, +3.51]	−0.92 (3.09)	[−7.03, +5.20]
40	+1.91 (0.61)	[+0.71, +3.10]	[+0.66, +3.16]	[+0.51, +3.31]	−0.67 (2.77)	[−6.15, +4.81]
50	+1.84 (0.56)	[+0.74, +2.94]	[+0.69, +3.00]	[+0.55, +3.14]	−0.73 (2.48)	[−5.64, +4.18]
60	+1.68 (0.52)	[+0.65, +2.70]	[+0.60, +2.76]	[+0.45, +2.90]	−0.93 (2.26)	[−5.41, +3.55]
67.1†	+1.56 (0.50)	[+0.57, +2.54]	[+0.52, +2.60]	[+0.37, +2.75]	−0.92 (2.15)	[−5.20, +3.36]
70	+1.50 (0.49)	[+0.53, +2.47]	[+0.47, +2.52]	[+0.33, +2.67]	−0.83 (2.11)	[−5.03, +3.37]
76.6†	+1.40 (0.48)	[+0.47, +2.34]	[+0.41, +2.40]	[+0.27, +2.54]	−0.59 (2.05)	[−4.67, +3.48]
80	+1.35 (0.46)	[+0.44, +2.26]	[+0.38, +2.32]	[+0.23, +2.46]	−0.60 (1.96)	[−4.50, +3.31]
90	+1.26 (0.44)	[+0.40, +2.11]	[+0.34, +2.18]	[+0.19, +2.32]	−0.52 (1.80)	[−4.13, +3.08]

Notes: Outcome: application denied with debt-to-income cited, percentage points; above-AMI sample. Each pool–period cell is an `RDHonest` loca linear fit (triangular kernel, jump at the +$2,500 donut cutoff) on the outcome residualised on the control set and lender and year effects; the respon is the post-reform cell less the pre-reform cell, its standard error clusters per-lender scores across cells, and its bias bound is the sum of the tw cells' worst-case biases. Intervals are 95%; Bmax is the largest extrapolation error allowed at distance h, equivalently a departure of Bmax/4 from straight line over any interval of length h on one side of the cutoff (Kolesár and Rothe, 2018). Conventional: Bmax = 0. †: MSE-optimal bandwidt of Table OA25. Standard errors in parentheses.

`RDHonest`'s rule-of-thumb bound, which reads the $5,000 reporting brackets as curvature, allows 11 times that and yields an interval of $[-7.4, +10.5]$ pp that excludes nothing; on a heaped support the rule of thumb violates its own premise, that local smoothness is no worse than smoothness at large scales.

Second, the per-group jumps are not the reassurance an earlier draft of this appendix took them for. On the common window at h = $20k the cosigned jump is +0.81 pp $[+0.21, +1.41]$ — but the *solo* jump is +0.57 pp $[+0.28, +0.86]$, also bounded away from zero, and bounded away from zero at three of the six windows against the cosigned jump's one. Both pools move under bias-aware estimation, and both move by three to four times full pass-through of the entire reform fee. That is the case for the difference-in-discontinuities, since whatever inflates both jumps is common and differences out. It is not a case for reading either level as pass-through, and it rules out the account in which cosigned borrowers responded to the reform and solo borrowers did not.

OA5.13 A Temporal Placebo: The False Reform Year

The event-study estimates (Figure OA7) and the spatial placebo-cutoff grid (Appendix OA5.6) both speak to parallel trends; I add a direct temporal placebo. Restricting the sample to the pre-reform years and assigning a counterfactual reform in 2020 — two years before the genuine g-fee increase — I re-estimate the design. As Table OA32 shows, the per-group level coefficients capture ordinary pre-period dynamics — the non-cosigned level is +0.05 and statistically significant, reflecting rate movements common to the pool around the false date — but the object the design identifies, the cosigned-minus-solo *differential*, is small and insignificant under both constructions (−0.01, $p = 0.93$ preserve; +0.03, $p = 0.79$ pooled). No differential discontinuity arises at a date with no cost shock, reinforcing that whatever the retired specification measured was tied to the

2022 window rather than to a pre-existing divergence between the two pools.

TABLE OA32. Placebo Reform Year — Difference-in-Discontinuities at a False (Pre-Reform) Treatment Date

Estimate	$\hat{\beta}_3$ **(above × false-post)**	**SE**	p**-value**
Non-cosigned	+0.0502	0.0211	0.018
Cosigned	+0.0405	0.1138	0.722
DiDisc (preserve)	−0.0097	0.1151	0.933
DiDisc (pooled)	+0.0270	0.1005	0.788

Notes: Pre-reform sample with a counterfactual reform in 2020. An insignificant cosigned-minus-solo differential confirms no differential discontinuity at a date with no cost shock.

OA5.14 Fee-Intensity Balance and LTV Reweighting

Because the fee is LTV-tiered, a difference in the LTV composition of cosigned and solo loans could mechanically generate a differential shock. Table OA33 rules this out. Cosigned high-balance loans sit at slightly *lower* LTV than solo (mean 83.2 vs. 85.0) — an imbalance the non-occupant co-borrower overlay produces in part, since it caps LTV/CLTV at 95 (90 on the manual track) against 97 for a solo transaction (Fannie Mae, 2020) — so under the tiered fee they received a marginally *smaller* fee shock (−3.23 bps): the imbalance runs against, not toward, the result. The retired specification's positive differential was nonetheless present *within* LTV bins where the reform fee is equalized, and inverse-propensity reweighting of solo borrowers to the cosigned LTV mix leaves both margins essentially unchanged (rate +0.350 → +0.367; rejection −0.051 → −0.049). Whatever the retired specification measured was therefore not an LTV-dose artifact, and the rejection insulation — which survives the year controls — is not an artifact of cosigned borrowers holding safer collateral. Under the year-FE primary the same reweighting is a ladder row and changes nothing either (−0.03–+0.02 pp; Table OA20).

OA5.15 The Retired Pooled-Intercept Specification: Diagnosis and Archive

Everything below describes the pooled-intercept specification an earlier draft used as its headline. The diagnosis that retired it comes first, and the common-window comparison it turned on follows. No positive rate differential quoted here describes the year-FE primary, whose own ladder is Appendix OA5.2.

OA5.15.1 Year Composition: The Diagnosis

An earlier draft of Section 5 estimated the difference-in-discontinuities with a single *Post* intercept per pool and reported a cosigned-minus-solo rate differential of +0.67 pp ($p = 0.041$) at $h = \$20$k. This subsection documents why that specification is retired and that table with it. The setting stacks three features no single one of which is fatal but whose combination is: the conforming limit

TABLE OA33. Fee-Intensity Balance — Within-LTV-Tier and IPW-Reweighted Difference-in-Discontinuities

Panel A: LTV mismatch and the implied reform fee dose (pre-reform, above-CLL, in-window)			
	Solo	**Cosigned**	**Difference**
Mean LTV	85.01	83.20	−1.82
Implied $E[\Delta g]$ (bps)	62.17	58.94	−3.23
Within-LTV-bin interest-rate DiDisc (pooled)			
LTV ≤ 80		−0.0278 (0.2861)	
LTV 80–90		+0.3066 (0.4719)	
LTV > 90		+1.0731 (0.5904)	
Panel B: IPW-reweighted DiDisc (solo reweighted to the cosigned LTV mix)			
	Unweighted	**IPW-weighted**	p **(unw. → IPW)**
Interest rate (pp), preserve	+0.350	+0.367	0.155 → 0.140
Application rejection, preserve	−0.051	−0.049	0.263 → 0.276

Notes: Panel A compares the LTV distributions behind the tiered 2022 fee (Kolmogorov–Smirnov $p = 2e{-}08$); Panel B reweights solo borrowers by inverse propensity to the cosigned LTV mix. Standard errors clustered by lender. The reading of both panels is in the paragraph above.

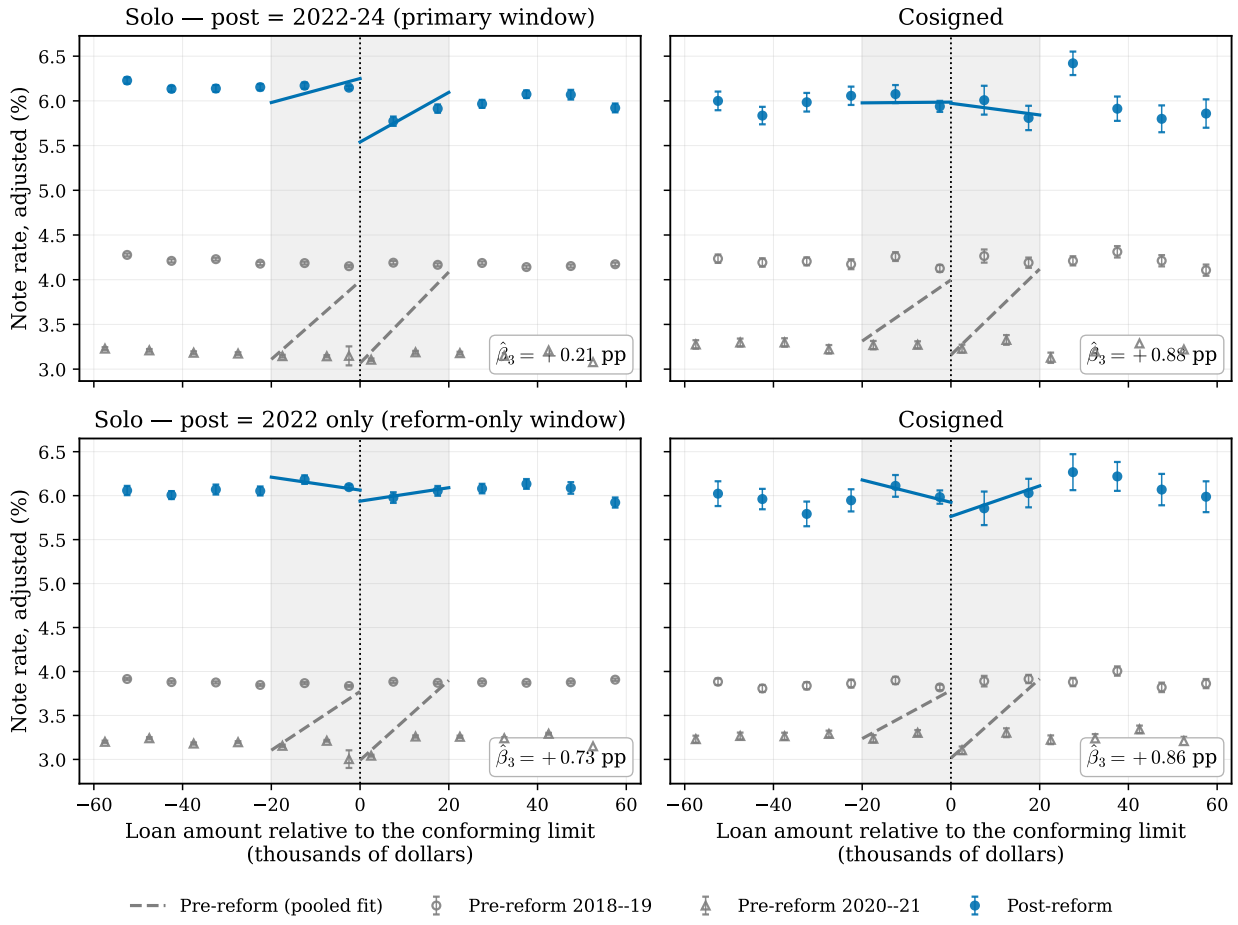
is reset annually, so the pooled design is a multi-cutoff RD with one cutoff per year (Cattaneo et al., 2016); mortgage rates moved three hundred basis points across the sample, so outcome *levels* differ across cutoffs by two orders of magnitude more than the treatment; and the running variable heaps on $5,000 reporting brackets whose offset from the cutoff changes with each reset, so the composition of the estimation window is year-dependent — the above-cutoff share ranges from 17% (2024) to 52% (2020). A pooled intercept then books year-level differences into the fitted jumps wherever the year mix differs across sides of the cutoff.

Table OA34 collects the diagnostics. Panel A holds the estimator fixed and adds year controls: year fixed effects alone flip the differential's sign and tighten its standard error, and year-specific slopes change little more, converging on the primary estimates of Table C.1. Panel B diagnoses the retired specification directly. The placebo row runs it on an outcome that cannot respond to the fee — each loan's origination-year mean below-cutoff rate, constant within year by construction — and recovers most of the retired differential with conventional significance: pure composition leakage, measured under the estimator's own weights. The reweighting row equalises the year distribution above and below the cutoff within each pool and removes the share-level component. Two further facts complete the picture. Estimated one pre-year against one post-year — the retired estimator itself, with nothing to pool — the twelve possible comparisons span -0.42 to +0.23 pp (median -0.03; 9 of 12 negative), so the retired coefficient lies outside the range of every year-consistent comparison and is therefore not a weighted average of them. And a maximally year-saturated event study (Figure OA7) puts the differential at −0.12, −0.07 and −0.13 pp in 2022, 2023 and 2024 against a pre-period mean of −0.00 — flat, and on the opposite side of zero from the retired estimate.

For completeness, the retired specification's own estimates are reproduced below, both margins in one table (Table OA35). The rejection differential, unlike the rate differential, is insensitive to the year controls — the same sign and order of magnitude under both specifications — which is why the body reports its sign with some confidence and its magnitude with none.

Figure OA10 shows the discontinuity itself, pool by pool and before against after, which is the picture the retired specification was reading a single intercept off. The pre-reform period is drawn in its two rate regimes, 2018–19 near 4.2% and 2020–21 near 3.2%, but fitted pooled, because pooled is what the retired equation does: its only time control is the *Post* dummy, where the primary specification (15) gives each year its own intercept and slopes. Neither regime shows a visible step at the cutoff on its own and the pooled pre-reform fit runs between them, which is the year-composition problem in a picture. The drawn jumps reproduce the reported coefficients of Table OA35 (+0.21 against +0.21 solo, +0.83 against +0.88 cosigned, top row).

FIGURE OA10. The rate discontinuity at the conforming limit, before and after the reform



Binned means of the note rate against the loan amount centred on the conforming limit, in $5,000 bins (the HMDA reporting bracket), for solo applications (left) and cosigned (right), on the primary above-AMI sample. The top row codes 2022–2024 as post-reform, the bottom row 2022 alone. Lines are local-linear fits estimated either side of the cutoff over the shaded $h = \$20$k window, on rates adjusted for the control set of Table OA35.

TABLE OA34. Retiring the Pooled-Intercept Specification: Year Composition and the Old Differential

	Cosigned − solo differential, by bandwidth ($000s)			
Specification / test	**18**	**20**	**22**	**24**
Panel A: the differential under progressively year-honest time controls				
Pooled intercepts (retired headline)	+0.63* (0.32)	+0.67** (0.33)	+0.70** (0.34)	+0.74** (0.32)
Pooled triple-difference, no year controls	+0.43 (0.29)	+0.45 (0.29)	+0.48 (0.30)	+0.55 (0.29)
+ year fixed effects	−0.13 (0.20)	−0.13 (0.20)	−0.09 (0.20)	−0.09 (0.19)
+ year-specific slopes	−0.15 (0.20)	−0.15 (0.20)	−0.15 (0.20)	−0.15 (0.20)
Year-FE primary (per-group, cov.-corrected)	−0.01 (0.23)	−0.06 (0.23)	−0.06 (0.23)	−0.05 (0.23)
Panel B: diagnostics of the pooled-intercept specification				
Placebo outcome: year-level mean rate	+0.56** (0.24)	+0.61** (0.24)	+0.62** (0.24)	+0.66*** (0.24)
Year-reweighted (IPW, as Table OA33)	+0.44 (0.32)	+0.48 (0.32)	+0.53 (0.33)	+0.52 (0.32)

Notes: All rows: above-AMI sample, interest-rate margin, lender-clustered standard errors. Panel A adds year controls to a fixed estimator; Panel B's placebo outcome is the origination year's mean in-window below-cutoff rate, constant within year and so unable to respond to the fee. Each row is read in the paragraph above.

TABLE OA35. Differential Pass-Through on Both Margins — Retired Pooled-Intercept Specification (Above AMI)

h **($000s)**	**Solo** $\hat{\beta}_3$	**Cosigned** $\hat{\beta}_3$	**DiDisc** $\hat{\Delta}$	N_{non}	N_{cos}
Panel A. Interest rate (percentage points)					
18	+0.0986 (0.1764)	+0.7264** (0.3501)	+0.6278* (0.3243)	22,453	2,433
20	+0.2137 (0.1796)	+0.8805** (0.3505)	+0.6669** (0.3270)	23,985	2,586
22	−0.0280 (0.1807)	+0.6762* (0.3534)	+0.7041** (0.3353)	25,682	2,768
24	−0.0190 (0.1767)	+0.7181** (0.3393)	+0.7371** (0.3241)	26,882	2,880
26	+0.0875 (0.1416)	+0.4430* (0.2580)	+0.3556 (0.2572)	30,751	3,328
28	−0.0141 (0.1373)	+0.2573 (0.2479)	+0.2714 (0.2558)	32,037	3,476
Panel B. Application rejection (probability units)					
62	+0.0146* (0.0081)	−0.0394 (0.0477)	−0.0540 (0.0512)	67,413	7,350
82	+0.0100 (0.0074)	−0.0345 (0.0430)	−0.0445 (0.0445)	85,640	9,503
102	+0.0119 (0.0073)	−0.0153 (0.0338)	−0.0272 (0.0358)	102,721	11,598
123	+0.0107* (0.0063)	−0.0016 (0.0272)	−0.0123 (0.0287)	119,499	13,772

Notes: The retired pooled-intercept specification (Appendix OA5.15.1; year-FE primaries: Tables C.1 and OA26) on applications above 120% of area median income, post period 2022 onwards (HMDA reports no origination month). Covariates enter the local regression (Calonico et al., 2019): county fixed effects, house-price index, log income, applicant and co-applicant demographics, lender fixed effects absorbed; standard errors clustered by lender, the DiDisc column cross-cluster-corrected. Panel B's grid is wider because the cosigned rejection cell is thin at narrow windows, and its minimum detectable effect at 80% power is 8–14 pp: uninformative intervals, not precise zeros. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

OA5.15.2 Both Margins on a Common Window

The contrast an earlier draft drew from the retired specification — the price moves and the rationing does not — needed one more thing before it can be read that way. The rate differential is estimated at $h \leq$ \$28k and the rejection differential at $h \geq$ \$62k, because each margin's own bias–variance trade-off points there. Read literally, the contrast therefore sets a local estimate against a global one. Table OA36 re-estimates both margins at the same six bandwidths inside the same \$125k window, under both covariate conventions, and reports the number of lender clusters behind each fit — 959 on the rate margin at $h =$ \$20k, rising with the window. Under the convention of record the rate differential runs +0.25–+0.67 pp, positive at all 6 windows and separated from zero at 4; under the residualise-first convention it runs +0.07–+0.24 pp, positive at all six and separated from zero at none. The rejection differential is negative at all six under both (−0.088–−0.025 conventional, that is two to nine percentage points) and separated from zero under neither. What survives across conventions is the ordering rather than either magnitude: price positive, rationing negative, at every window. The rejection differential also attenuates as the window widens, which is consistent with dilution of a local effect. I do not press that reading. The six windows are nested, so the estimates are strongly correlated and a near-monotone attenuation path is close to mechanical; the rate differential's own path across the same windows is not monotone. The two margins also remain estimated on nested but different populations, originated loans against all applications, and no choice of window can change that.

Table OA36. Both Margins on a Common Bandwidth Grid and a Common Data Window, under Both Covariate Conventions

	Conventional (CCFT)		*Bias-aware (AK)*			
h **(\$000s)**	$\hat{\Delta}$	**95% CI**	$\hat{\Delta}$	**95% CI**	**MDE**	G
Panel A. Interest rate (percentage points)						
20	+0.667** (0.327)	[+0.026, +1.308]	+0.241	[−0.351, +0.833]	0.916	959
30	+0.287 (0.254)	[−0.210, +0.784]	+0.074	[−0.399, +0.547]	0.710	1,036
40	+0.487*** (0.182)	[+0.130, +0.843]	+0.114	[−0.299, +0.527]	0.510	1,091
50	+0.463*** (0.167)	[+0.135, +0.791]	+0.195	[−0.182, +0.572]	0.469	1,123
62	+0.369** (0.177)	[+0.022, +0.716]	+0.211	[−0.137, +0.560]	0.496	1,169
75	+0.248* (0.148)	[−0.042, +0.538]	+0.198	[−0.122, +0.518]	0.415	1,209
Panel B. Application rejection (probability)						
20	−0.088 (0.079)	[−0.242, +0.067]	−0.097	[−0.308, +0.114]	0.221	965
30	−0.078 (0.064)	[−0.205, +0.048]	−0.089	[−0.268, +0.091]	0.180	1,047
40	−0.063 (0.058)	[−0.178, +0.051]	−0.076	[−0.240, +0.089]	0.163	1,103
50	−0.049 (0.051)	[−0.149, +0.051]	−0.064	[−0.218, +0.090]	0.143	1,134
62	−0.054 (0.051)	[−0.154, +0.046]	−0.058	[−0.204, +0.088]	0.143	1,183
75	−0.025 (0.047)	[−0.116, +0.066]	−0.047	[−0.189, +0.095]	0.131	1,223

Notes: Both margins at the same bandwidth h inside the same \$125k window on the above-AMI sample, each estimator beside *its own* interval. *Conventional (CCFT)* puts covariates inside the local regression (Calonico et al., 2019); *bias-aware (AK)* residualises the outcome first, so the two estimate different objects (Appendix OA5.12). MDE is the minimum detectable effect at 80% power, G the lender clusters.

Within the retired specification, then, the price margin returned an apparent estimate and

the rationing margin a sign, and the common grid showed the contrast was not an artifact of the windows. The year-composition diagnostics of Appendix OA5.15.1 subsequently retired the price-margin estimate; the rationing sign carries over to the year-FE primary unchanged.

OA6 Quantitative Model: Computational Detail

OA6.1 Mapping the Stylized Game into the OLG Economy

Table OA37 states the object-by-object correspondence between the asymmetric-information game of Section 3 and the quantitative economy of Section 6.

TABLE OA37. Mapping the asymmetric-information game (Section 3) into the OLG economy.

Section 3 object	**Quantitative counterpart**	**Parameters / equations**
Two types $\theta \in \{G, B\}$, repayment $\pi_G > \pi_B$	Same types; π_θ prices the pool, but the default *decision* is endogenized through a type-specific stigma and limited commitment	π_B, $\Delta\pi$; $\bar{\xi}$, $\Delta\xi$ with $\xi_\theta = \bar{\xi} \pm \Delta\xi/2$; V_j^{PD}
Income-ordered prior $\lambda(y)$, $\lambda'(y) > 0$	Prior on the child's type, increasing in own productivity	$\lambda(z_c)$, $\partial\lambda/\partial z_c > 0$
Guarantee $o \in \{0, 1\}$ with offer propensity $q_\theta(y, y_p)$	Two-sided: discrete help choice by the parent at $j = 43$ times the child's take-up at $j = 18$. Both are equilibrium objects under rational expectations	$q_\theta \leftrightarrow \pi_\kappa(z_p) \cdot \Pr(\text{accept} \mid \theta)$; $\kappa \in \{1, 2, 3, 4\}$; σ_{menu}
Taste shock $\eta_p \sim F_\eta$ (full support)	Non-financial help cost plus Type-I extreme-value taste shocks: *location* of $F_\eta \to$ mean cost, *dispersion* $\to$ logit temperature	$\eta_\kappa \leq 0$ (location); σ_{help} (dispersion)
Marginal Cost Dominance (Assumption 1) giving $q_G > q_B$	Not assumed: parents observe θ and internalize a *type-specific* expected liability, larger for Bad children, so $q_G > q_B$ arises endogenously	$L_\theta \approx (1 - \pi_\theta)m$, eq. (OA44)
Bayesian posterior $\Lambda_1(y)$; pool quality μ_{co}	Realised Good-type share of each (h, z_c) pool, read off the equilibrium measure; reported, not priced, under the pricing grid	$q_G^k(h, z_c)$
Rule-constrained pricing $1 + r(y) = 1/\bar{\pi}(y)$, computed on the occupant's own file	Rate *level* from the observable schedule, the same for solo and cosigned loans under the pricing grid; in the counterfactuals a cosigned premium that breaks even on the lender's book	$r_{base}(z_c)$; $P(h, z_p)$, eq. (24)
Guarantee fee g (the FHFA shock)	Above-conforming-limit rate shock — the lever of the natural experiment	rate shock on $h \geq 2$
Wedge: price off y, accept off $Y_{co} = y + y_p$, DTI cap κ	Base rate from the child's own-income risk bucket; eligibility on *joint* income	$\bar{\omega}_{solo} = 0.40$, $\bar{\omega}_{joint} = 0.42$; Ω_{cos}
Three-region cutoffs $y_F < \tilde{y}_G < \tilde{y}_B$ (Prop. 2)	Cross-section of (h, z_c) cells; a region is read off which types take the guarantee	take-up by type, $\Pr(\text{cosign} \mid \theta, z_c)$

Does the estimated economy reproduce the three-region structure? The last row of Table OA37 reads the regions off which types take the guarantee, and that can be checked rather than asserted. In the estimated economy every income state with buyers is in Region 1: both types

cosign. Adverse selection is a matter of degree. It is strongest in the two states just above the lowest with buyers, where Good types take the guarantee at about a third of the rate of Bad types, and weaker toward the top. Region 2 shows in the gains but not in take-up: from income state 9 up only the risky child gains from a cosigned loan (Figure 7), but the choice noise keeps both types taking it. Region 3 never forms. Nor can the rate signature of Proposition OA1, the jump to a separating rate, appear: under the pricing grid the cosign premium is zero in every cell. Priced on losses, in the reform of Section 8, the cosigned loan draws Good types back, its premium is negative at every income, and it jumps nowhere.

OA6.2 When the Fixed Point is Declared Solved

The joint fixed point of Appendix A.8 is reached iteratively, and the stopping rule is not a single tolerance. It is worth stating in full, because the map is not a contraction and a naive rule certifies the wrong thing.

Convergence requires the max-norm of the change across every block — default policy, premium, help probabilities, cash cost and pool composition — to fall below a tolerance. Three qualifications are layered on that. First, a cell carrying no mass can flip on a knife edge forever without the equilibrium being unsettled, so the criterion is also satisfied by an *inhabited-cell* norm together with a mass-weighted default change an order of magnitude tighter; which of the two certified a given solve is recorded rather than assumed. Second, the criterion must hold on *consecutive* iterations, because the pool-composition block cycles when undamped and a criterion met once is as consistent with a cycle passing through the tolerance band as with a fixed point. Third, the state residuals settle well before the reported moments do — the cosigned share can still be drifting when the premium has been inside tolerance for a dozen iterations — so certification is withheld while any reported moment is still moving relative to its own tolerance. Against these, a plateau rule aborts a solve whose inhabited norm stops improving, reporting failure rather than exhausting the iteration budget. The inner inheritance loop carries its own separate tolerance, and the composition block a separate one again. Every certified solve reports the residual $\|BR(q_G) - q_G\|$ at the halt, so the tightness of a fixed point is a number in the log rather than a claim.

OA6.3 Housing Supply and the Construction Sector

In the baseline stationary equilibrium the house price p_{own} is calibrated to match the homeownership rate, which is equivalent to a perfectly elastic supply of owned housing at that price. To evaluate the general-equilibrium incidence of the counterfactual pricing reform of Section 8, I close the model with an explicit construction sector that delivers an upward-sloping supply curve. A representative competitive construction firm produces new housing by combining a fixed factor (developable land) with a variable input through a Cobb-Douglas technology; profit maximisation yields an iso-elastic supply schedule

$$H_s(p) = C\,p^{\varepsilon}, \qquad \varepsilon = \frac{\alpha_c}{1-\alpha_c}, \tag{OA43}$$

where α_c is the variable-input share in construction and ε is the price elasticity of housing supply. This is the construction technology of Castellanos et al. (2024), who use it to study the Irish loan-to-value and loan-to-income reform of 2015 in a model with endogenous house prices and rents. The scale coefficient is calibrated to pass the supply curve through the baseline equilibrium (p^*, H^*), $C = H^*/(p^*)^{\varepsilon}$, so the baseline equilibrium is left exactly unchanged and ε governs only the counterfactual price response. The owned- and rental-housing markets each carry their own supply curve, $H_s^{own}(p_{own})$ and $H_s^{rent}(p_{rent})$, calibrated to their respective baseline prices and quantities.

I set $\varepsilon = 2.2$. Under Cobb-Douglas with competitively priced factors the exponent is $(1-s)/s$, where s is land's share of the value of housing, so the parameter can be read off a land-share series. The usual source is the national accounts: Davis and Heathcote (2007) find that residential land accounts for 36 percent of the value of the aggregate housing stock on average over 1975–2006, rising to 46 percent by 2006, implying elasticities of 1.78 and 1.17. But the aggregate stock is the wrong population for this model. Every buyer here is a first-time buyer, first-time buyers purchase cheaper houses than the average home, and cheaper houses are less land-intensive, so the aggregate land share overstates the fixed factor in the technology that actually constrains them.

That distinction can be measured rather than assumed. Davis et al. (2021) estimate residential land prices and land shares for 54,515 census tracts, applying the assessed-to-market correction and covering the large majority of the U.S. single-family stock. I weight those tracts by the 831,000 young-buyer originations in the 2021 HMDA data of Section 4 and evaluate them at the price point the model's buyers occupy, a price-to-income ratio of 3.56 that the model reproduces at its house-size floor. That gives a land share of 0.306, and hence $\varepsilon = 2.27$. The figure is stable at 0.303–0.310 ($\varepsilon = 2.22$–2.30) across the age definitions and price-to-income targets that could reasonably stand in for the model's buyer. I use $\varepsilon = 2.2$, the round value at the least elastic end of that range, so the calibration understates rather than overstates the ownership gain that survives in general equilibrium.

The gap from the aggregate is entirely a composition effect, and a large one. Sorting tracts by value, the land share climbs monotonically from 0.183 in the decile of tracts where the cheapest tenth of young buyers purchase to 0.577 in the most expensive, a range far wider than the disagreement in the literature, and the reason the population matters more here than the data source. Weighting the same tracts by purchase volume without conditioning on price gives 0.399 ($\varepsilon = 1.51$), close to the national accounts figure and to the 0.356 that Davis et al. (2021) report for all single-family parcels, because mortgage-financed young buyers are not in fact concentrated in cheap markets: the median young-buyer purchase sits at the 67th percentile of the national tract-value distribution. What separates 0.306 from 0.399 is not where these buyers live but which house they buy once there.

The result is corroborated from a second and unrelated direction. Saiz (2010) imposes a common slope across metropolitan areas and estimates $\varepsilon = 1.54$ (his Table III, column 1), placing it inside the range of one to three that the literature proposes at the national level; his population-weighted metropolitan average is 1.75. The value used here sits above both but well inside that

range, which is what one expects given that Saiz estimates an average across all housing rather than the starter segment. For reference, Castellanos et al. (2024) set a land share of 0.33 for Ireland, close to the figure obtained here for U.S. first-time buyers. Every general-equilibrium economy of Section 8 is solved at $\varepsilon = 2.2$. The partial-equilibrium steps of the same ladder are the opposite endpoint, $\varepsilon \to \infty$, at which supply absorbs the shift and the price does not move — which is why the baseline calibration of p_{own} to the homeownership rate is equivalent to perfectly elastic supply. Note that $\varepsilon = 0$ would be the *most* price-responsive case, not the least: with the housing stock fixed the entire demand shift falls on the price.

Two caveats bound what this parameter can be asked to do. First, the empirical literature does not estimate the object in (OA43). Saiz (2010) derives his elasticities from a monocentric-city model in which construction costs are constant and land rents rise as the city expands outward, with no substitution between land and structures; the elasticity he estimates therefore falls as a city builds out, and treating it as the constant exponent here is a local approximation — defensible in this application because the reform moves the house price by less than one percent. Other frequently cited figures are further still from the present object: Aastveit et al. (2023) estimate the response of building *permits*, a construction flow rather than the stock governed by ε, and report it falling from 2.63 over 1996–2006 to 1.75 over 2012–2017. I take from that literature the direction of travel — supply responsiveness has declined — rather than a level. For the same reason I use land's share of the value of the housing *stock* rather than the roughly 11 percent that raw land represents in the cost of a newly built house (Davis and Heathcote, 2007): the fixed factor in (OA43) stands for the scarcity of desirable locations, which is capitalised into the stock and largely absent from acreage at the margin of new construction.

Second, elasticities are considerably lower in the land-constrained metropolitan areas from which the natural experiment of Section 5 is identified: Saiz (2010) puts them below one in coastal California, Miami, New York, Boston and Chicago. Since a lower ε implies a larger price response and hence less surviving ownership gain, the national calibration makes the reported homeownership effect an upper bound for precisely those markets.

The liability the parent *decides* on is not the one their budget pays. Equation (A.16) is the transfer the parent's budget constraint carries, and it is type-blind: once the guarantee is invoked the parent pays, whatever the child turned out to be. The object entering the *help decision* at $j = 43$ is different, and the difference is what generates Marginal Cost Dominance. A parent observes the child's type θ, so the liability they price ex ante is

$$L_\theta(z_p, z_c) \;=\; \sum_h \pi^h_{cos,\theta}(z_p, z_c, h)\; \varphi\,(1-\pi_\theta)\; p_{cover}(h, z_p, z_c)\; m(h, z_p, z_c), \tag{OA44}$$

where $\pi^h_{cos,\theta}$ is the equilibrium distribution of house sizes among cosigned children of type θ. Two features are deliberate. The default probability is the *structural* rate $1-\pi_\theta$ rather than the behavioural one: a parent with private type information expects type-consistent repayment, not the

inflated rate that includes the guarantee's own effect on the child's default incentive, so the moral-hazard channel is absent from the ex-ante expectation while remaining in the realized transfer. And the house-size weights are type-specific, because the two types buy differently.

Because $\pi_G > \pi_B$, equation (OA44) gives $L_G < L_B$: guaranteeing a Bad child is more expensive to the parent than guaranteeing a Good one. That is exactly Assumption 1, which the stylized model of Section 3 imposes by fiat, and here it is an implication of the parent's problem rather than a restriction on it. A reader who took (A.16) alone as the parent's decision object would find the liability identical across types and $q_G > q_B$ unobtainable; the two equations are different objects and both are needed.

The aggregate adding-up condition,

$$\int_j \tau_{j,p}(\mathbf{s}_{j,p})\, d\mathcal{M}_j(\mathbf{s}_{j,p}) = \int_j d_{j,c}(\mathbf{s}_{j,c})\, m_{j,c}(\mathbf{s}_{j,c})\, d\mathcal{M}_j(\mathbf{s}_{j,c}) \tag{OA45}$$

— aggregate parental transfers $\tau_{j,p}$ equal the aggregate payments made on behalf of defaulting children, $d_{j,c} \in \{0,1\}$ being the child's default indicator — then holds by construction rather than as a separate market-clearing requirement, since $\bar{d}$ is itself the equilibrium default policy. What the certainty-equivalent treatment gives up is not the level of the transfer but its dispersion across parents, and hence any precautionary response to having cosigned.

OA7 Structural Estimation: Further Identification and Validation

OA7.1 Parameters Set Outside the SMM

Tables OA38 and OA39 list every parameter fixed before estimation, grouped by how it was fixed: Panel A external evidence, Panel B contract and policy facts, Panel C internal calibration, Panel D structural taste dispersion, Panel E numerical. Every value is read off the solved economy the estimates come from. Section 7 discusses the one that is a normalisation rather than a calibration, the level of the default penalty.

OA7.2 The Type Gap $\Delta\pi$ Against External Evidence

The type gap is estimated (Section 7 and Table D.2). A Bad type repays over the life of the loan with probability $\pi_B = 0.878$ and a Good type with $\pi_G = 0.986$, so the annual involuntary hazards $1 - \pi_\theta^{1/M}$ are 0.43% and 0.05%. Three strands of evidence speak to a gap of this kind. Gupta and Hansman (2022) attribute 30 to 40 percent of the cross-sectional leverage–default relationship to unobserved-type selection. Gerardi et al. (2018) put strategic, able-to-pay default under 1% of defaults, so a type gap should run mostly through ability to pay. And the identified-selection magnitudes in consumer credit (Karlan and Zinman, 2009; Adams et al., 2009; Einav et al., 2012) bound the gap at a few percentage points.

The estimate meets the second and the third. Per year the two involuntary hazards differ by less than half a percentage point, inside the consumer-credit bound. And the gap runs through

Table OA38. Parameters set outside the SMM: external evidence and institutional design

Parameter	Symbol	Value	How set
Panel A: set outside the model (external evidence, or fixed with literature motivation)			
Altruism elasticity to parent income	γ	1.00	unit elasticity (the homothetic benchmark), fixed in the SMM. De Nardi (2004); Lockwood (2012) motivate income-dependent altruism; their luxury finding implies $\gamma > 1$, so $\gamma = 1$ is the conservative choice
Discount factor	β	0.960	annual: 0.96 in Cocco (2005), 0.9606 estimated by Nakajima and Telyukova (2020)
Return on savings	r	0.030	annual, riskless in the model and read as the return on the household's portfolio: the top of the 1–3% peacetime range of real safe returns in Jordà et al. (2019), below their returns on risky wealth; Li et al. (2016) use 2.7%. Mortgage rates sit 1.3–1.8 points above r (data mortgage–T-bill spread: 2.9), so debt costs more than savings earn
Risk aversion	σ	1.50	exactly the 1.5 of Bacher (2022); the modal value in life-cycle housing models is 2 (Kiyotaki et al., 2011; Garriga and Hedlund, 2020; Kaplan et al., 2020), with portfolio-choice settings higher. Low end of the range
Housing expenditure share	θ	0.35	rent and utilities over total expenditure, renters aged 30–40, PSID family files 2019–2021: weighted median 0.35, flat across income quintiles (0.33–0.41). The 0.24 of Davis and Ortalo-Magné (2011) is rent over income (0.26 in the same sample); the model has no income tax, so its utility weight is matched to the consumption share
Housing depreciation	δ_h	0.015	Harding et al. (2007): $\approx$2% net of maintenance (2.5% gross); the model has no maintenance choice, so the net figure is the comparison. Set conservatively below it
Income persistence	ρ	0.96	Heathcote et al. (2010), PSID: $\rho \approx 0.97$
Income innovation s.d.	σ_ε	0.110	Heathcote et al. (2010); at the lower edge of the 0.11–0.21 range in the PSID literature
Age profile of income	P_j	PSID	head labour income by 3-year age group net of year means, PSID family files 2003–2021, heads reporting no retirement; model age j = age $-$ 18. Relative to the purchase age (36): 0.24 at entry, 1.07 at 53, 0.92 at 66 (Appendix A.5)
Panel B: institutional design			
Loan-to-value limit, solo	λ_m	0.90	US conforming: 10% down, as in Benetton et al. (2022). NSMO first-time buyers: mean CLTV 88.1 (median 95)
Loan-to-value limit, cosigned	λ_m^{co}	0.90	**equal to the solo limit**: the NSMO shows no cosigned–solo divergence (median 95 both; means 87.5 vs 88.2; share >90 0.605 vs 0.612)
Recourse fraction	φ	1.00	the US non-occupant cosigner is jointly and severally liable from the first missed payment, not only after foreclosure: full recourse (Section 2)
Mortgage term (years)	M	30	30-year fixed
Conforming-limit house tier	h^{CLL}	2	tier at which the conforming limit binds

Notes: Panels are ordered by how each value was fixed, not by what it does; Panels C–E are in Table OA39. **A** is external evidence, each entry checked against the cited paper, with γ motivated rather than point-estimated there and two entries at or below their anchor range as the row states. **B** records contract and policy facts.

TABLE OA39. Parameters set outside the SMM: internal calibration, taste, and numerical settings

Parameter	Symbol	Value	How set
Panel C: calibrated to an internal target			
Default-penalty level	$\bar{\xi}$	4.40	*normalisation*: held in the SMM. Scaling every default penalty together leaves the targeted moments flat, so the data do not pin it (Appendix D.2)
Rent-to-price ratio	p_{rent}/p_{own}	0.090	gross rental yield on a *constant-quality* unit, so the price–rent ratio of 11.1 is not the quoted 15–20: Urban Institute (2025) find owner-occupied homes $\approx 37\%$ more valuable than rentals after quality adjustment, giving a comparable ratio of 15.2 and a 6.6% yield against a US average of 6.7%
Owner utility premium	ν	1.100	*internal target*: the homeownership rate. Kaplan et al. (2020) also calibrate this parameter internally, to 1.015; the present value is higher, which the rental-grid cap below partly stands in for
Cash-transfer size	ψ_{cash}	0.100	share of parent income at $j = 43$, scaled by house size: 6.0–10.8% of parent income, or 0.2–2.5% of house value (0.8% at the median parent). Sits between the median and mean of the PSID parental-transfer distribution in Brandsaas (2025) ($1,000 / $3,960 / $10,000 at the median / mean / p95, 2016 USD). The *incidence* is a separate targeted moment (24.8% against a target of 25.0%)
Rental-grid cap	h_{rent}^{max}	0.40	*internal target*: makes HO(z) monotone increasing. **Cost:** owner/renter size ratio 3.4× vs 1.5× in data (untargeted)
Type–income gradient scale	λ_α	2.545	normalisation: $\sqrt{1-\rho^2}/\sigma_\varepsilon$
Panel D: structural taste dispersion			
Parental-help taste dispersion	σ_{help}	0.50	*structural*: dispersion in the parent's willingness to help. Deliberately an order of magnitude above the numerical smoothers in Panel E — at 0.1 cosigning vanishes altogether, so the parameter is bounded below by the existence of the margin
Tenure taste dispersion	σ_{rb}	0.75	*structural*: taste dispersion over the rent/buy choice, set so that homeownership responds continuously to the house price; a probe along p_{own} at the estimate confirms it
Panel E: numerical (solution method, not economics)			
Default-choice smoothing	σ_{def}	0.100	logit smoothing; not calibrated
Menu smoothing	σ_{menu}	0.100	logit smoothing; not calibrated
Asset / income / house grid points	n_a, n_z, n_h	30, 11, 6	discretisation
Lifespan (periods)	J	70	ages 1–70

Notes: Continues Table OA38. **C** is internal calibration, chosen so the model matches a stated target and therefore not independent evidence for what the model then delivers; **D** is a preference parameter and **E** affects only the numerical solution. Estimated parameters: Table D.2.

ability to pay: a Bad type's involuntary hazard is 9.5 times a Good type's, more than the 4.1-fold gap in total default among solo mortgage holders, so default by choice, which both types exercise, narrows the type gap rather than creating it. The first strand measures a different object, the share of a leverage–default correlation, and is not reproduced here.

OA7.3 Untargeted Moments

Table 7 in Section 7 reports four moments the fitted model is asked to reproduce without having been fit to them: three held out of the SMM objective, and the wedge above a credit score of 750 (U_3), which enters it but which no parameter moves. This subsection gives the construction of each target; U_3 is built as M_7 is (Appendix D.1). Together they discipline the parts of the mechanism the seven targeted moments leave free: how much of the cosign–solo default gap the parental cure absorbs, what survives to terminal loss, what the gap is above a high credit score, and how much of the rate gap is composition.

U_1 is the recovery side of the cosign–solo wedge: the share of the cosign pool's excess early-stage risk that never reaches terminal loss. In the data it is one minus the terminal-to-pre-cure ratio of the NSMO within-cohort wedge, $1 - 0.08/1.36 = 0.94$ raw and $1 - 0.08/0.56 = 0.86$ with FICO/CLTV controls; both ingredients are attenuated by the administrative retention of delinquency flags, so the range is indicative rather than sharp. In the model it is the mass-weighted coverage probability p_{cover} over delinquent cosigned borrowers. U_2 completes the ladder with the wedge that survives the cure, cosigned delinquency $\times(1 - p_{cover})$ less solo default. At -0.12 pp against the NSMO's $+0.08$ pp the estimated economy resolves too many cosigned delinquencies in the parent's favour, and the attenuation in the data pushes the target up rather than down, so the gap is not a measurement artifact. Together U_1 and U_2 describe how much of the excess the guarantee absorbs.

U_4 is the unconditional cosign rate gap of Section 4 (Fact 3): the Fannie Mae note rate of cosigned loans less that of single-borrower loans, with no controls, $+0.23$ pp (s.e. 0.02). Its model counterpart is the mean contract rate of cosigned less solo loans over every purchase, $+0.07$ pp. Under today's pricing a cosigned loan pays its cell's solo rate, so the model's gap is composition alone: cosigners borrow in cells whose solo rate is higher. In the data, $+0.08$ pp survives the price cell; the model has no channel for that part.

OA8 Counterfactual: The Welfare Measure

OA8.1 Welfare: The Consumption-Equivalent Variation

Welfare in Section 8 is reported as the consumption-equivalent variation

$$\lambda = \left(\frac{V^{CF} + gB}{V^{BL} + gB} \right)^{1/\alpha} - 1, \tag{OA46}$$

with $g = 1/(1-\sigma)$ and $\alpha = (1-\theta)(1-\sigma)$ — at $\sigma = 1.5$ and $\theta = 0.35$, $g = -2$ and $\alpha = -0.325$ — and B the survival-discounted horizon. V is the lifetime value of an agent entering at $j = 18$, so

both stages of the life cycle sit inside it: the same agent reaches the parental stage at $j = 43$ and bears the offer cost and the contingent liability there. The aggregate is taken over the stationary income distribution at entry, and the comparison by income state holds that distribution fixed, so each state's value compares the same children in the two economies.

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